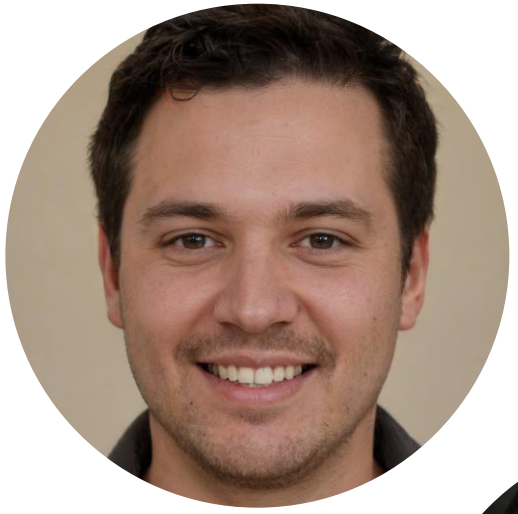
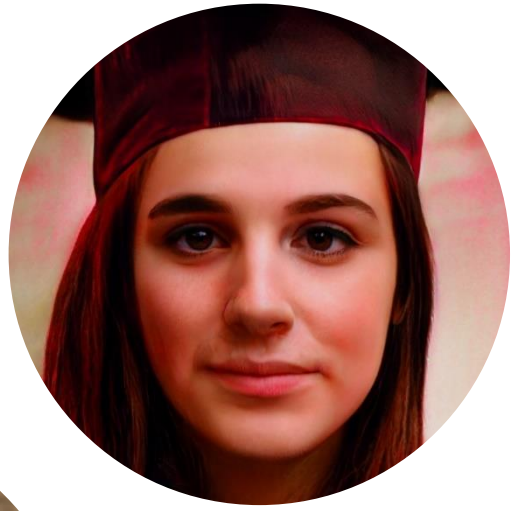
Introduction to Generative Models







Introduction to Generative Models

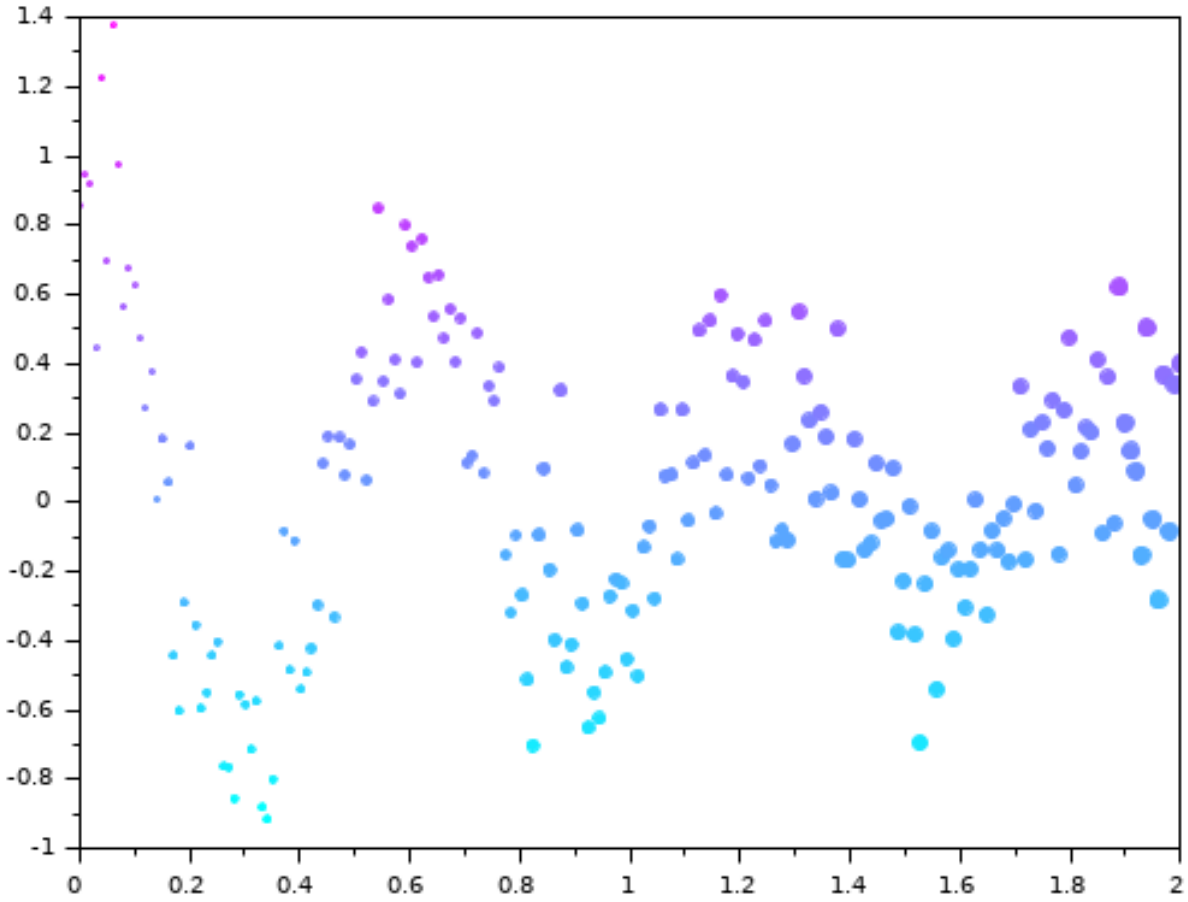


Agenda

Today: Basics, Next week: Advanced

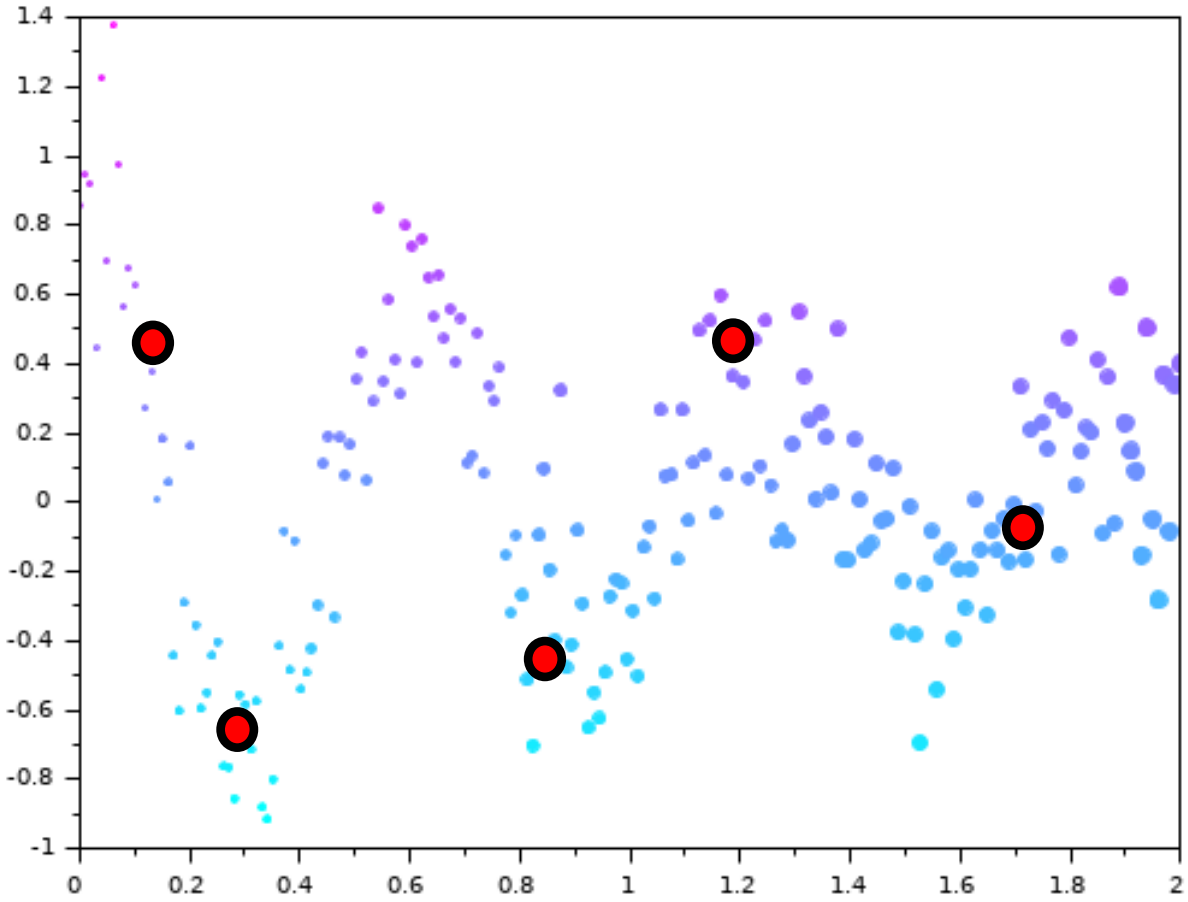
1. Goal, motivation, Basic methods
    1. Parametric methods
    2. Autoregressive methods
    3. Latent space mapping
2. Variational Auto Encoder (VAE)
3. Generative Adversarial Networks (GAN)
    1. Introduction (basic setup, intuition)
    2. Evaluation
    3. Image to image (pix2pix, ~~CycleGAN~~)
    4. Problems and how to improve GAN performance (losses, tricks etc.)
    5. StyleGAN
    6. Extras (GAN Dissection, Single Image)

Goal



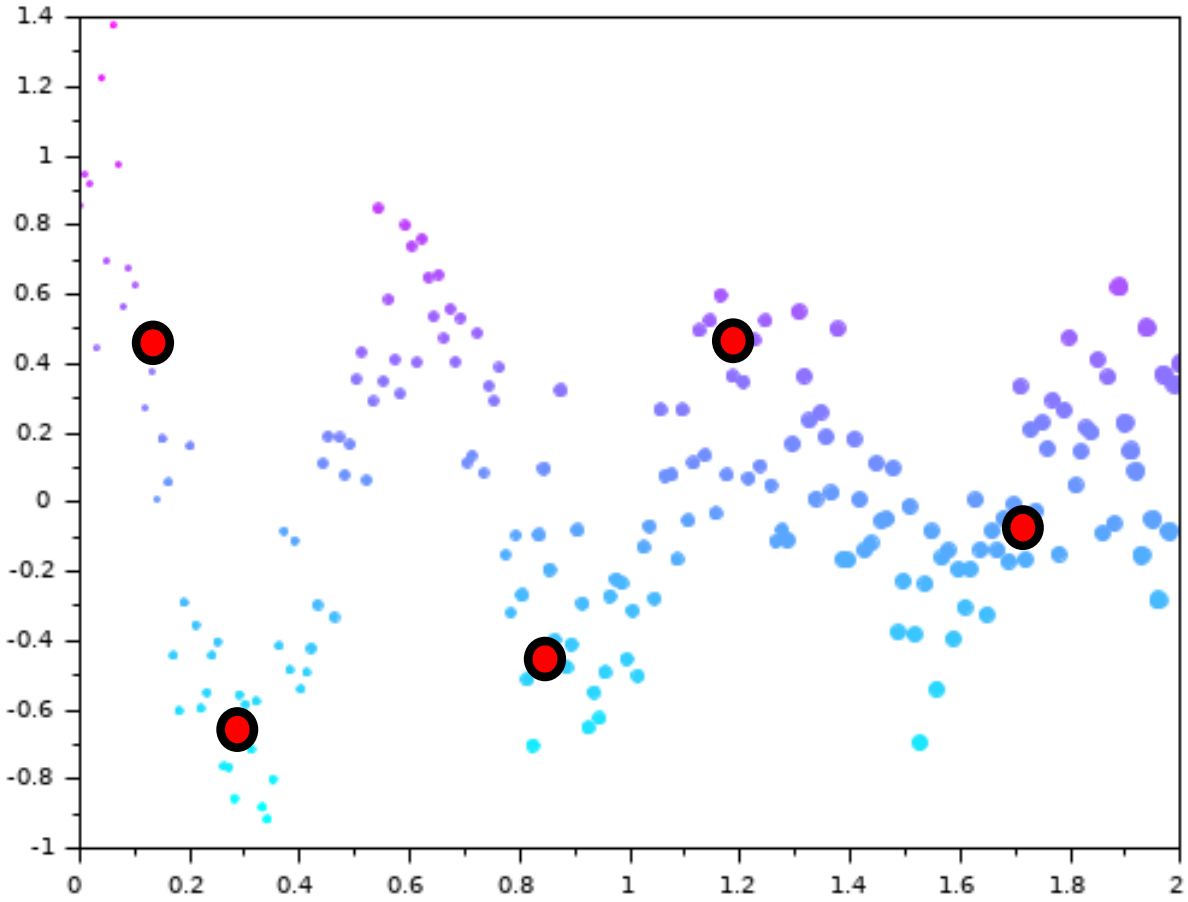
Goal

Images



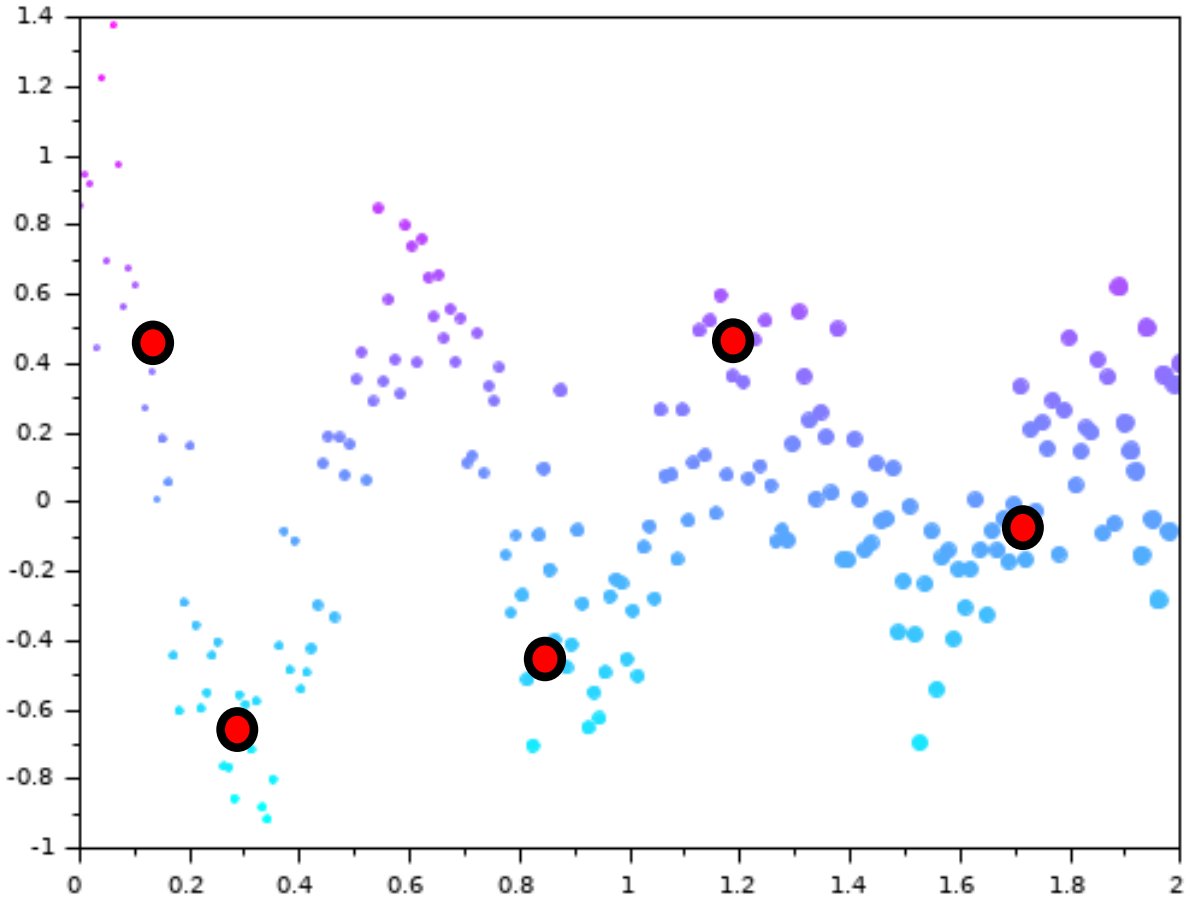
Goal

Images
Text



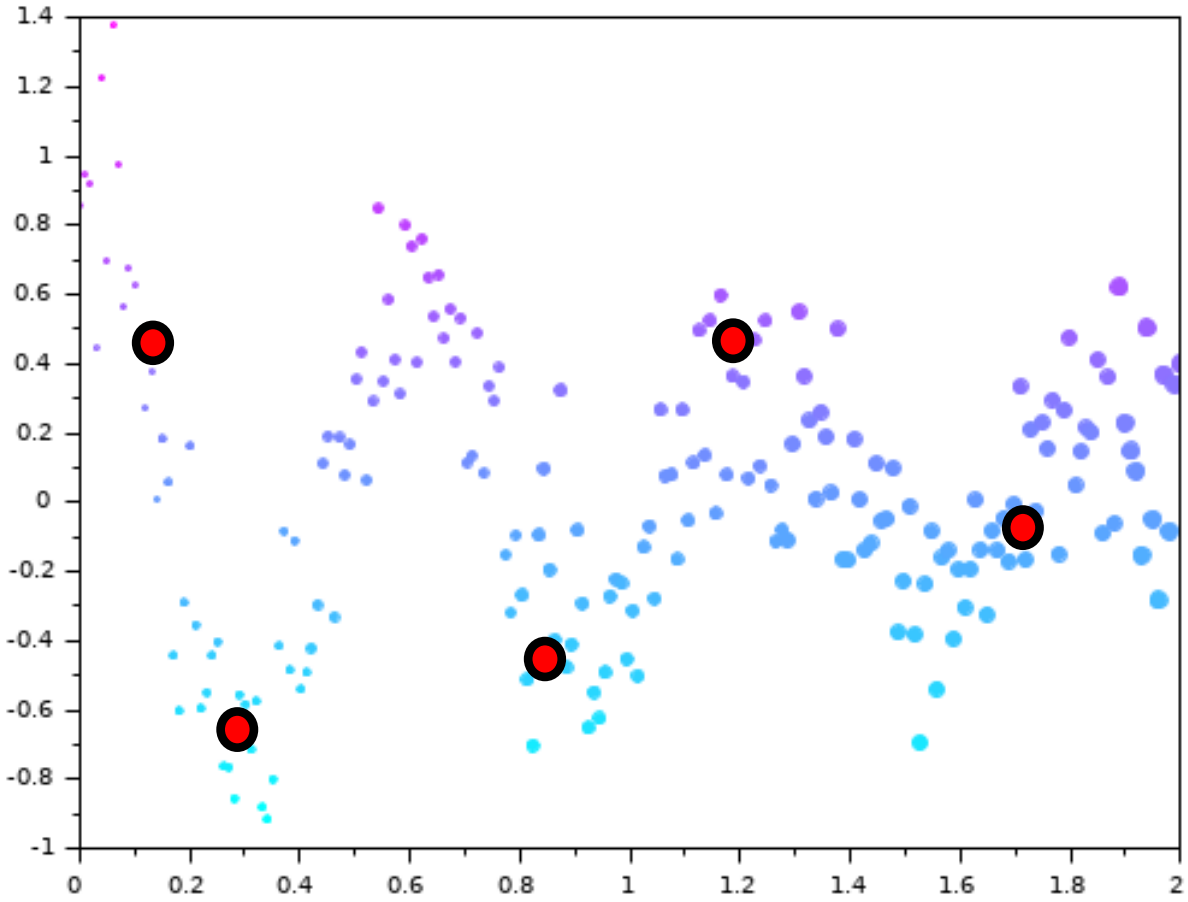
Goal

Images
Text
Audio



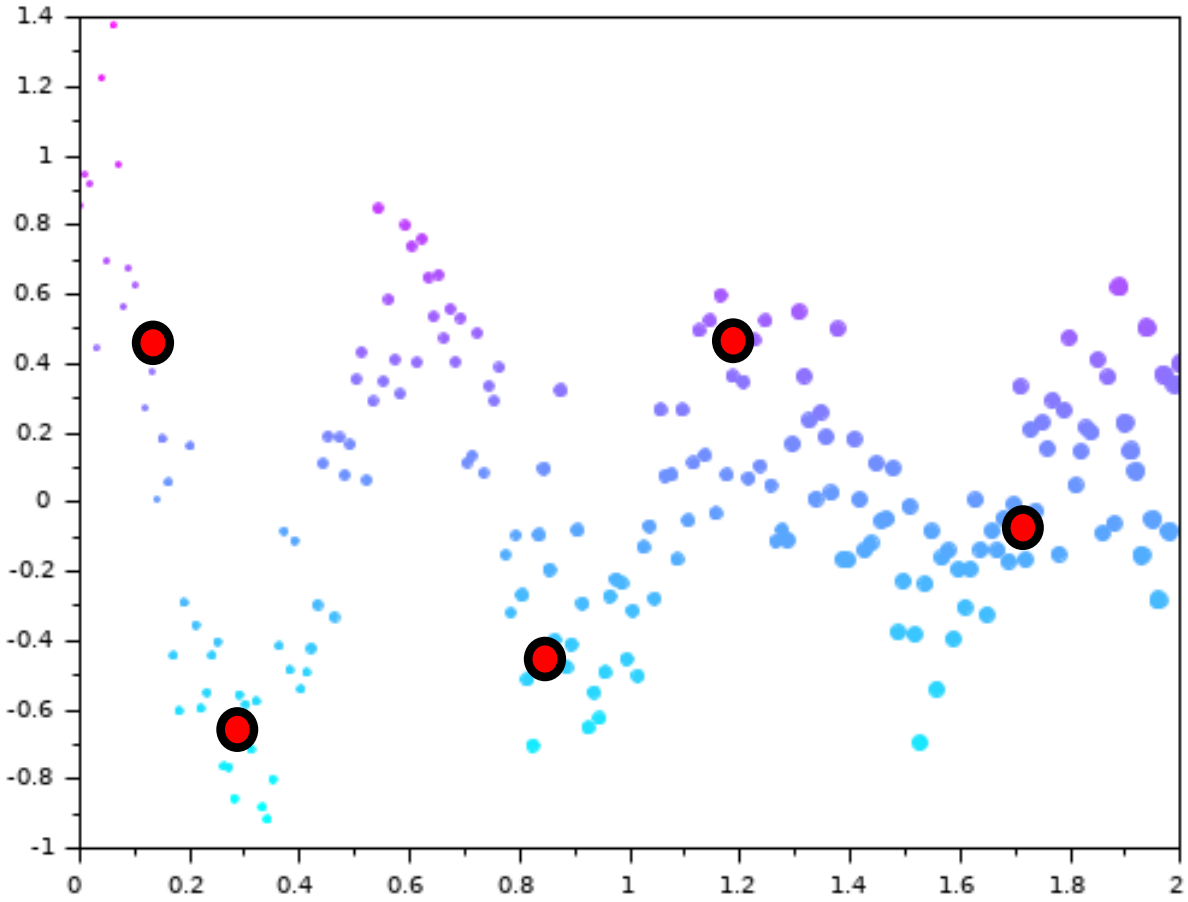
Goal

Images
Text
Audio
Video



Goal

Images
Text
Audio
Video
Whatever…

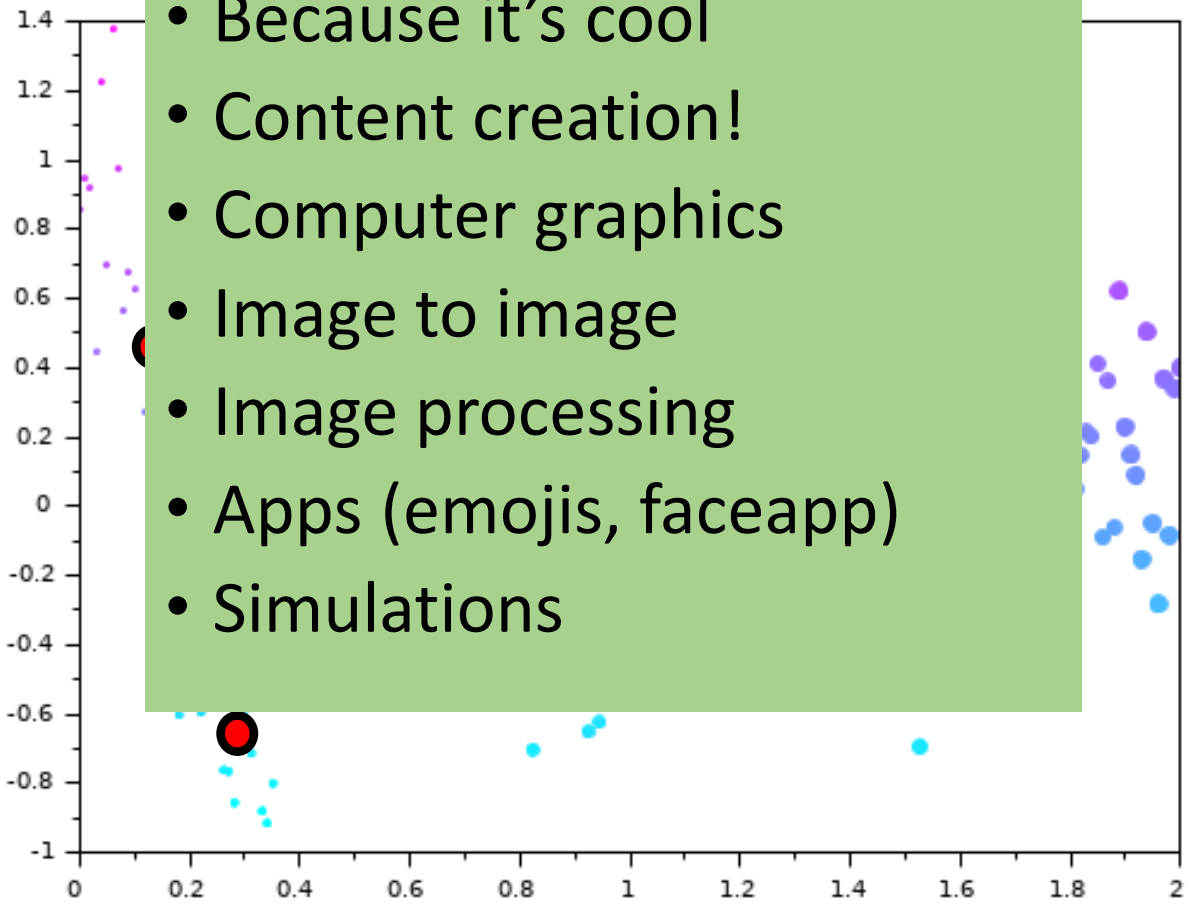


Goal

Images
Text
Audio
Video
Whatever…

Why?

- Because it’s cool
- Content creation!
- Computer graphics
- Image to image
- Image processing
- Apps (emojis, faceapp)
- Simulations

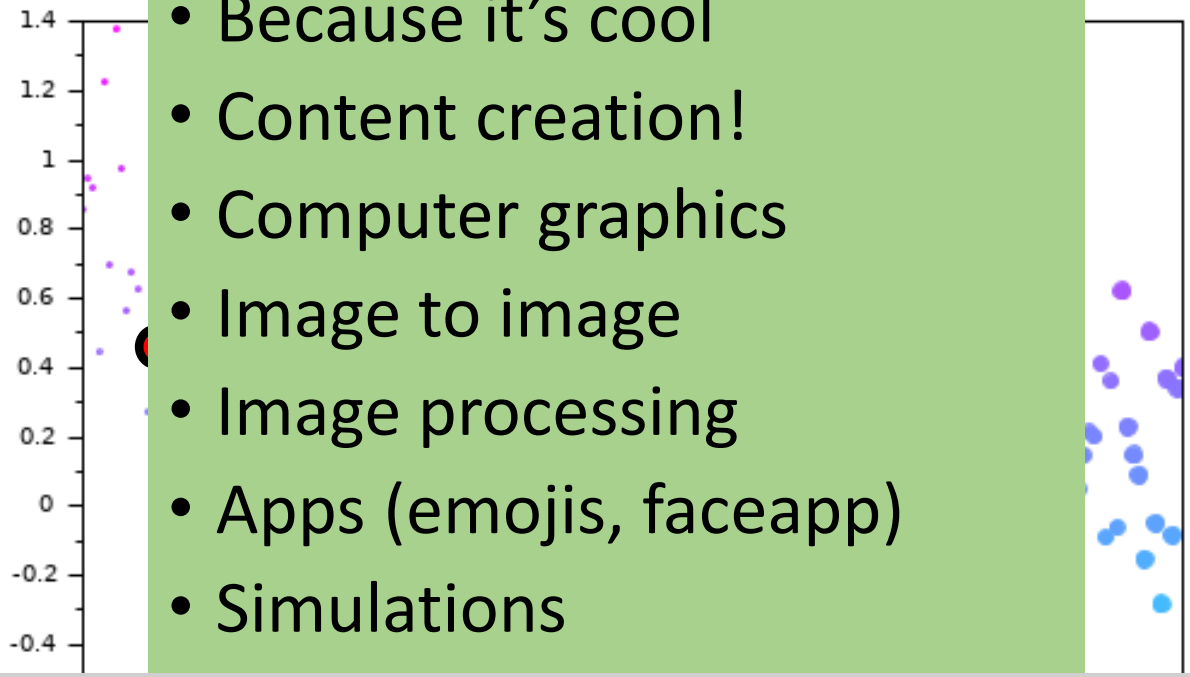


Goal

Images
Text
Audio
Video
Whatever…

Why?

- Because it’s cool
- Content creation!
- Computer graphics
- Image to image
- Image processing
- Apps (emojis, faceapp)
- Simulations



The correct approach when there is more than one valid solution!

Prompt: A bunch of students trying to figure out what generative models are.



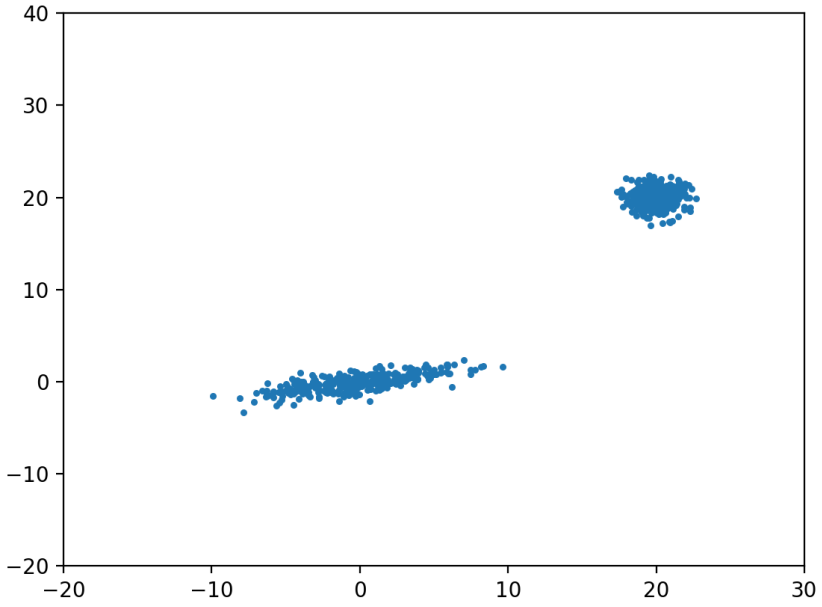


Generative methods:

- Parametric distribution estimation (e.g. GMM)
- Autoregressive models (e.g. RNN, Causal CNN, Diffusion)
- Latent space mapping (e.g. VAE, GAN)

Parametric Distribution Estimation Example: GMM

Step 1: observe a set of samples



Step 2: assume a GMM model

$$p(x|\theta) = \sum_i \pi_i \mathcal{N}(x|\mu_i, \Sigma_i)$$

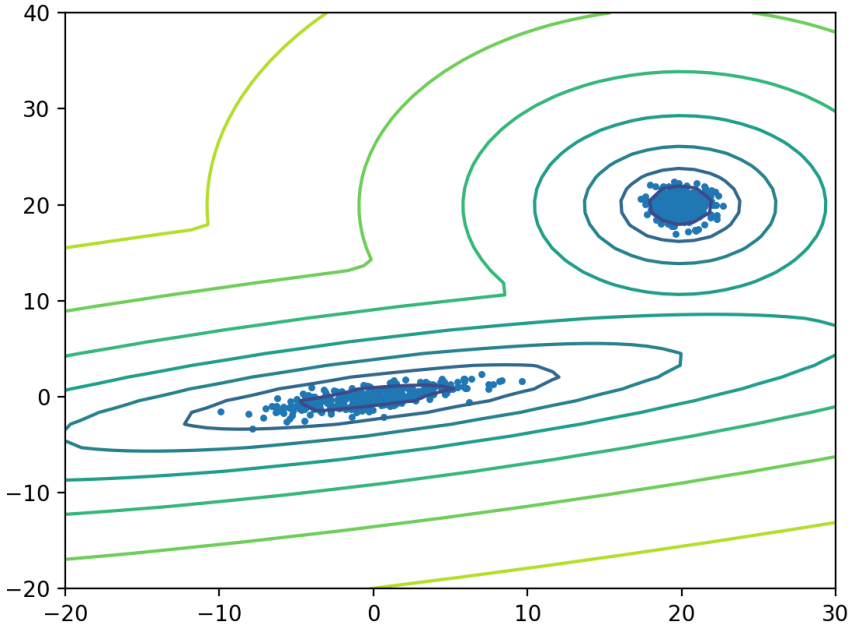
Step 3: perform maximum likelihood learning

$$\max_\theta \sum_{x^{(j)} \in \text{Dataset}} \log p(\theta|x^{(j)})$$

Elements in the slide taken from NVIDIA CVPR'18 GANs tutorial

Parametric Distribution Estimation Example: GMM

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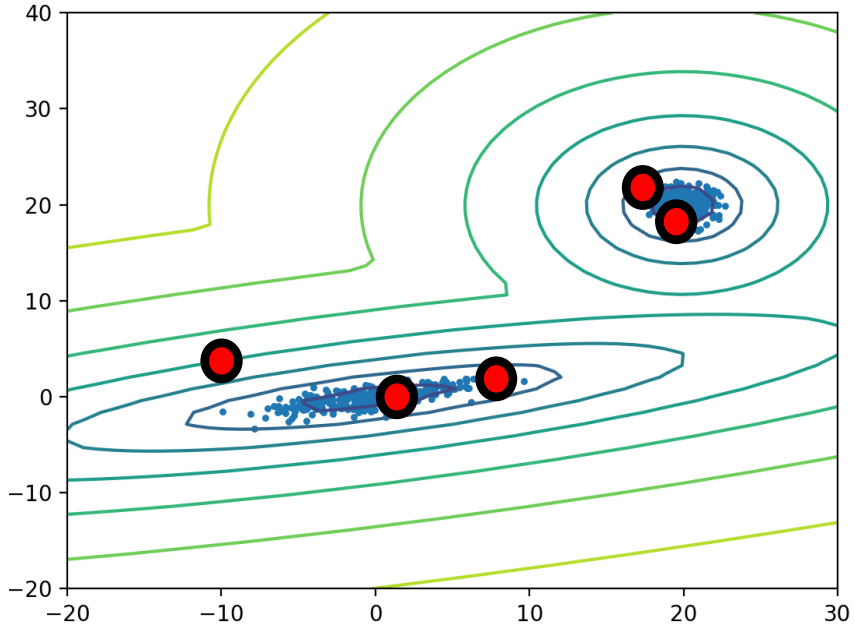
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Elements in the slide taken from NVIDIA CVPR'18 GANs tutorial

Parametric Distribution Estimation Example: GMM

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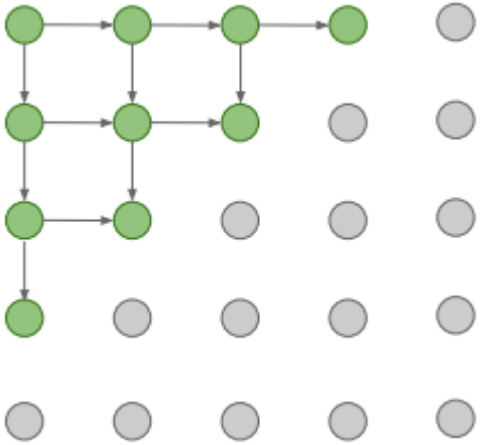
$$\max_\theta \sum_{x^{(j)} \in \text{Dataset}} \log p(\theta|x^{(j)})$$

Step 4: Sample

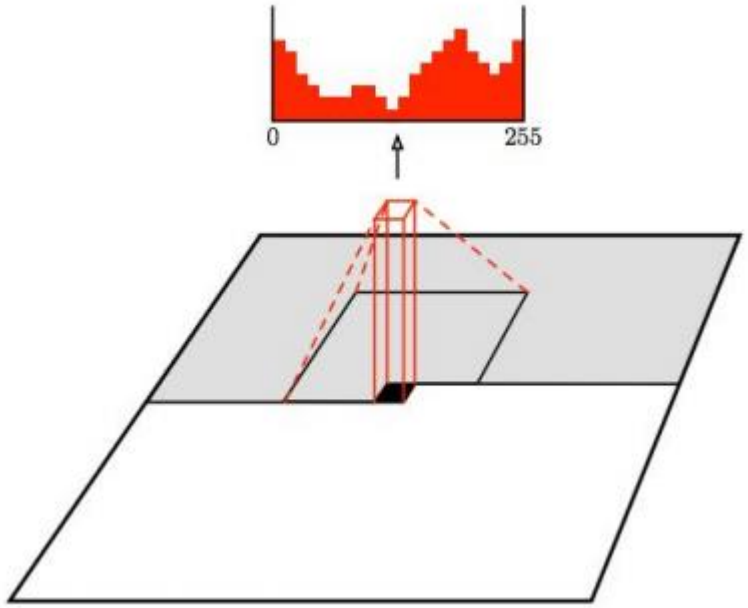
Elements in the slide taken from NVIDIA CVPR'18 GANs tutorial

Autoregressive image generation - Basic

PixelRNN



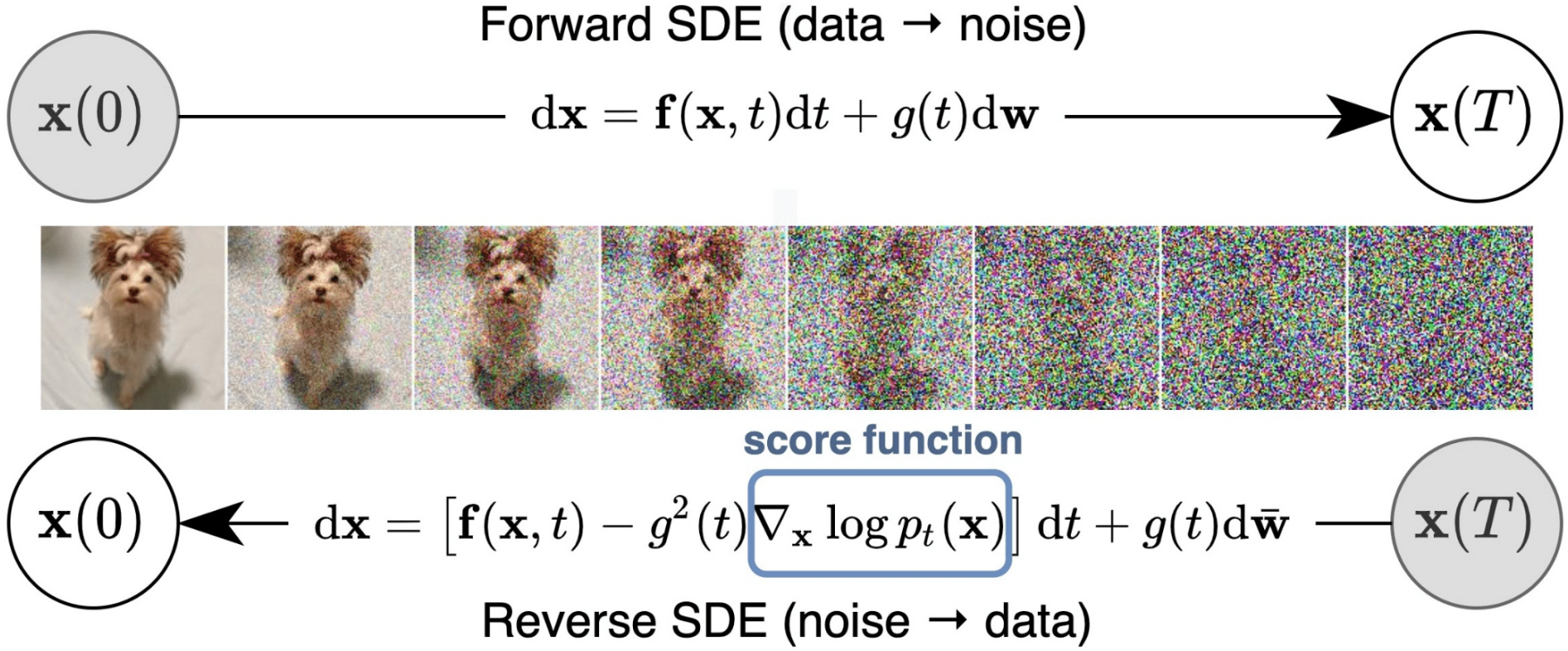
PixelCNN



Van der Oord 2016

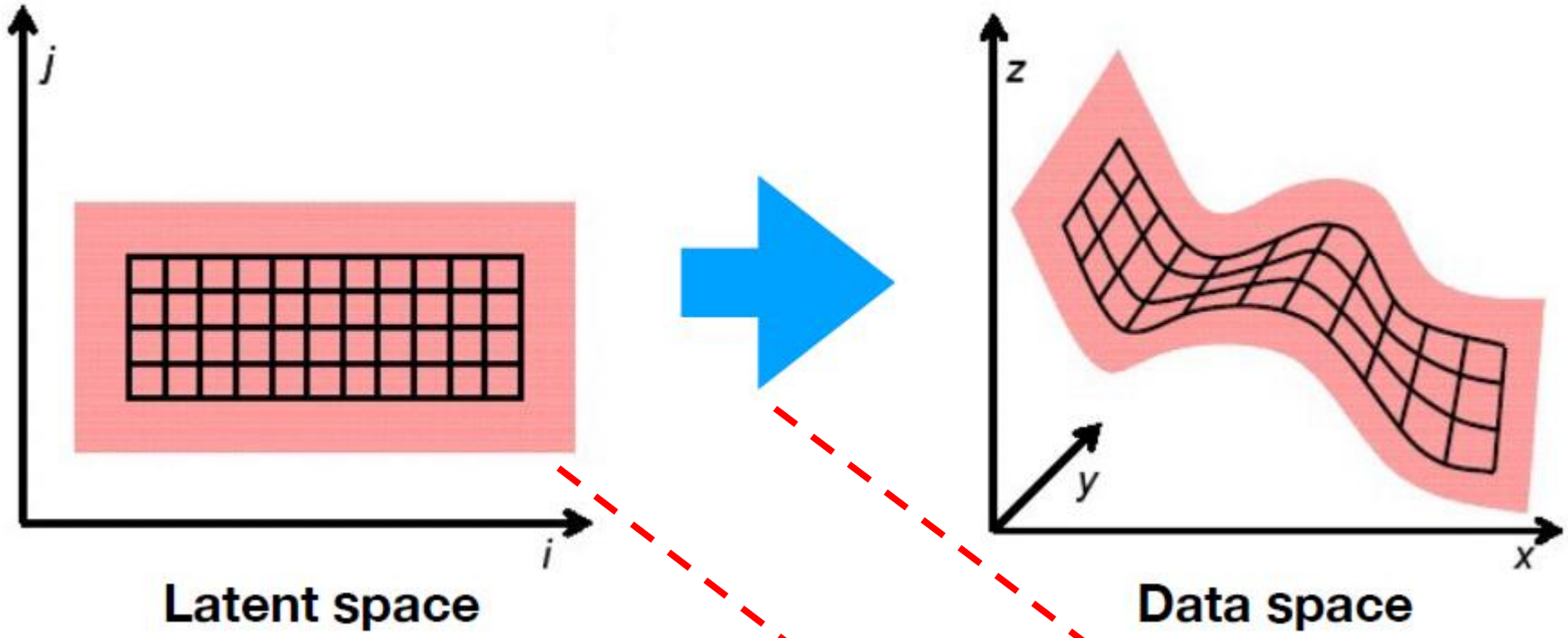
Autoregressive image generation - Recent

Diffusion models:



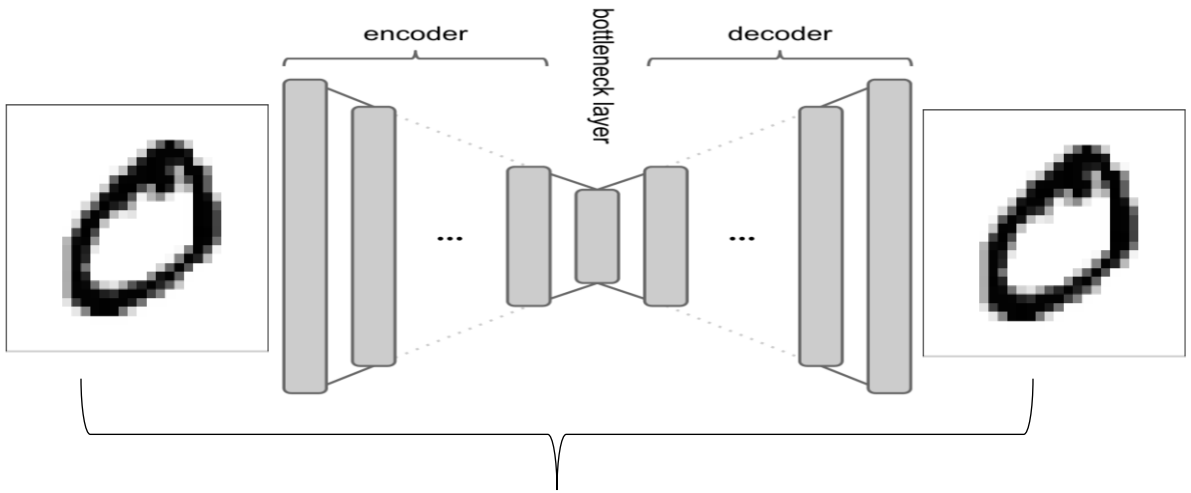
Wait for the advanced generative models class!

Latent space mapping approach

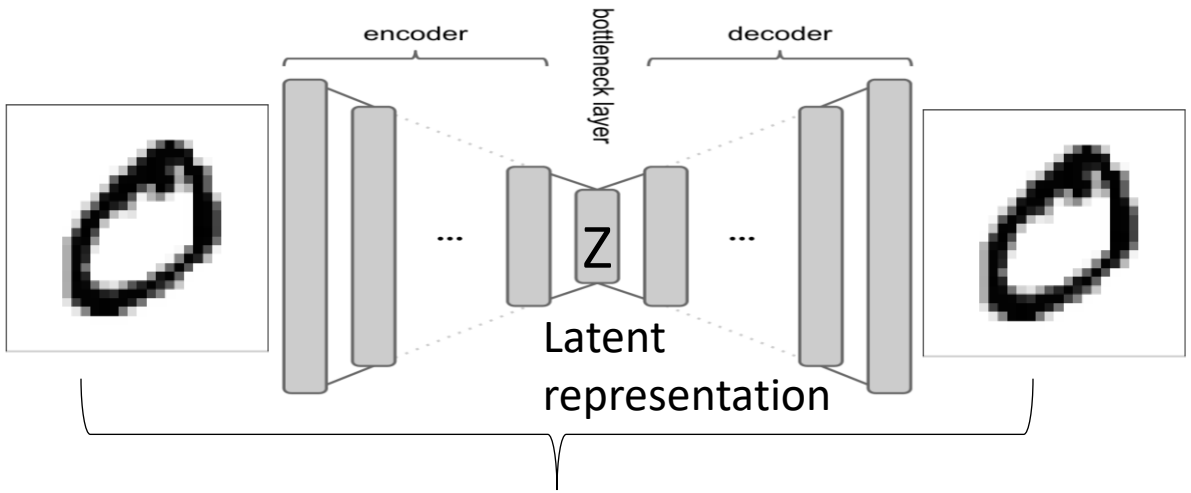


Data likelihood: $p_\theta(x) = \int p_\theta(z) p_\theta(x|z) dz$

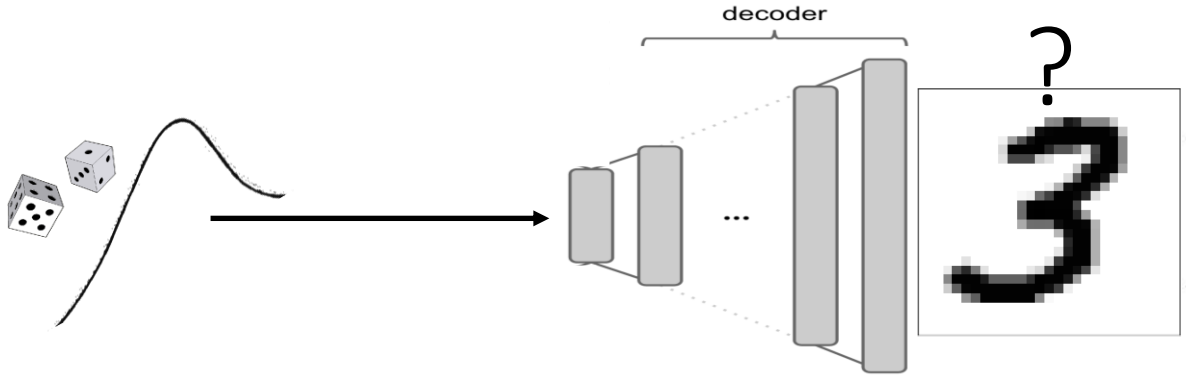
Autoencoders

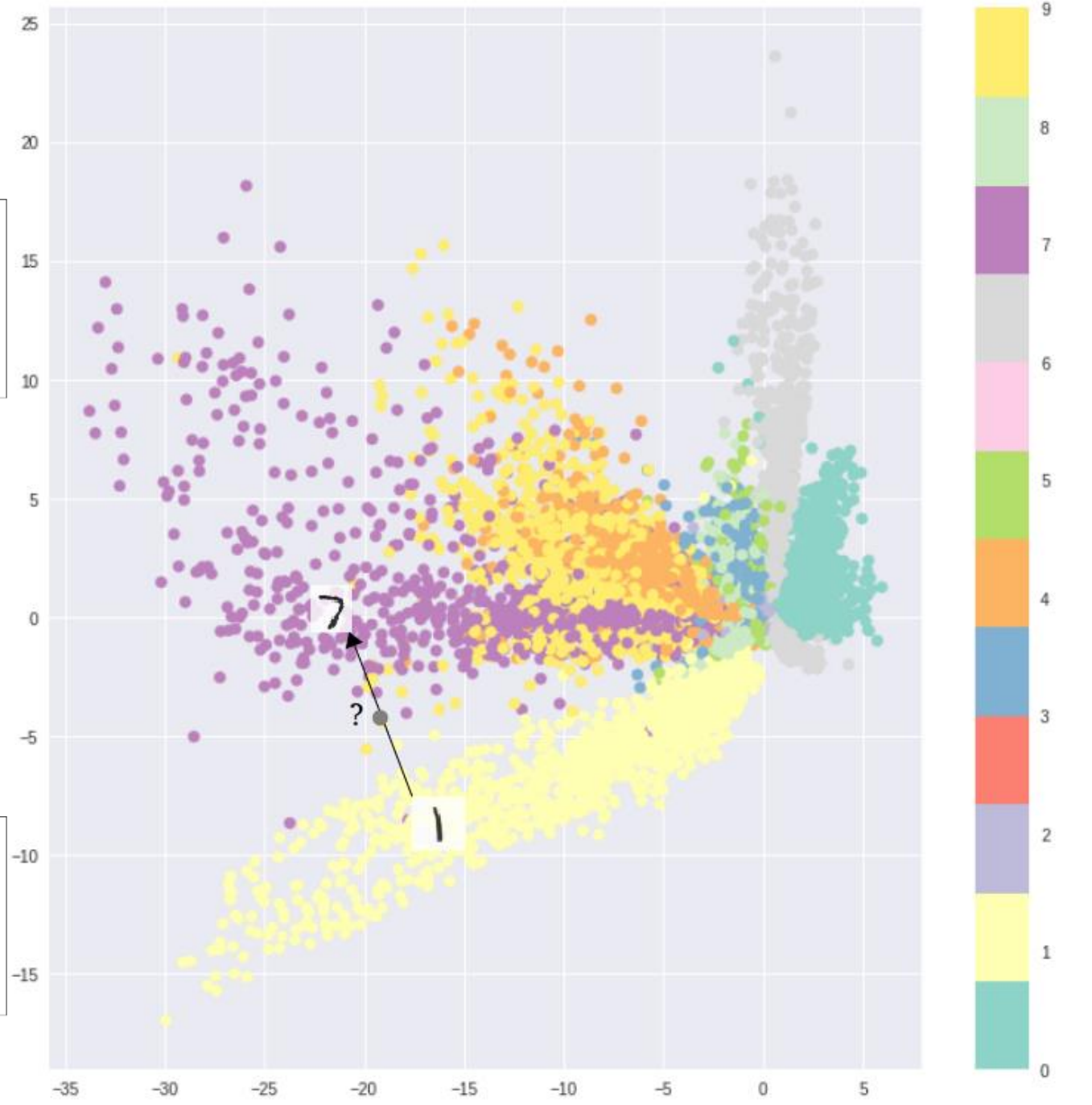


Autoencoders

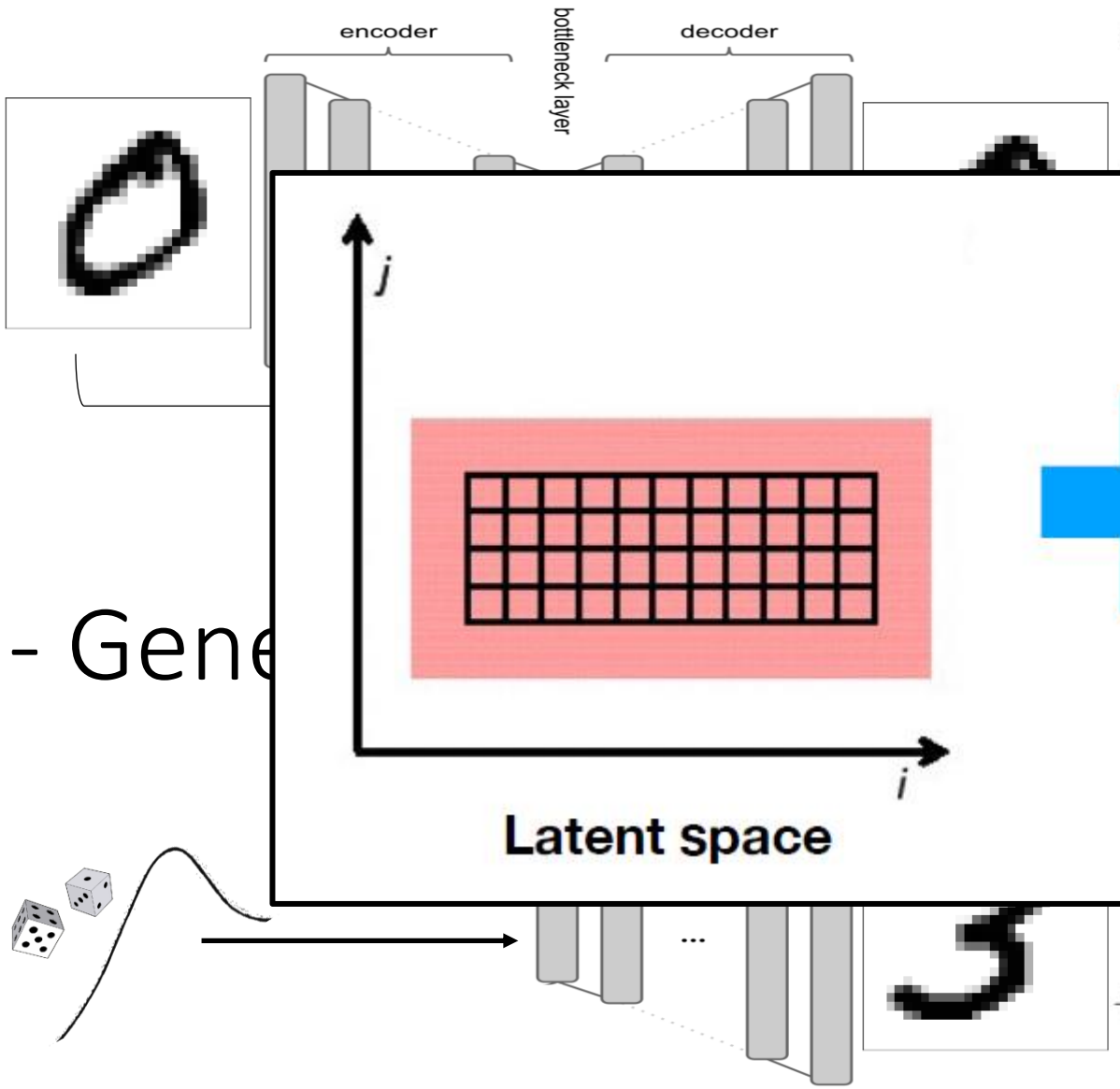


- Generative model?

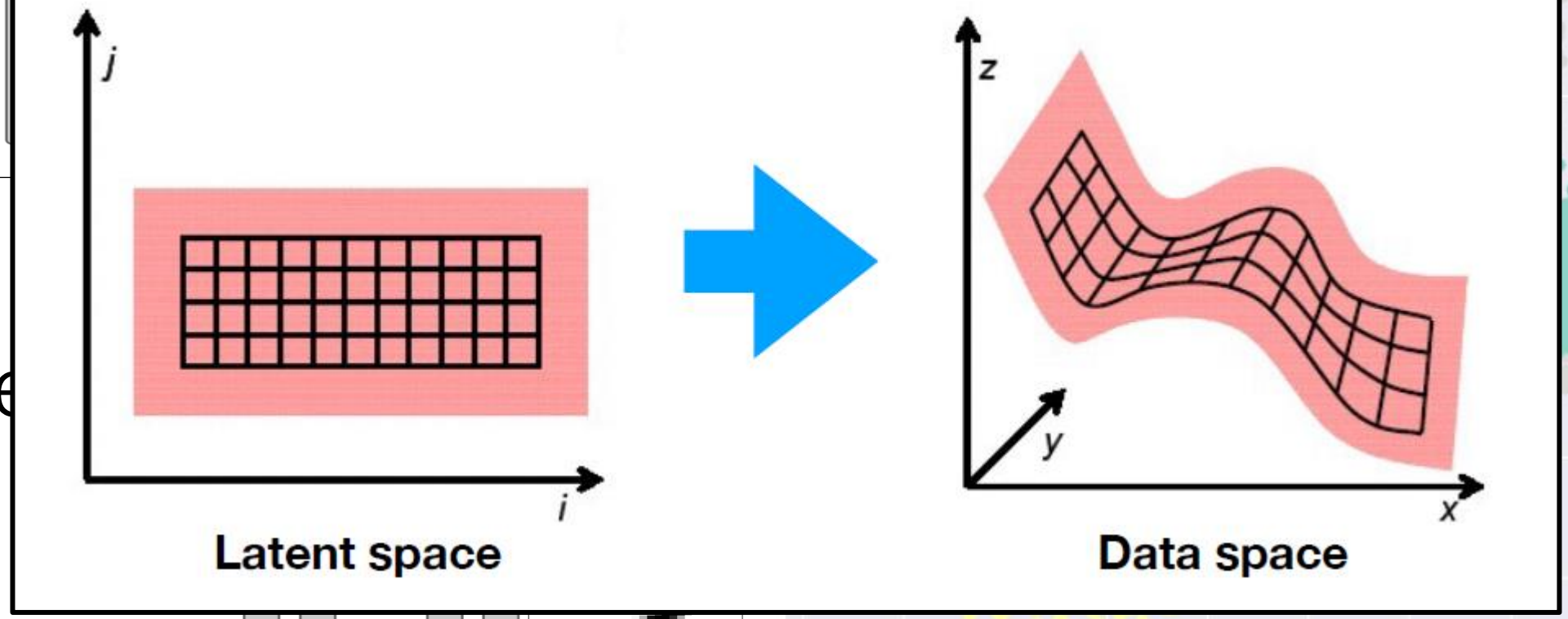




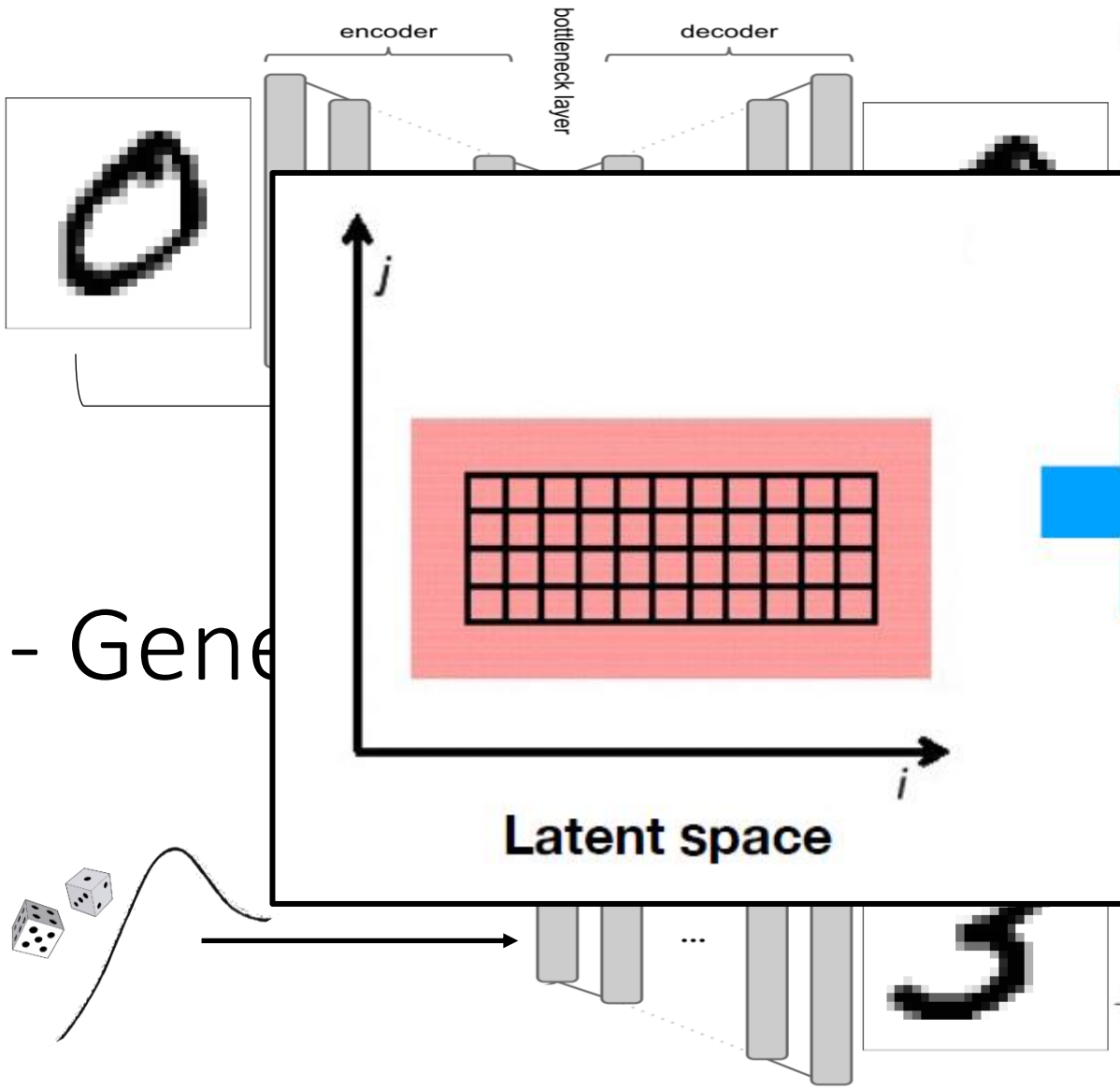
Autoencoders



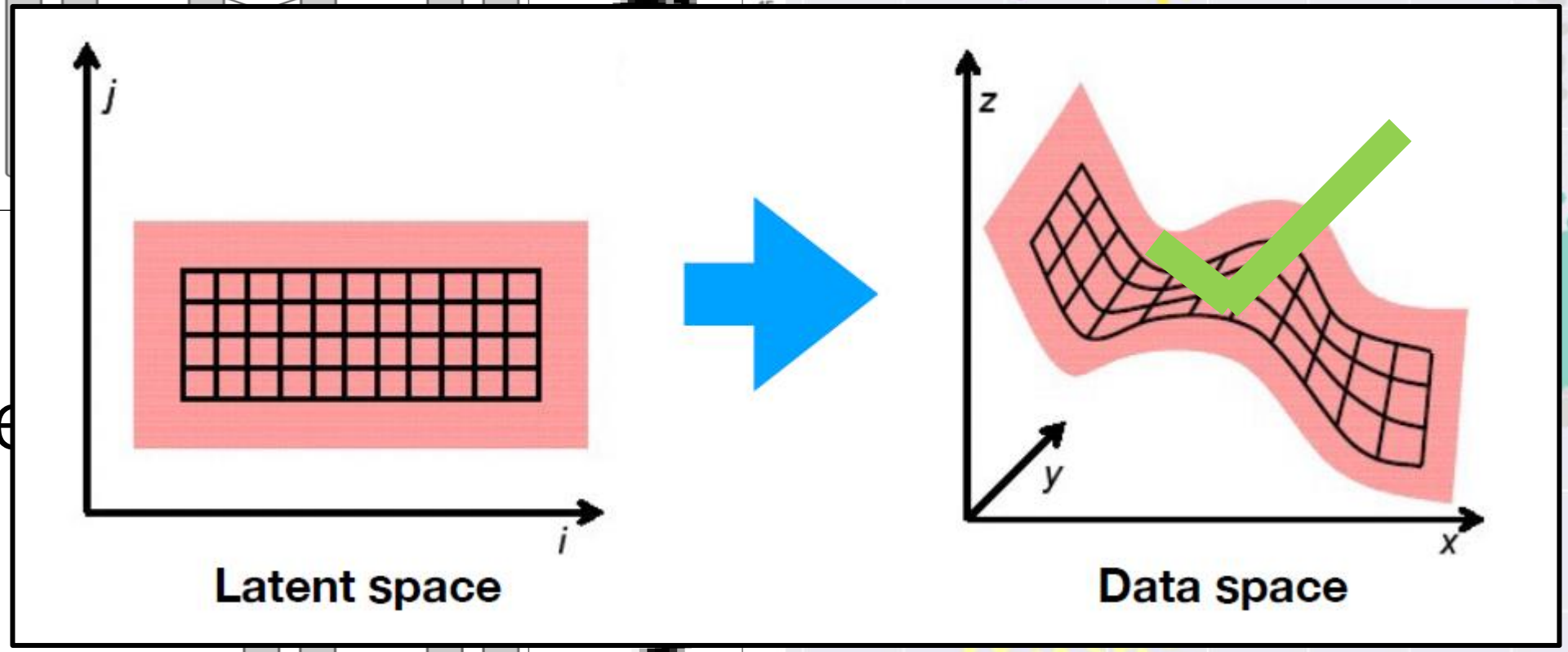
- Gene



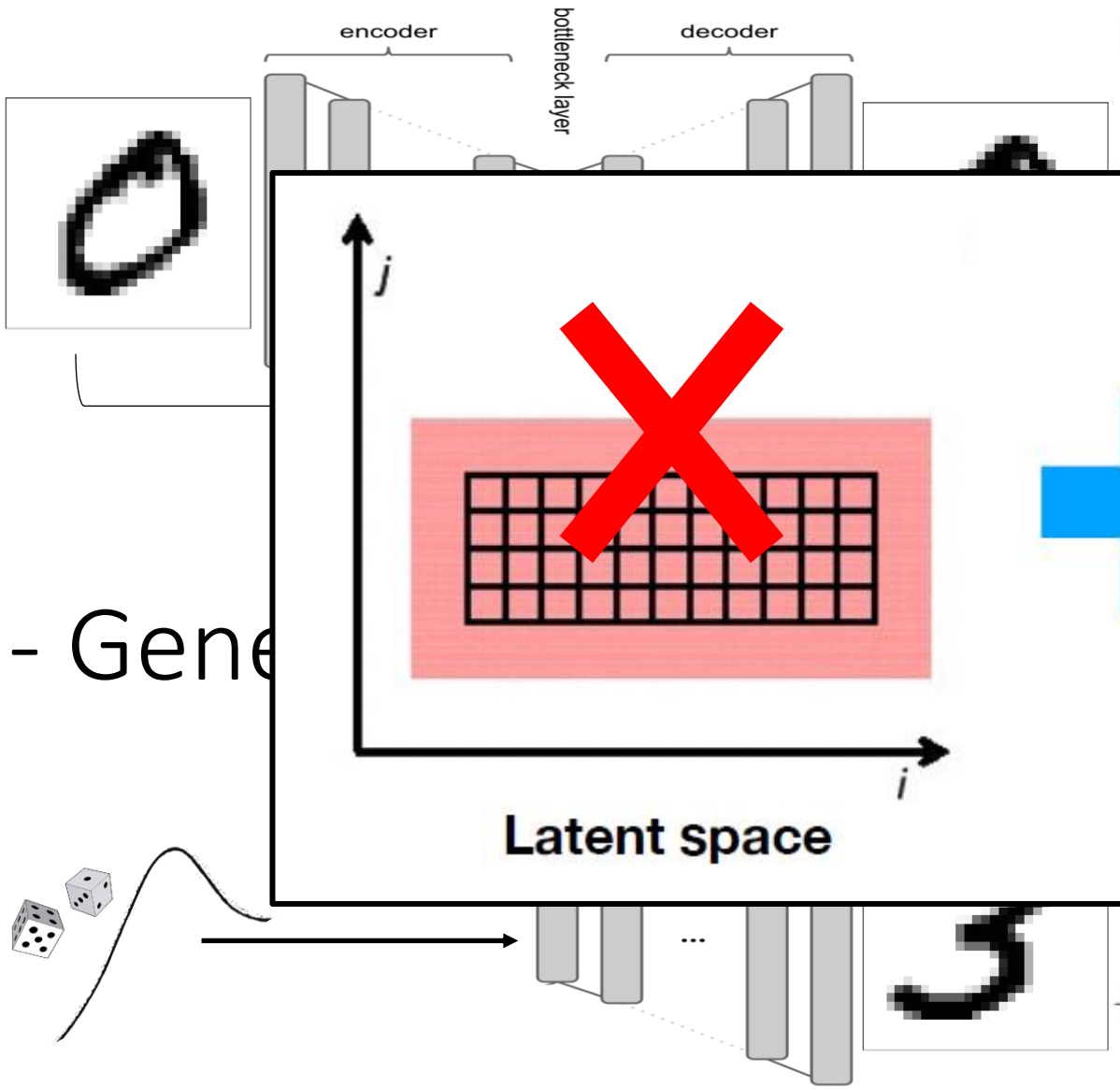
Autoencoders



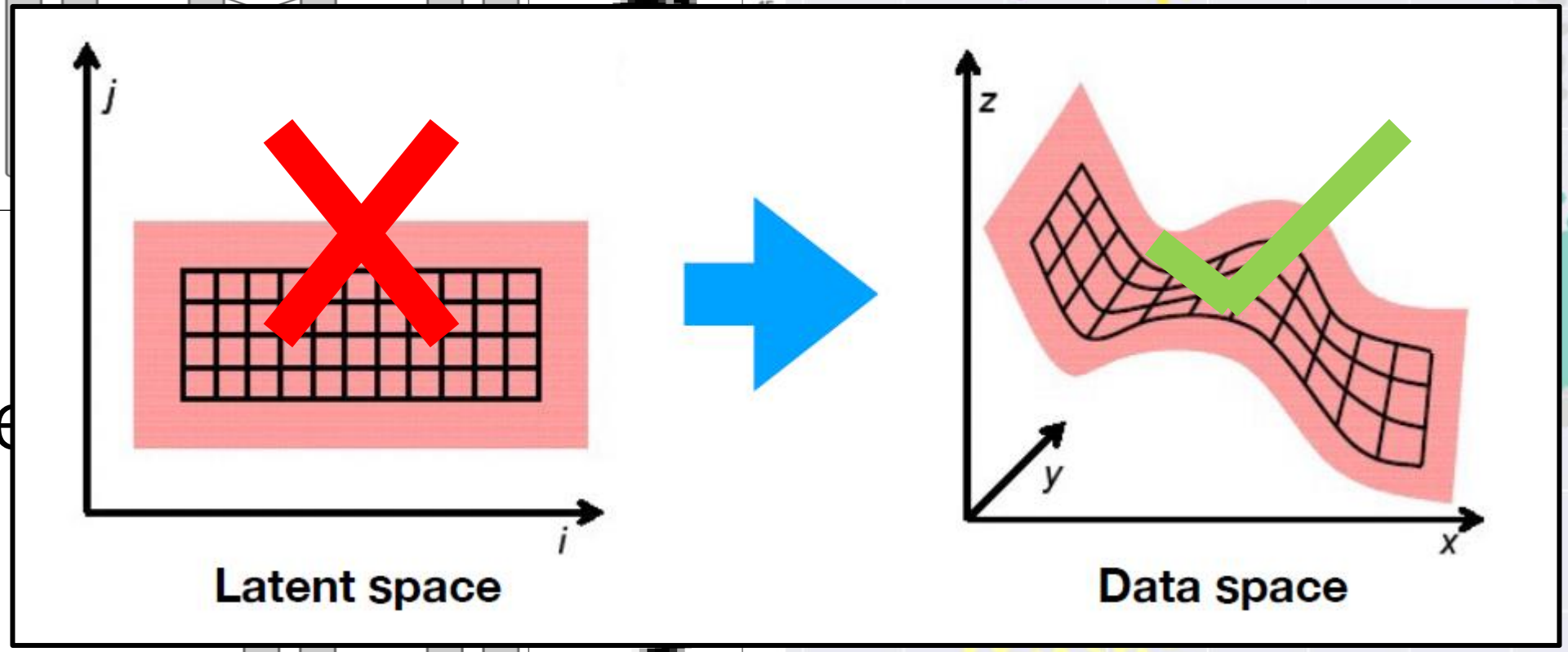
- Gene



Autoencoders

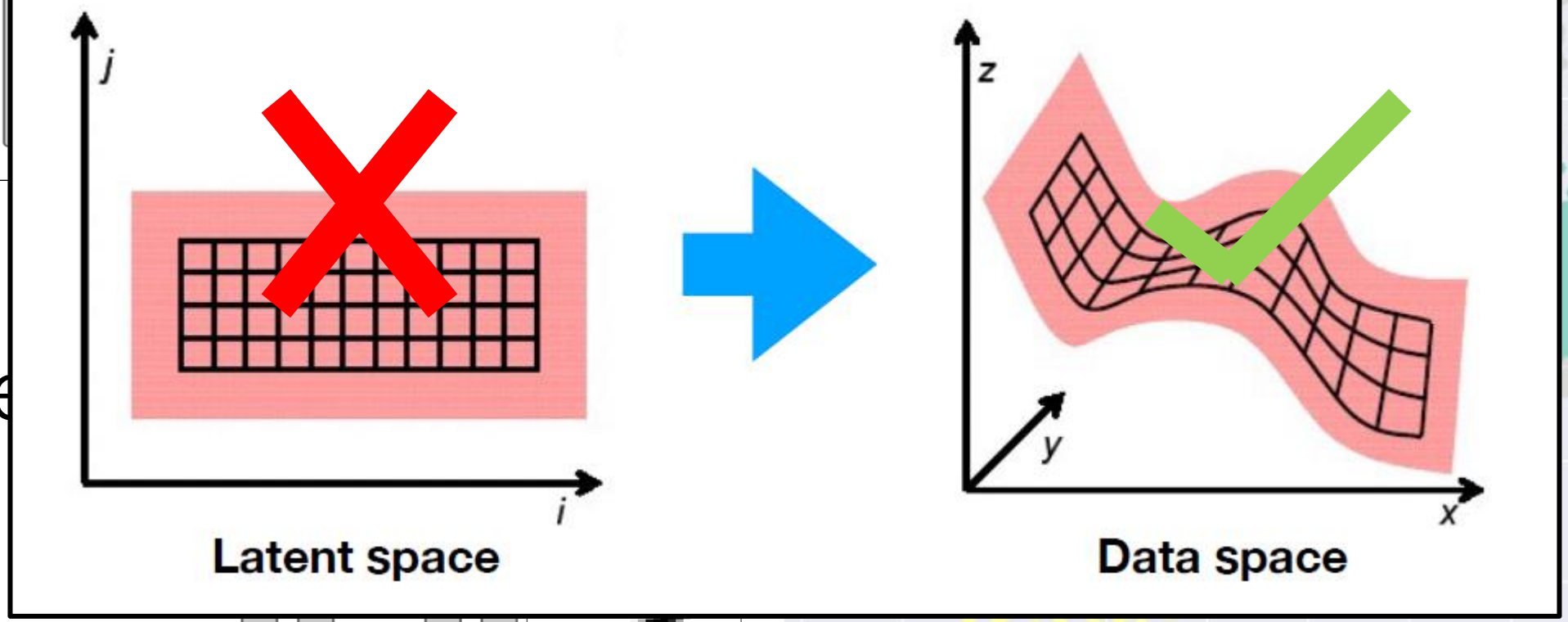


- Gene

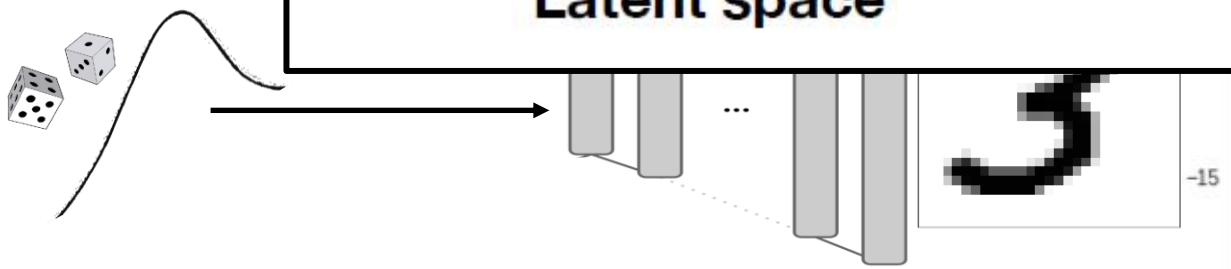


Autoencoders

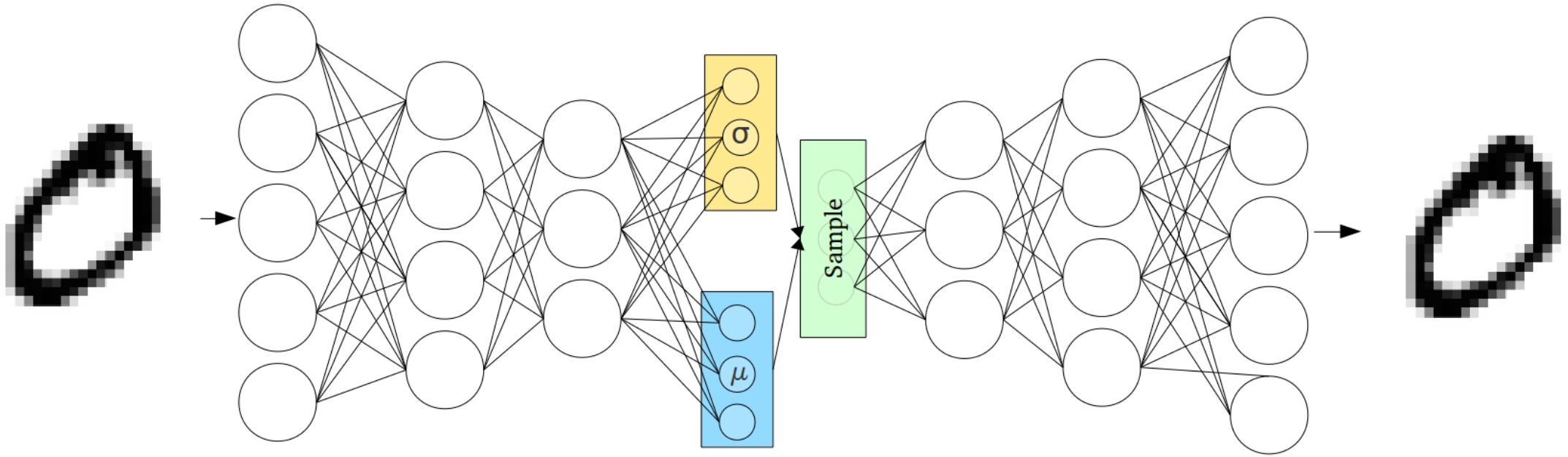
AE does not transform one *pre-determined* distribution to another!



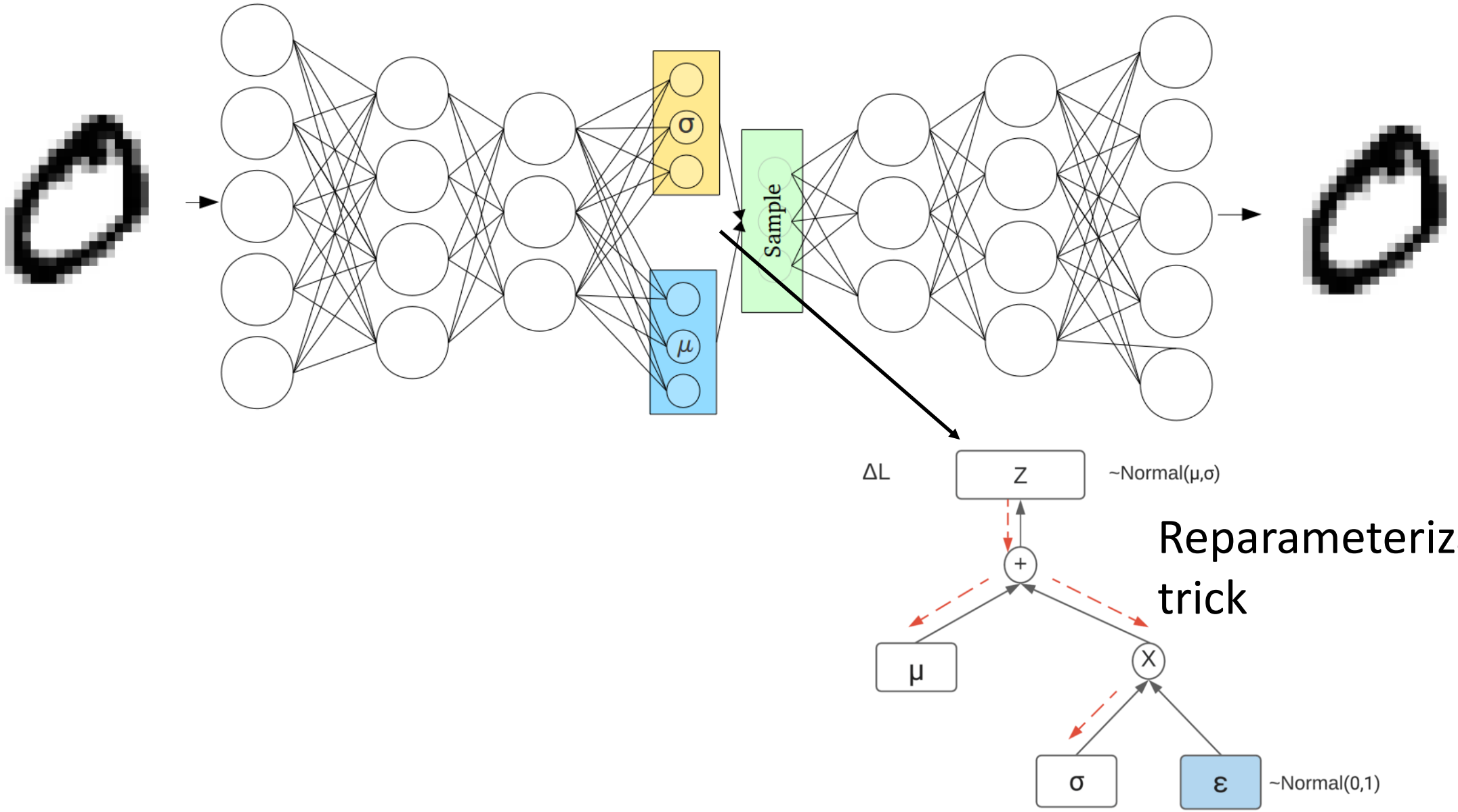
- Gene



Variational Autoencoders (Kingma&Welling 2014)

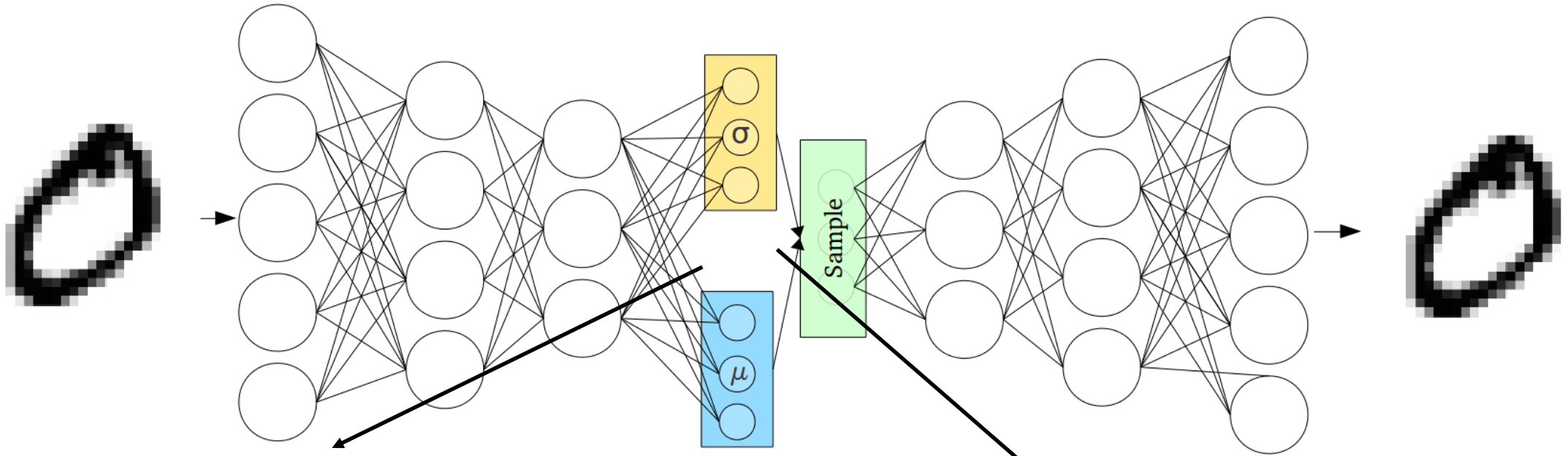


Variational Autoencoders (Kingma&Welling 2014)



Reparameterization trick

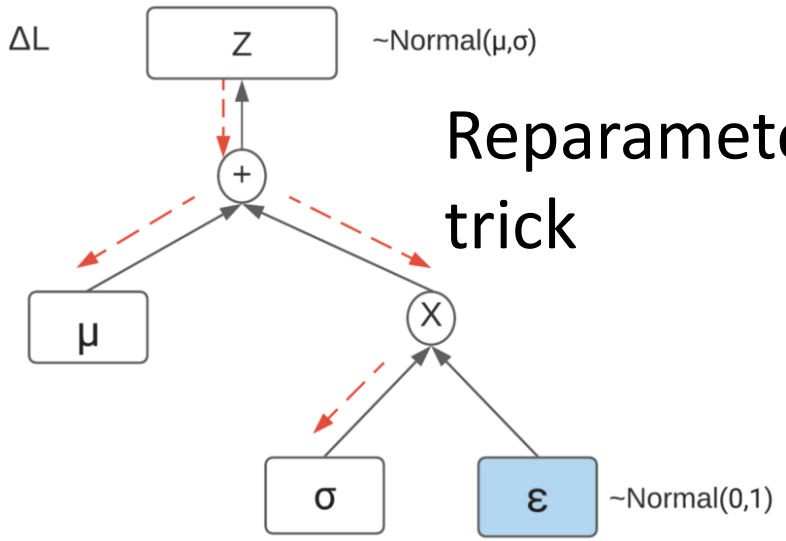
Variational Autoencoders (Kingma&Welling 2014)



Regularization:
encourage $p(z) \sim \mathrm{N}(0,1)$

by KL divergence:

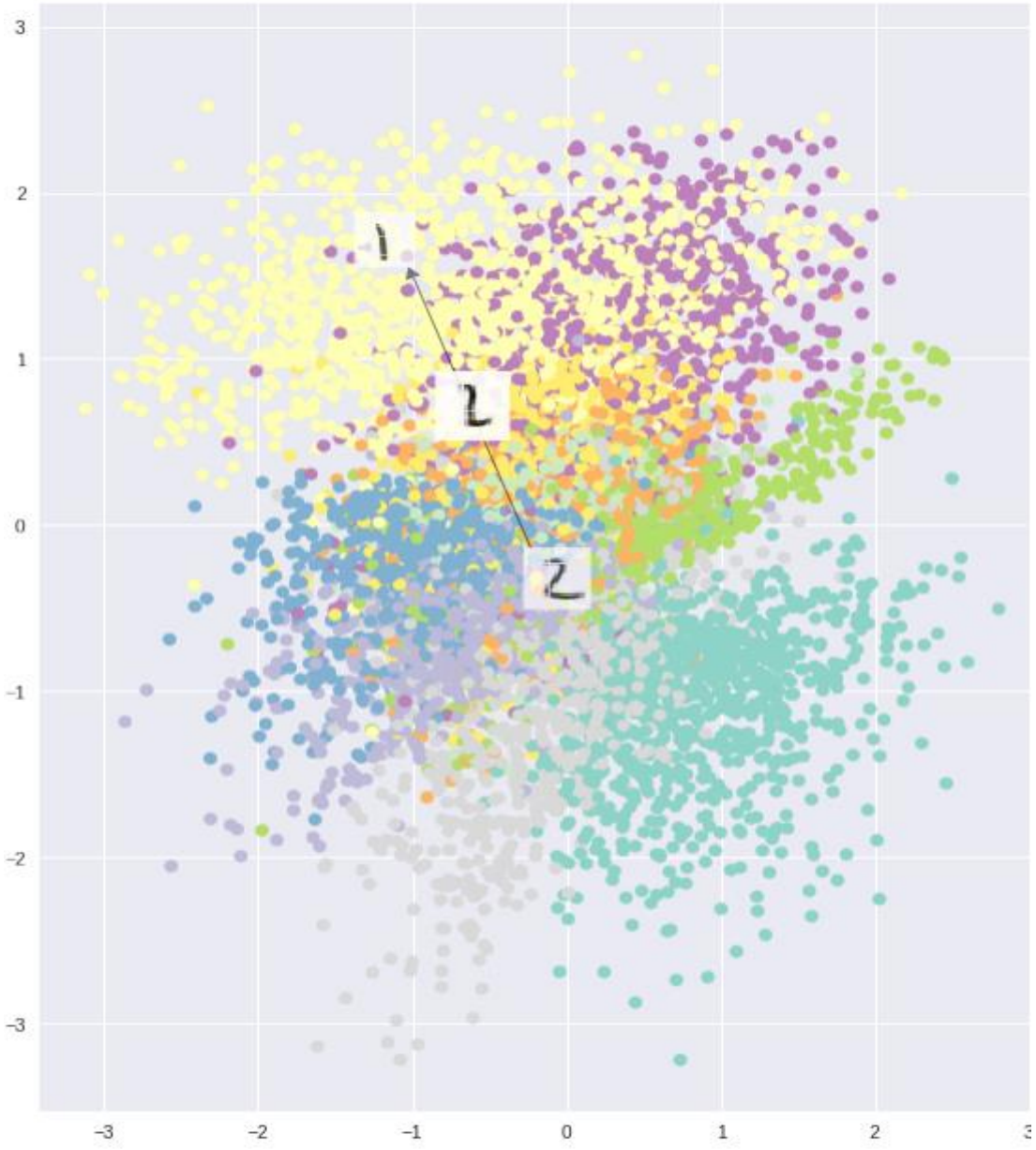
$$\sum_{i=1}^{n} \sigma_i^2 + \mu_i^2 - \log(\sigma_i) - 1$$

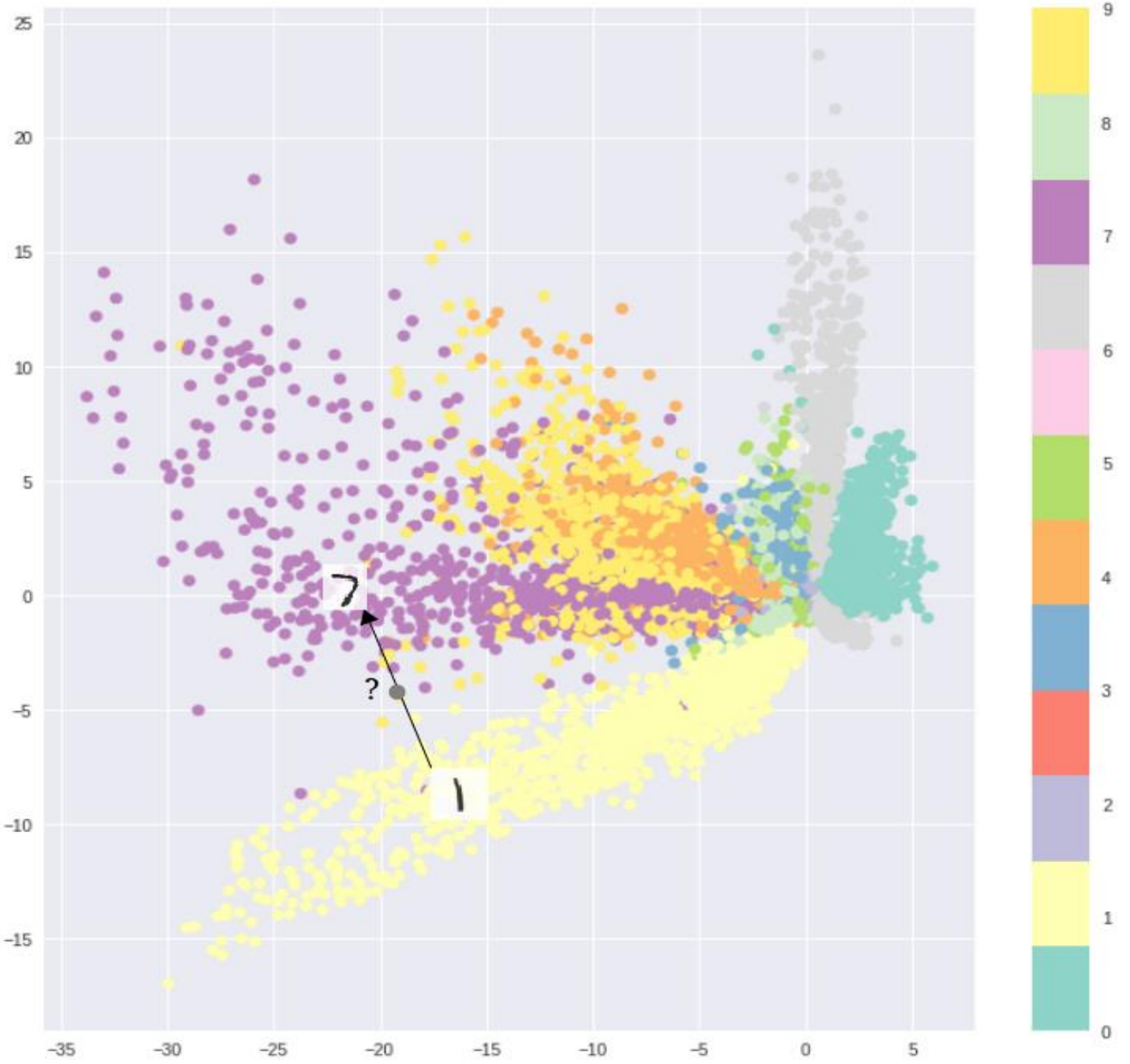


Reparameterization trick

VAE

AE


3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

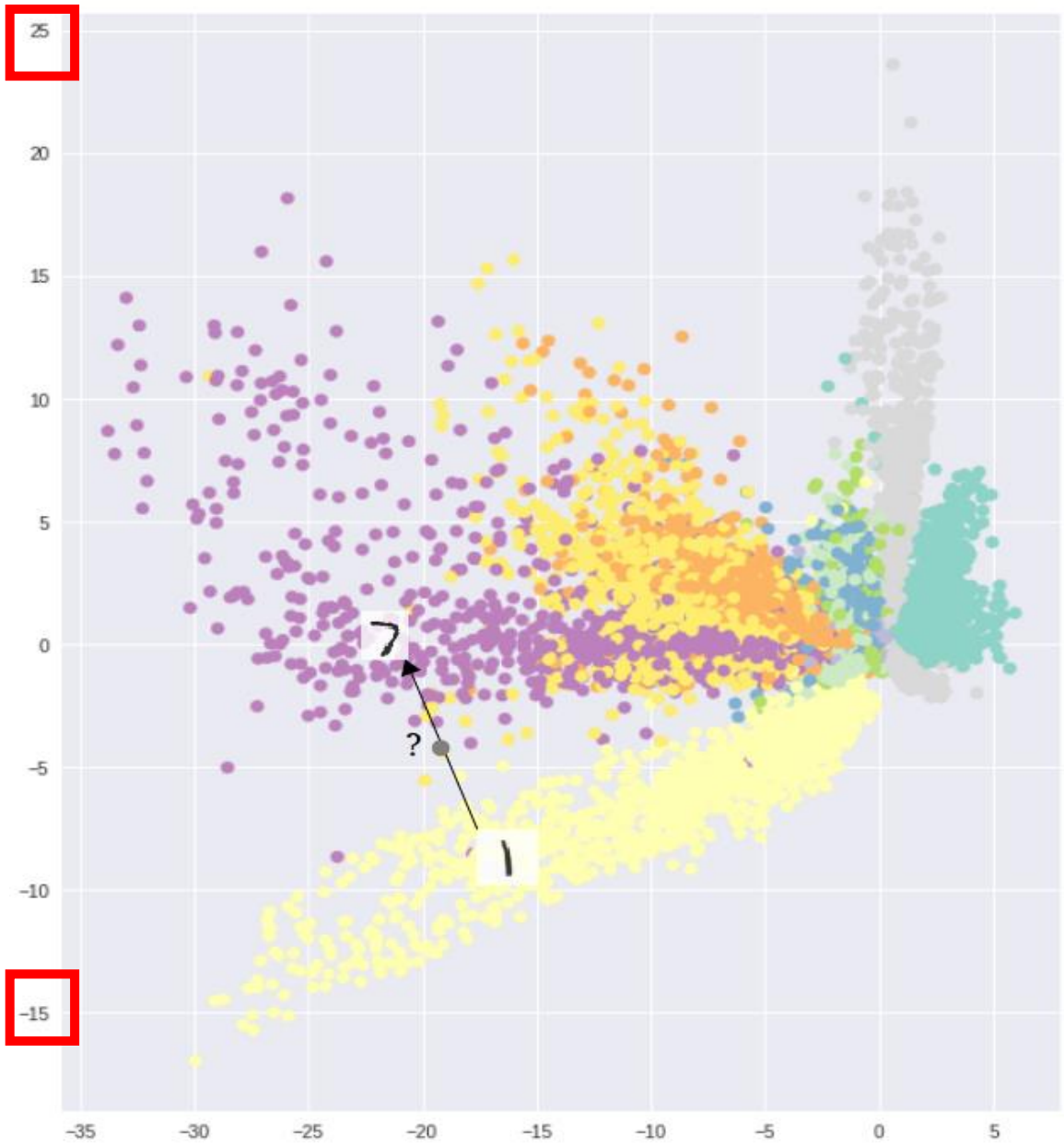

25
20
15
10
5
0
−5
−10
−15
−35
−30
−25
−20
−15
−10
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0
5
?
9
8
7
6
5
4
3
2
1
0

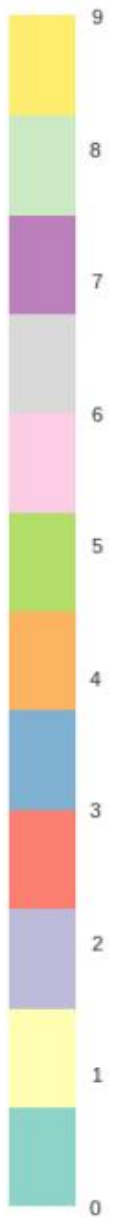
VAE

AE

Also check out the scale!



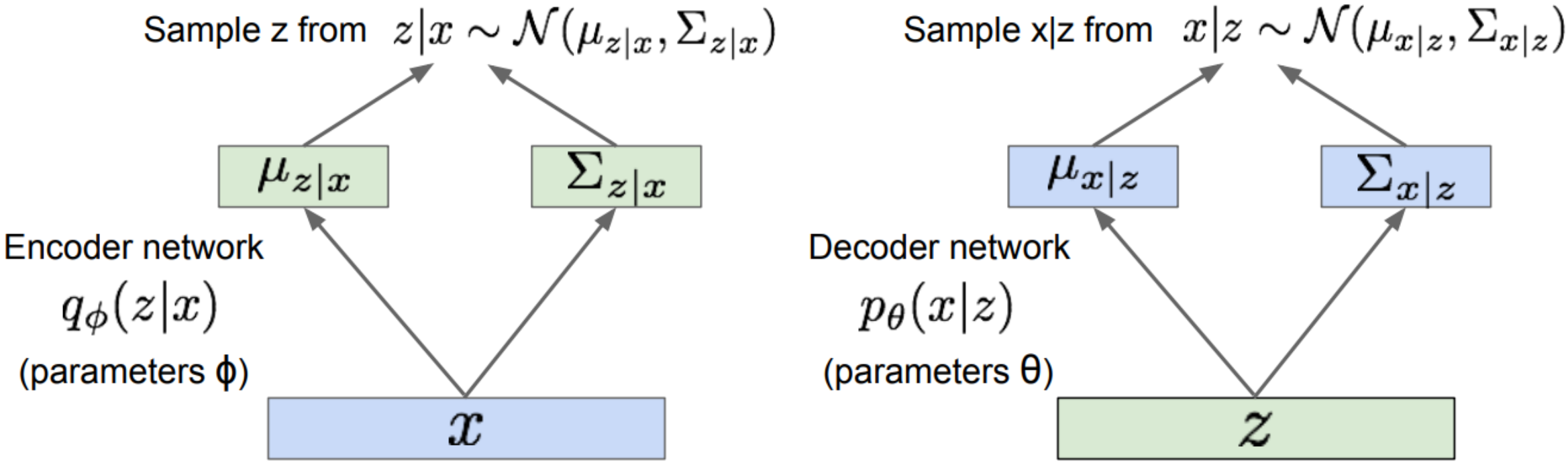




Probabilistic interpretation

Data likelihood: $p_\theta(x) = \int p_\theta(z) p_\theta(x|z) dz$

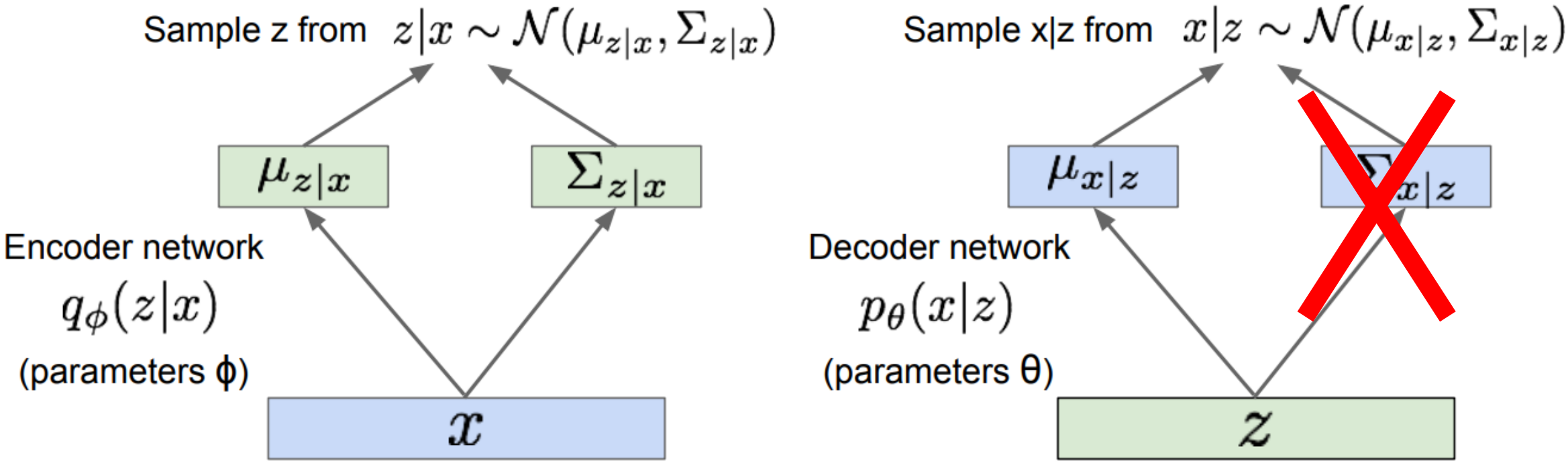
Goal: make $\log p_\theta(x^{(i)})$ as high as possible



Probabilistic interpretation

Data likelihood: $p_\theta(x) = \int p_\theta(z) p_\theta(x|z) dz$

Goal: make $\log p_\theta(x^{(i)})$ as high as possible

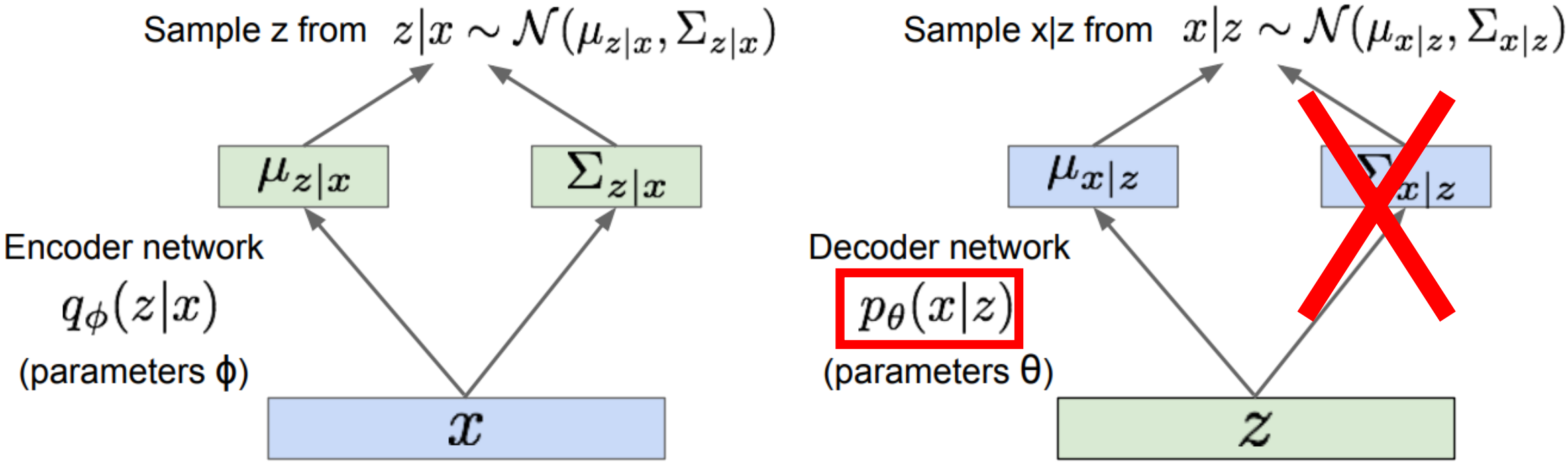


Probabilistic interpretation

N(0,1)

Data likelihood: $p_\theta(x) = \int p_\theta(z) p_\theta(x|z) dz$

Goal: make $\log p_\theta(x^{(i)})$ as high as possible



$$\log p_\theta(x^{(i)}) = \mathbf{E}_{z\sim q_\phi(z|x^{(i)})}\left[\log p_\theta(x^{(i)})\right] \quad (p_\theta(x^{(i)}) \text{ Does not depend on } z)$$

$$= \mathbf{E}_z\left[\log \frac{p_\theta(x^{(i)} \mid z)p_\theta(z)}{p_\theta(z \mid x^{(i)})}\right] \quad \text{(Bayes' Rule)}$$

$$= \mathbf{E}_z\left[\log \frac{p_\theta(x^{(i)} \mid z)p_\theta(z)}{p_\theta(z \mid x^{(i)})}\frac{q_\phi(z \mid x^{(i)})}{q_\phi(z \mid x^{(i)})}\right] \quad \text{(Multiply by constant)}$$

$$= \mathbf{E}_z\left[\log p_\theta(x^{(i)} \mid z)\right] - \mathbf{E}_z\left[\log \frac{q_\phi(z \mid x^{(i)})}{p_\theta(z)}\right] + \mathbf{E}_z\left[\log \frac{q_\phi(z \mid x^{(i)})}{p_\theta(z \mid x^{(i)})}\right] \quad \text{(Logarithms)}$$

$$= \mathbf{E}_z\left[\log p_\theta(x^{(i)} \mid z)\right] - D_{KL}(q_\phi(z \mid x^{(i)}) \,||\, p_\theta(z)) + D_{KL}(q_\phi(z \mid x^{(i)}) \,||\, p_\theta(z \mid x^{(i)}))$$

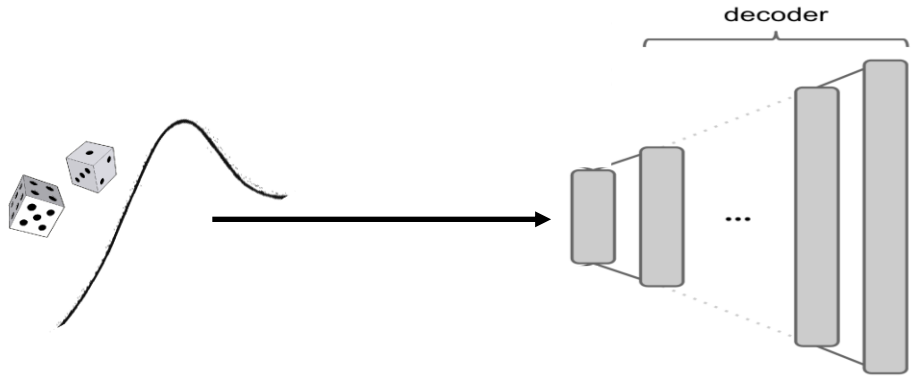
Decoder network gives $p_\theta(x|z)$, can compute estimate of this term through sampling. (Sampling differentiable through reparam. trick, see paper.)

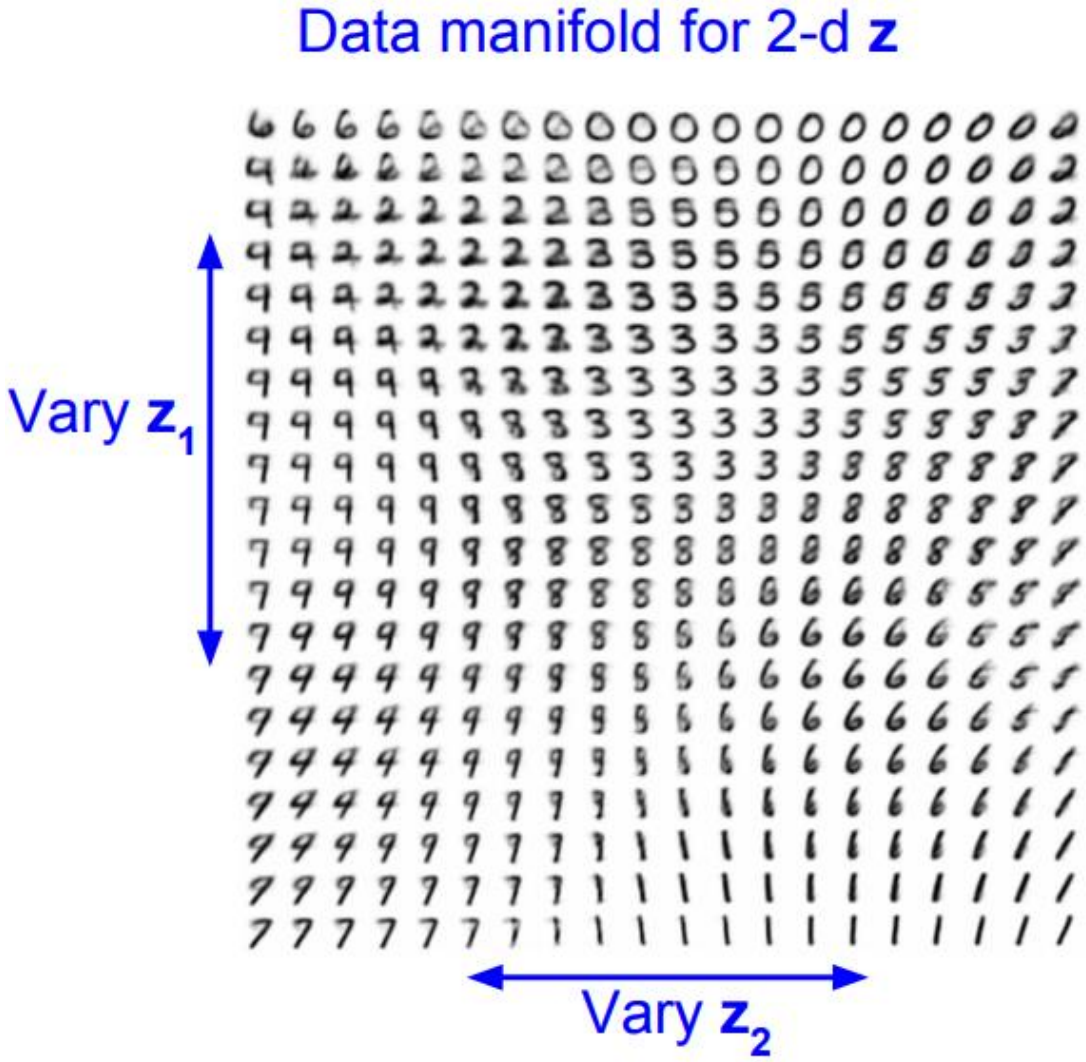
This KL term (between Gaussians for encoder and z prior) has nice closed-form solution!

$p_\theta(z|x)$ intractable (saw earlier), can't compute this KL term :(But we know KL divergence always >= 0.

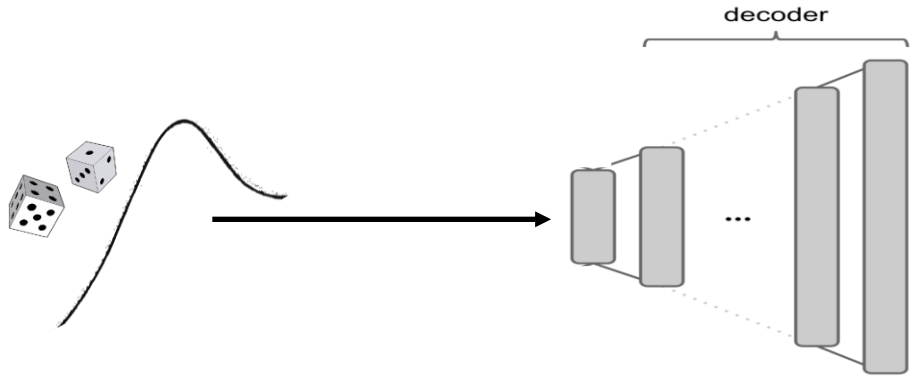
Slide credit: Stanford cs231n

Generate data



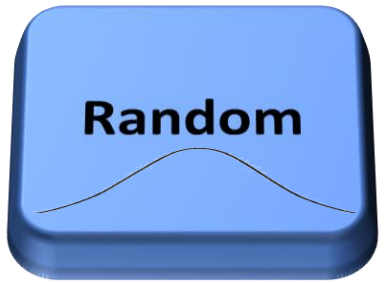


Generate data





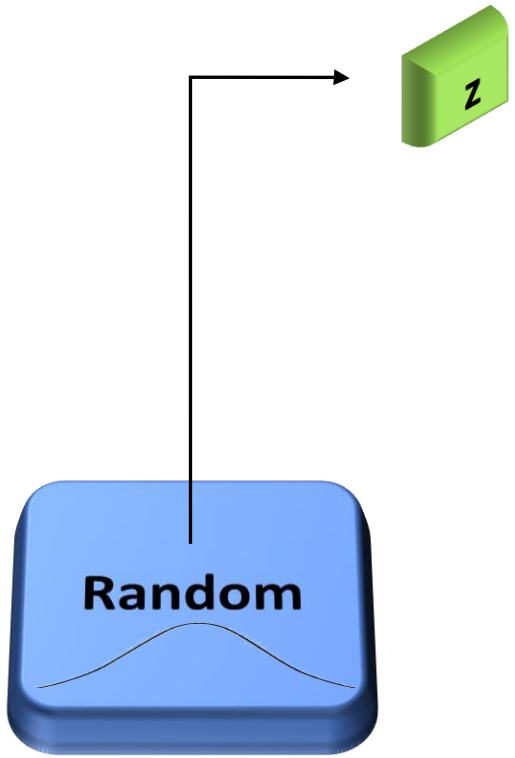
How about this idea for a generative model?



How about this idea for a generative model?



How about this idea for a generative model?



How about this idea for a generative model?



How about this idea for a generative model?



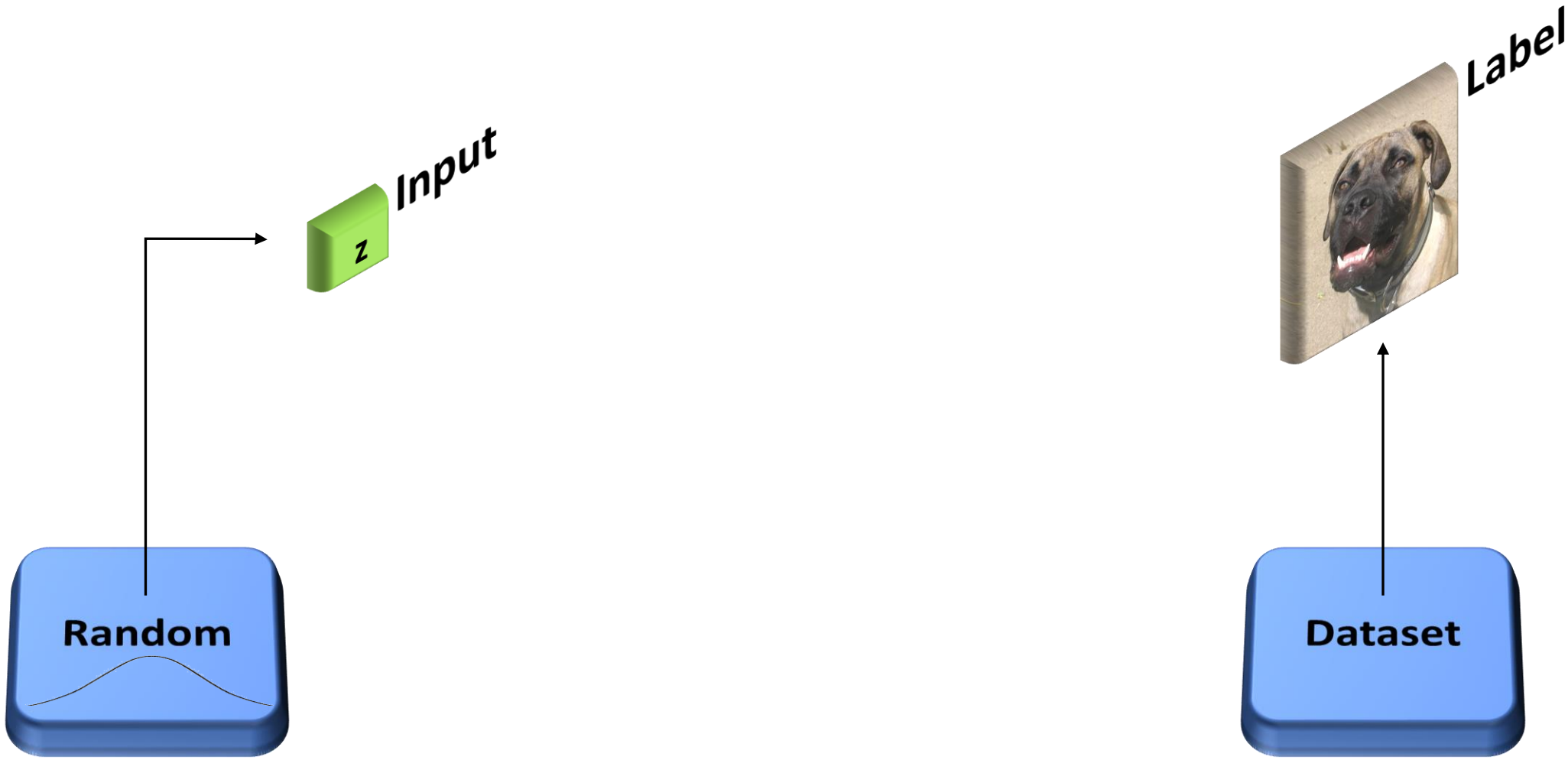
How about this idea for a generative model?



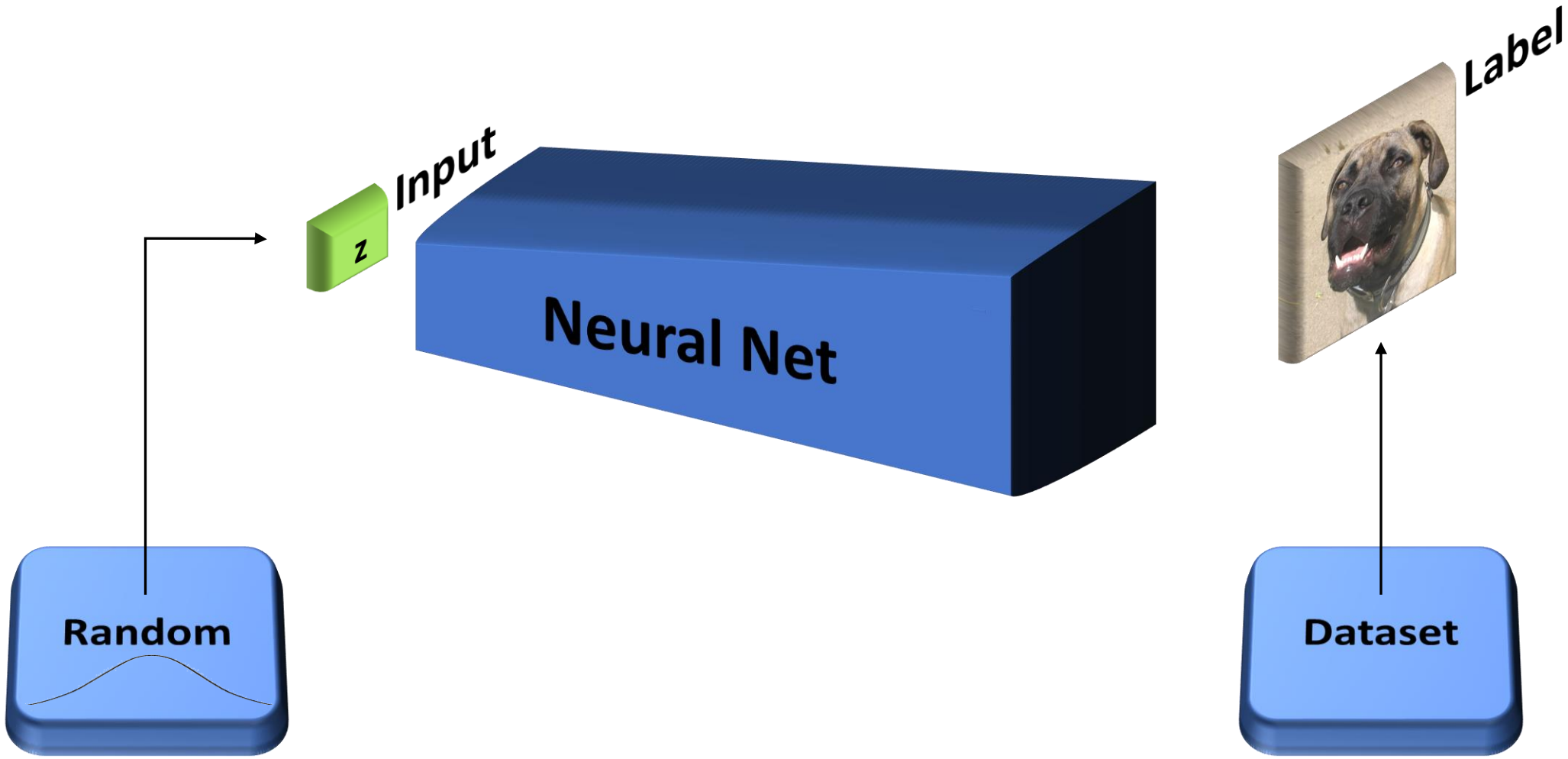
How about this idea for a generative model?



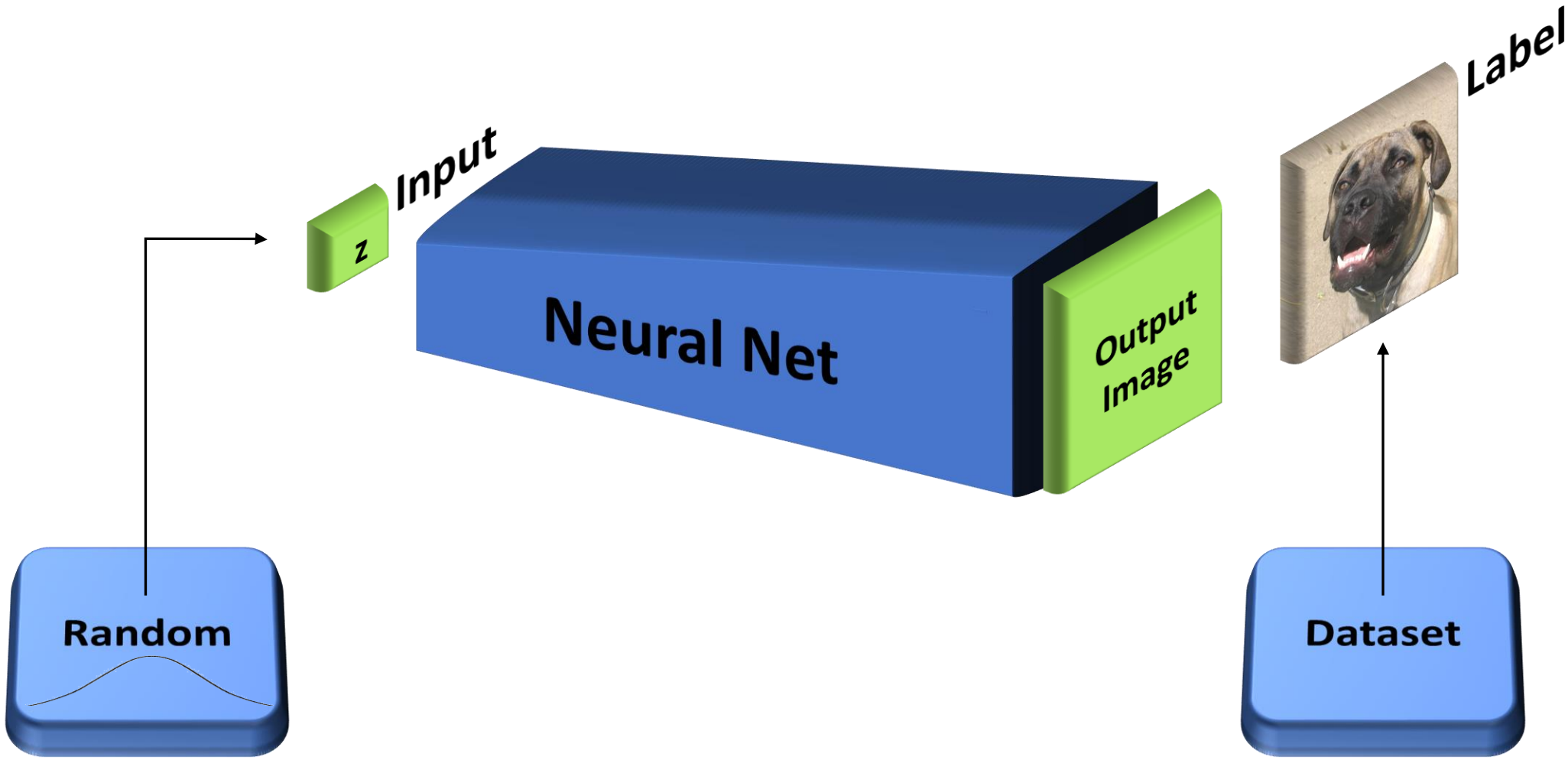
How about this idea for a generative model?



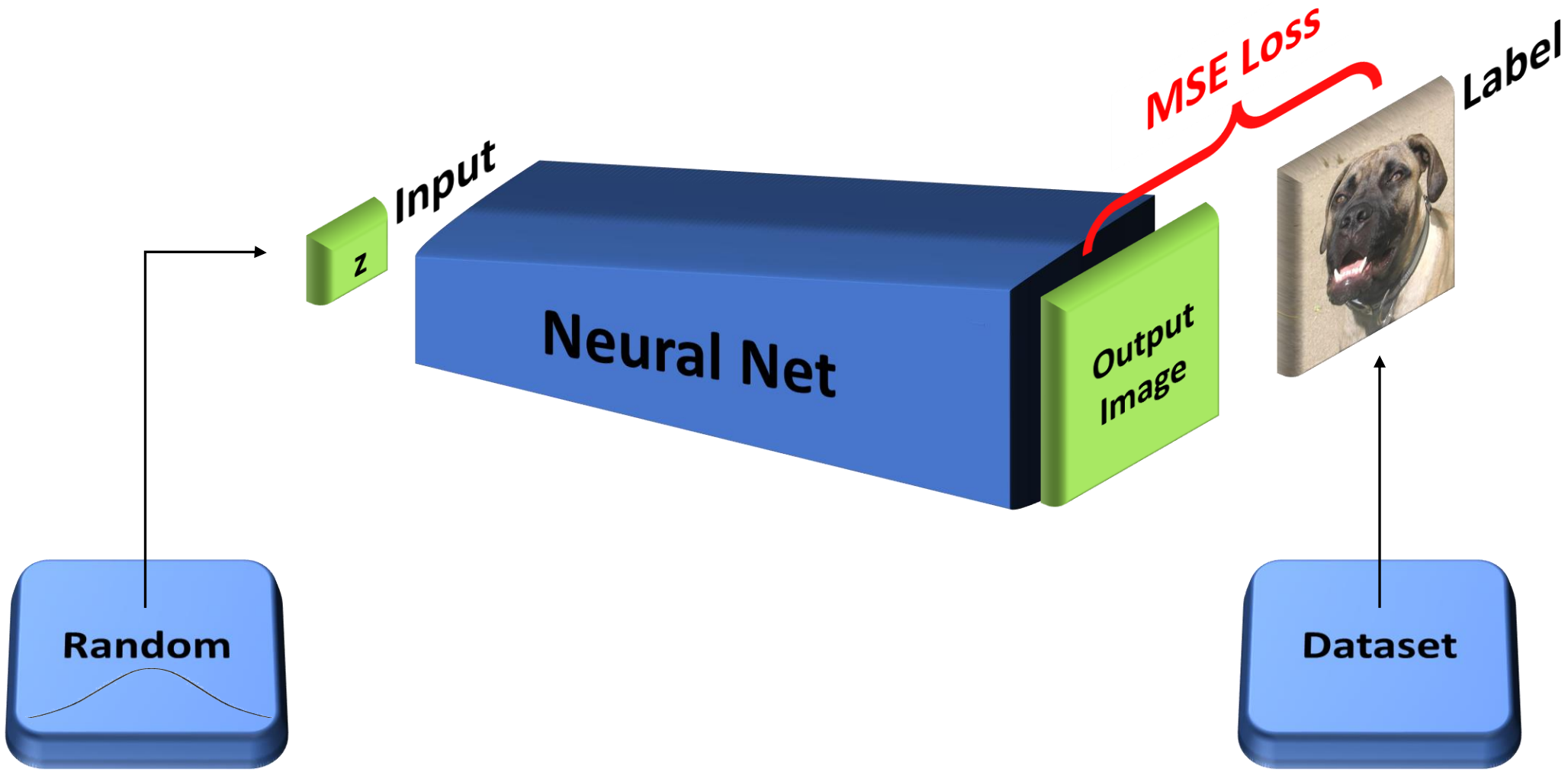
How about this idea for a generative model?



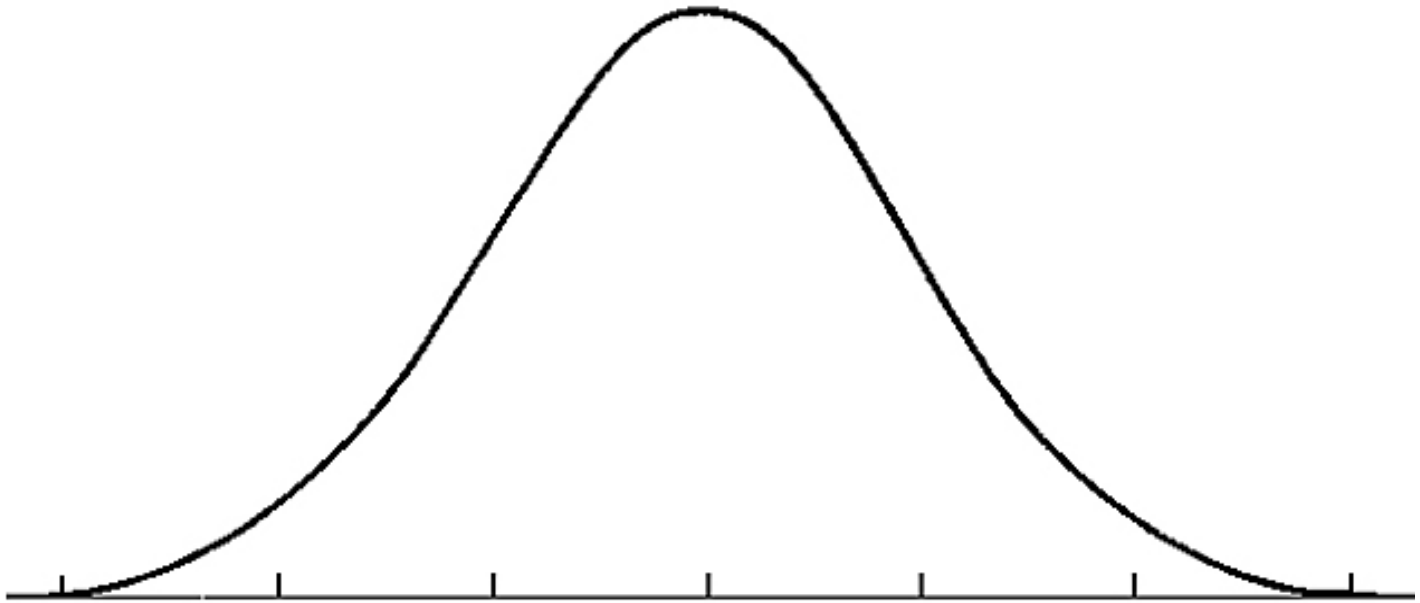
How about this idea for a generative model?

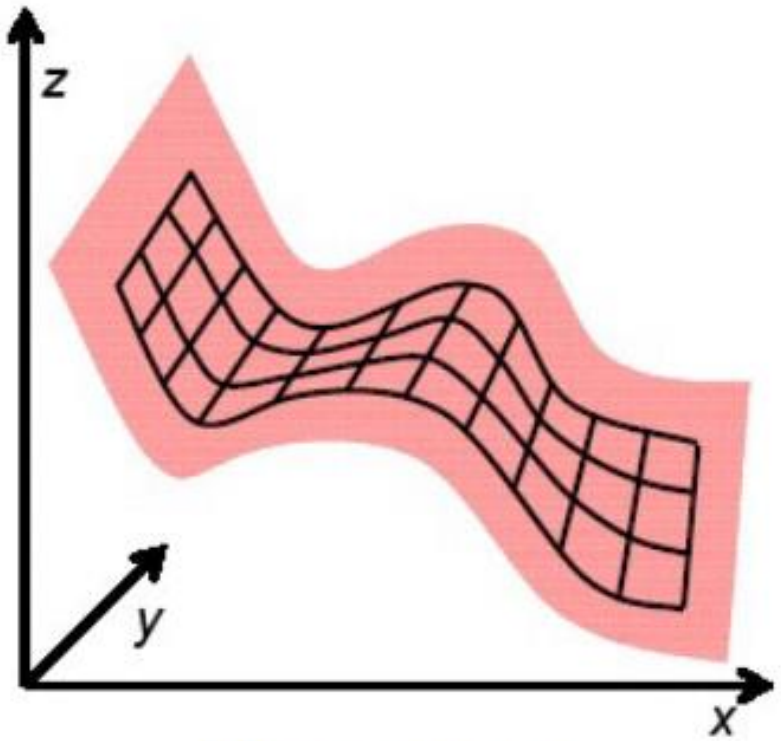


How about this idea for a generative model?



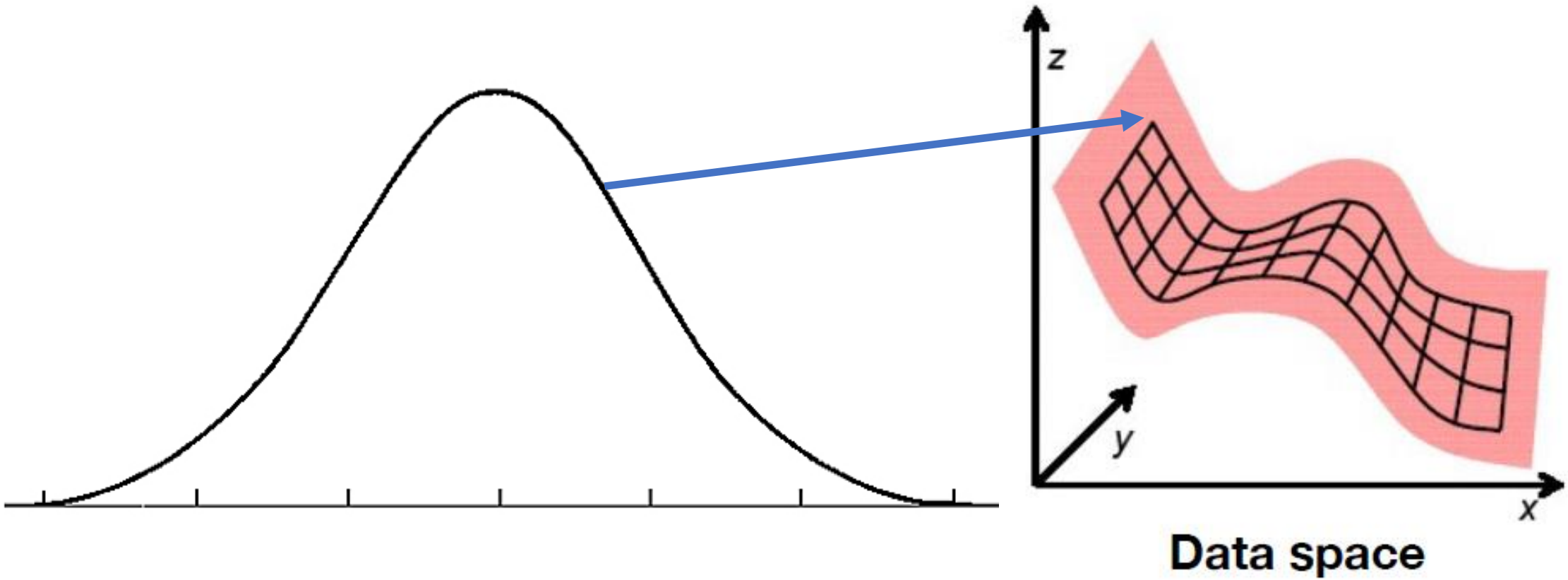
No good!



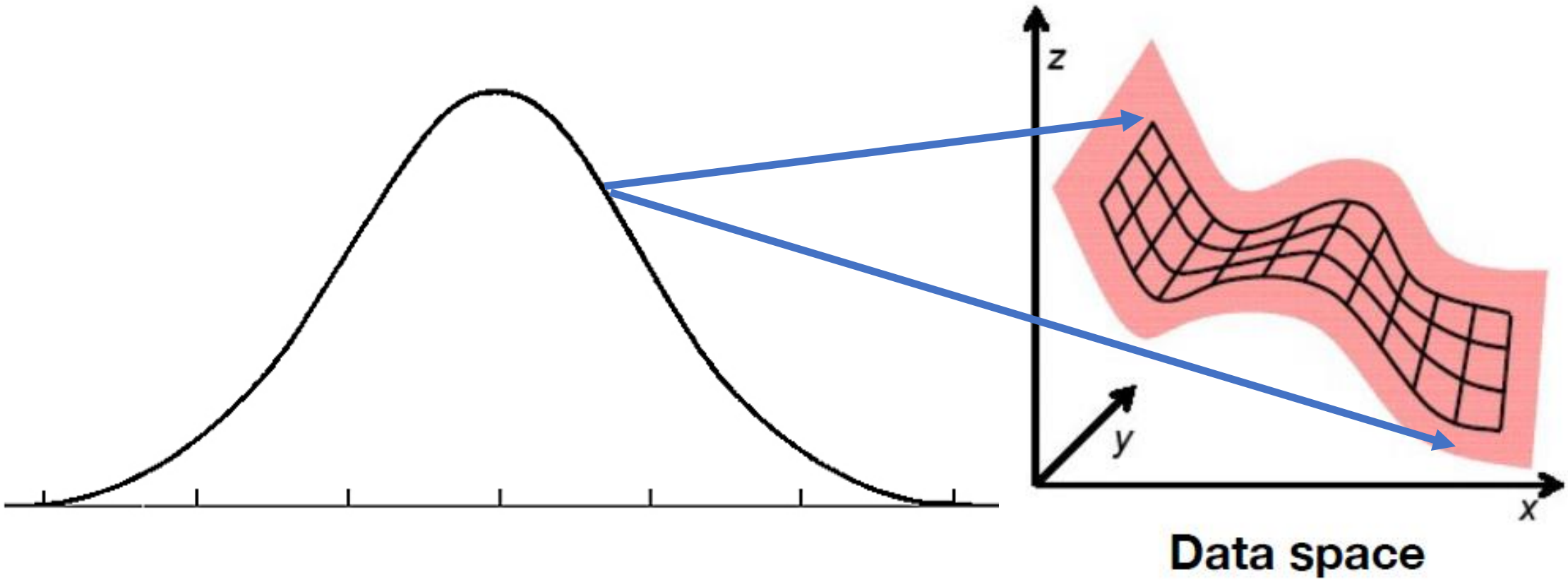


Data space

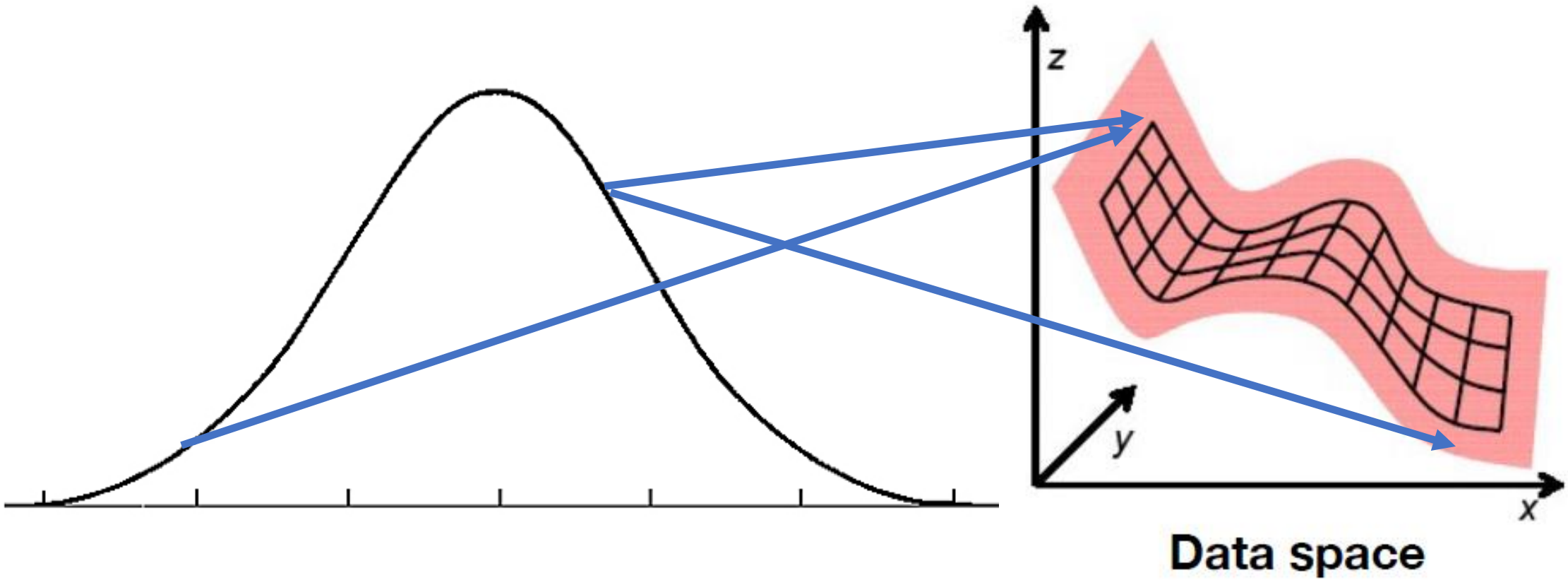
No good!



No good!

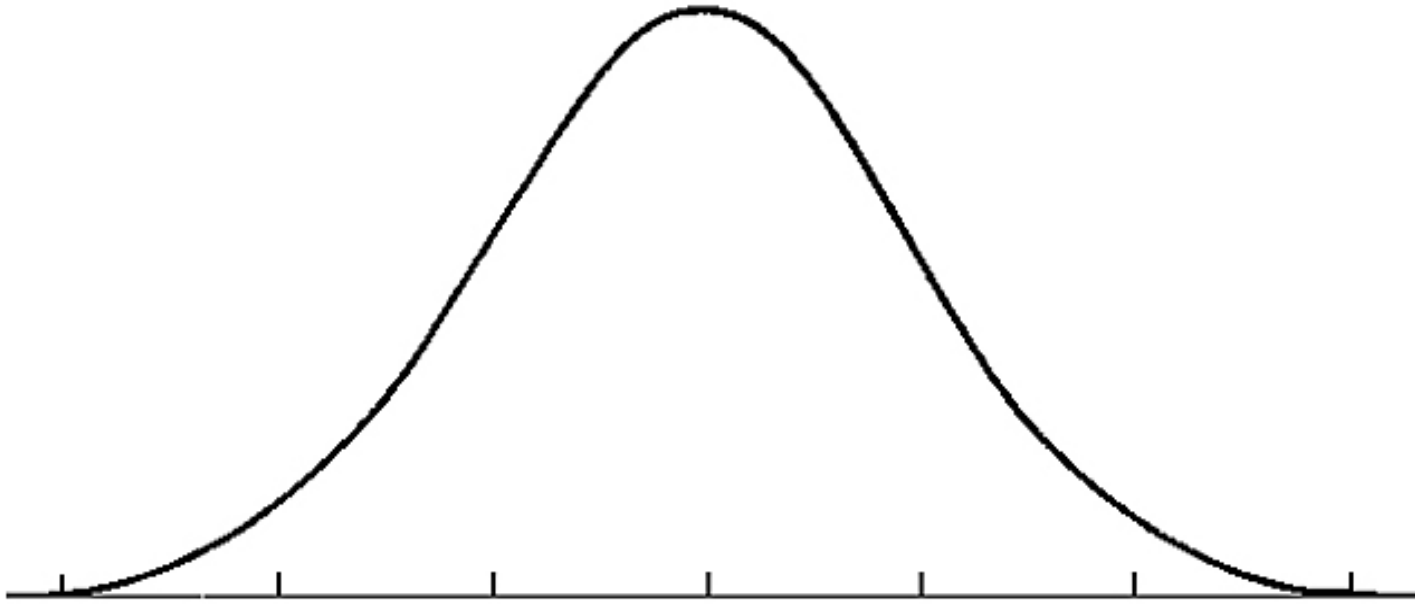


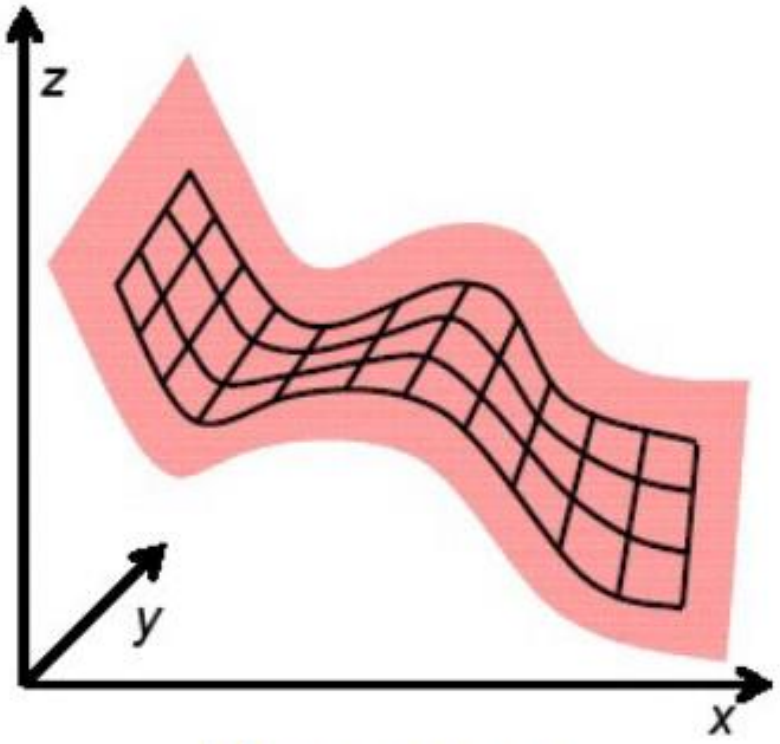
No good!



No good!

In expectation: every noise is mapped to every instance





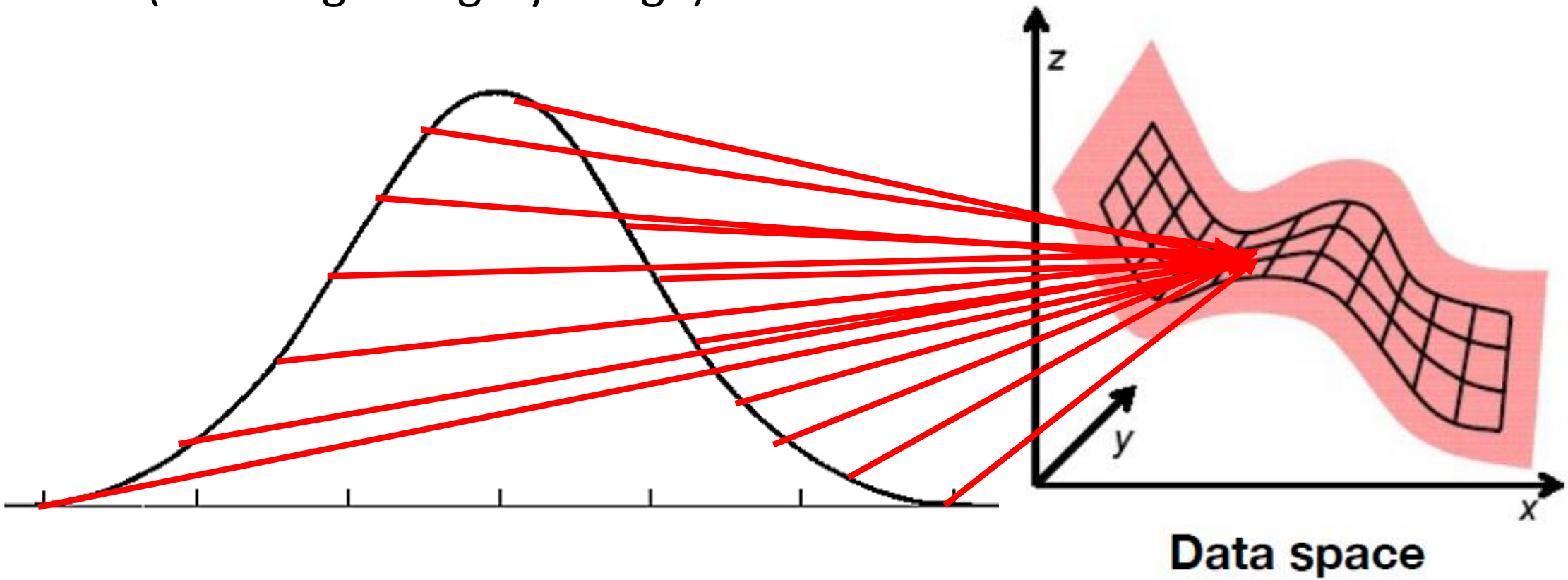
Data space

No good!

Multimodality not obtained!

In expectation: every noise is mapped to every instance

Best L2 solution: All noise is mapped to the mean
(For images: ~ grey image)

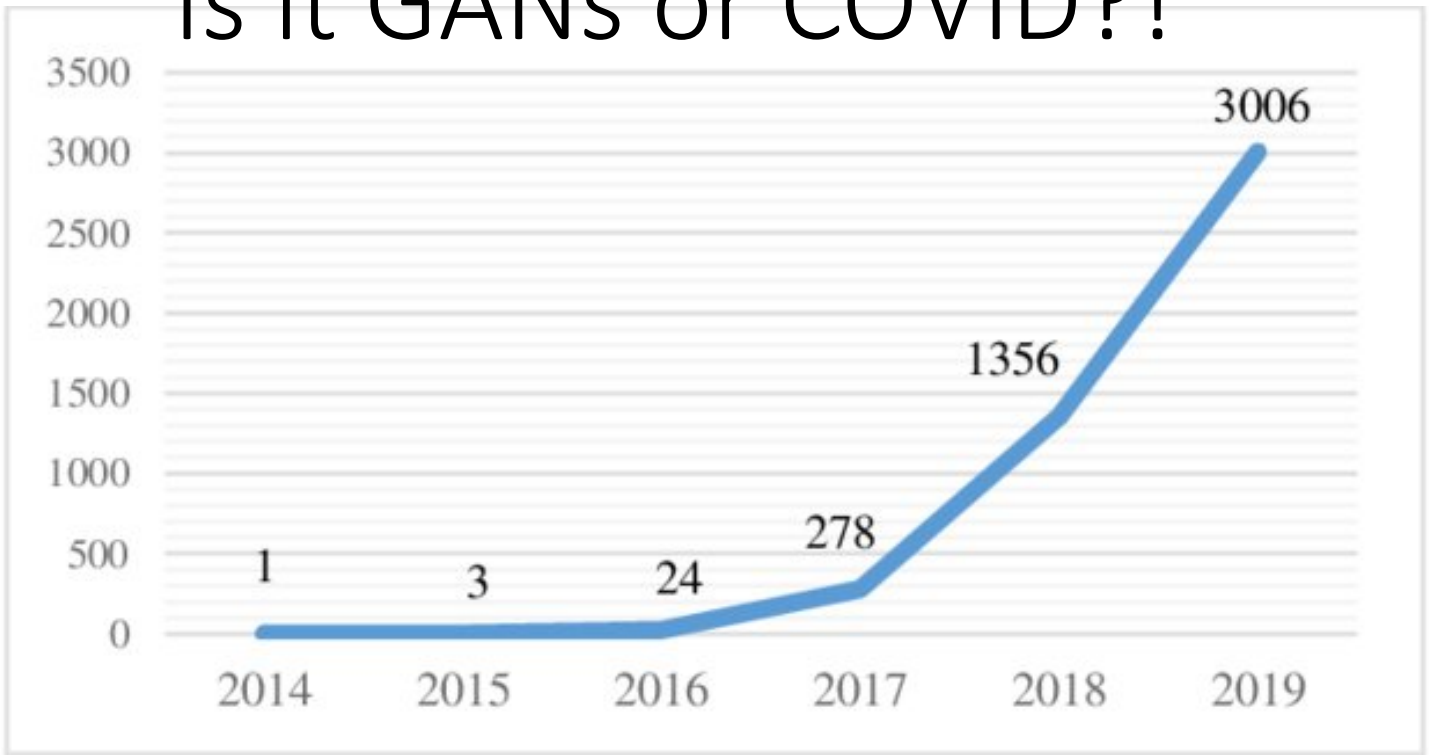


Generative Adversarial Networks



Generative Adversarial Networks

Is it GANs or COVID?!



of GAN related papers per year
(Salehi et al.)



Q: What makes a good counterfeiter?

A: Can fool a good cop



Generator





Discriminator

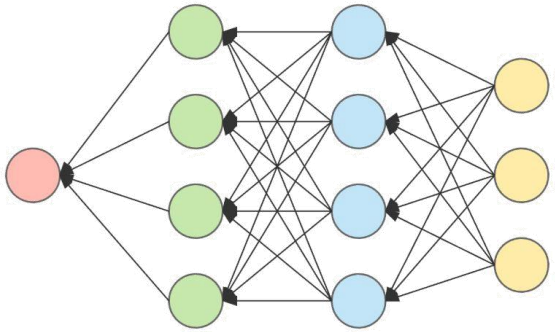


Train D



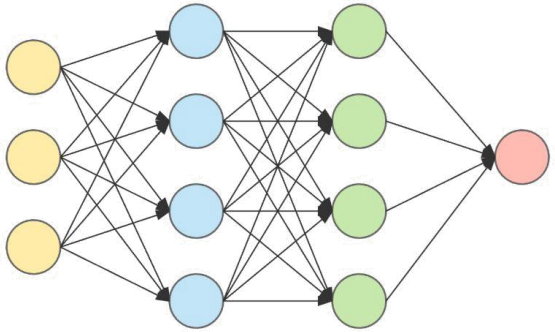


Generator





Discriminator

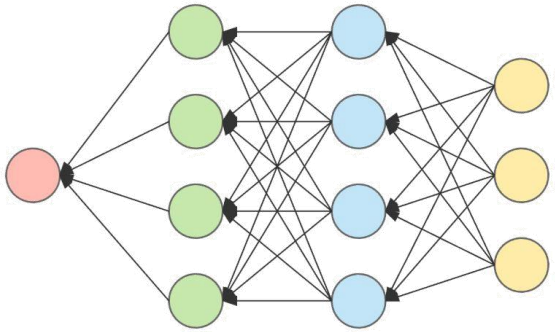


Train D



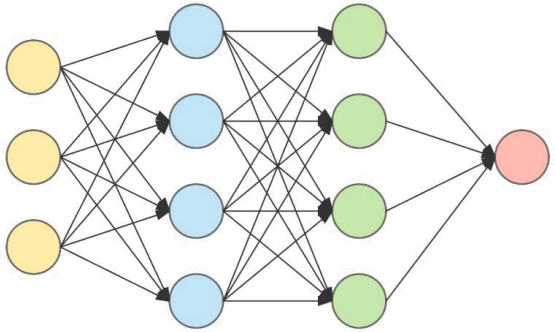


Generator

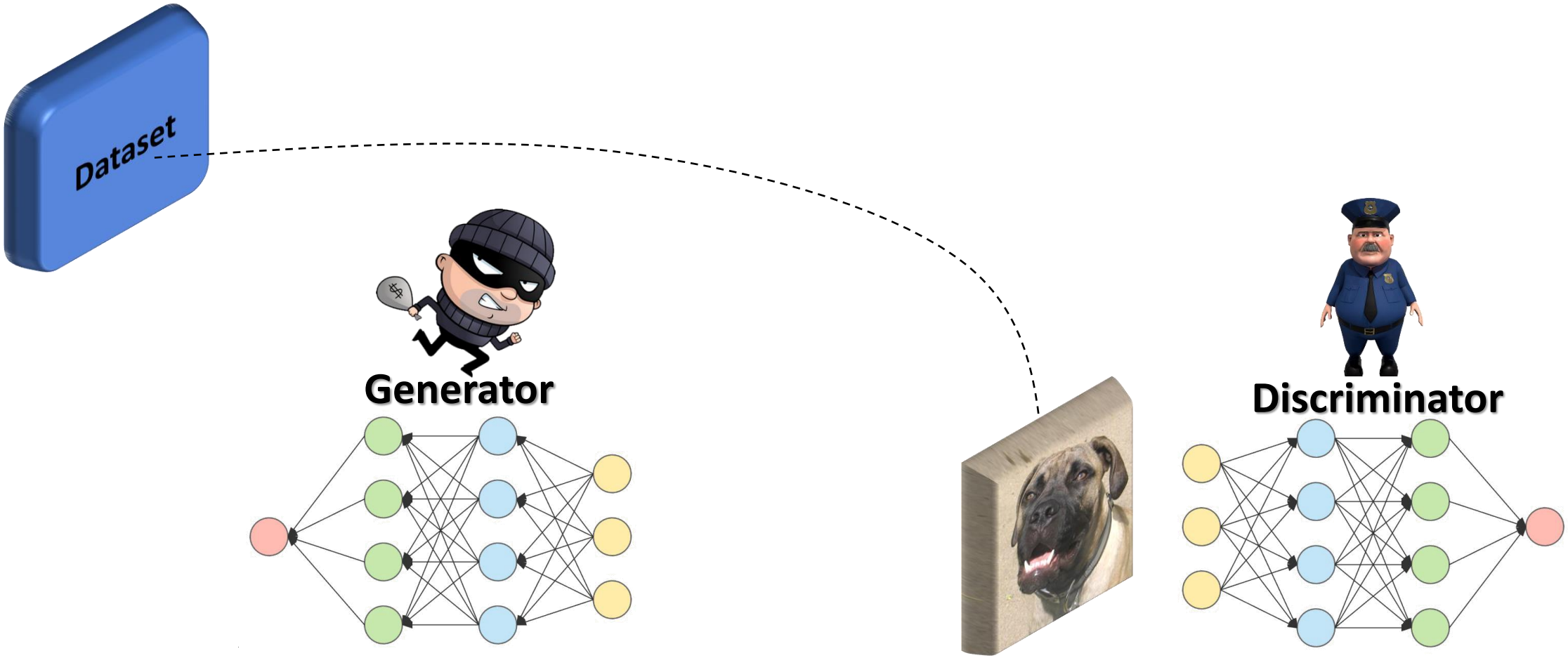




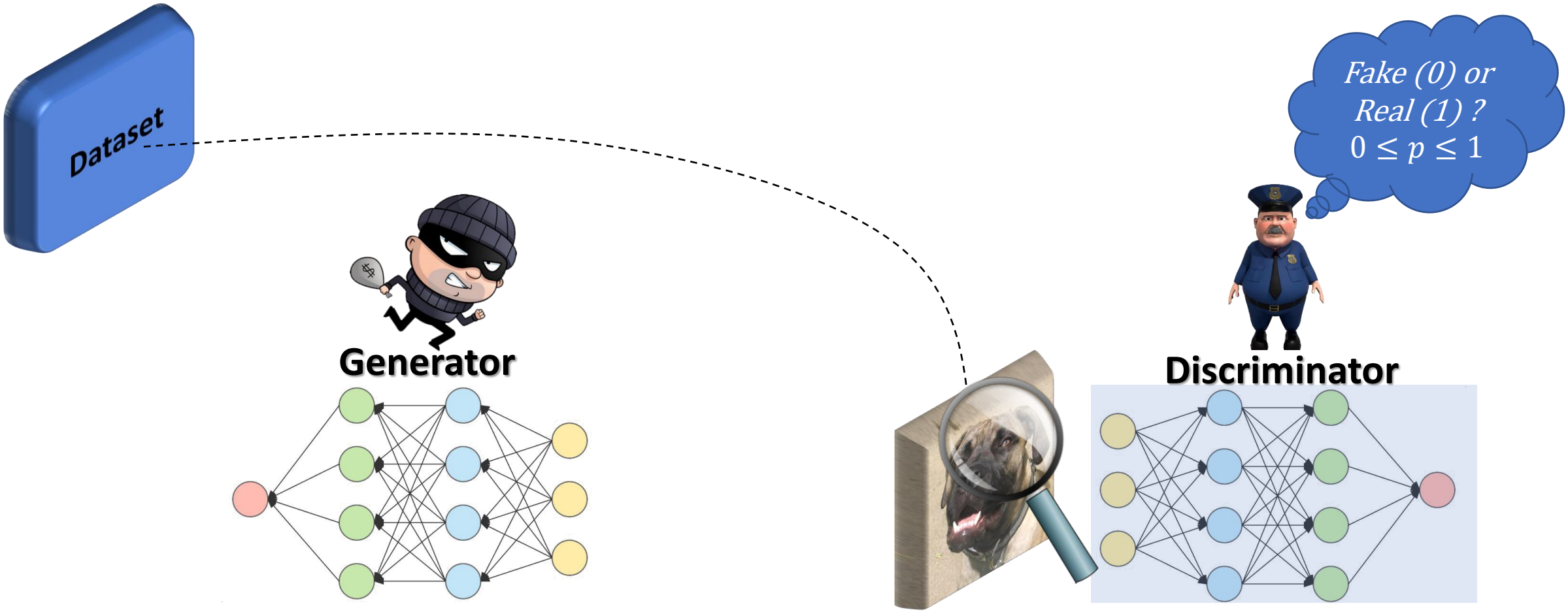
Discriminator



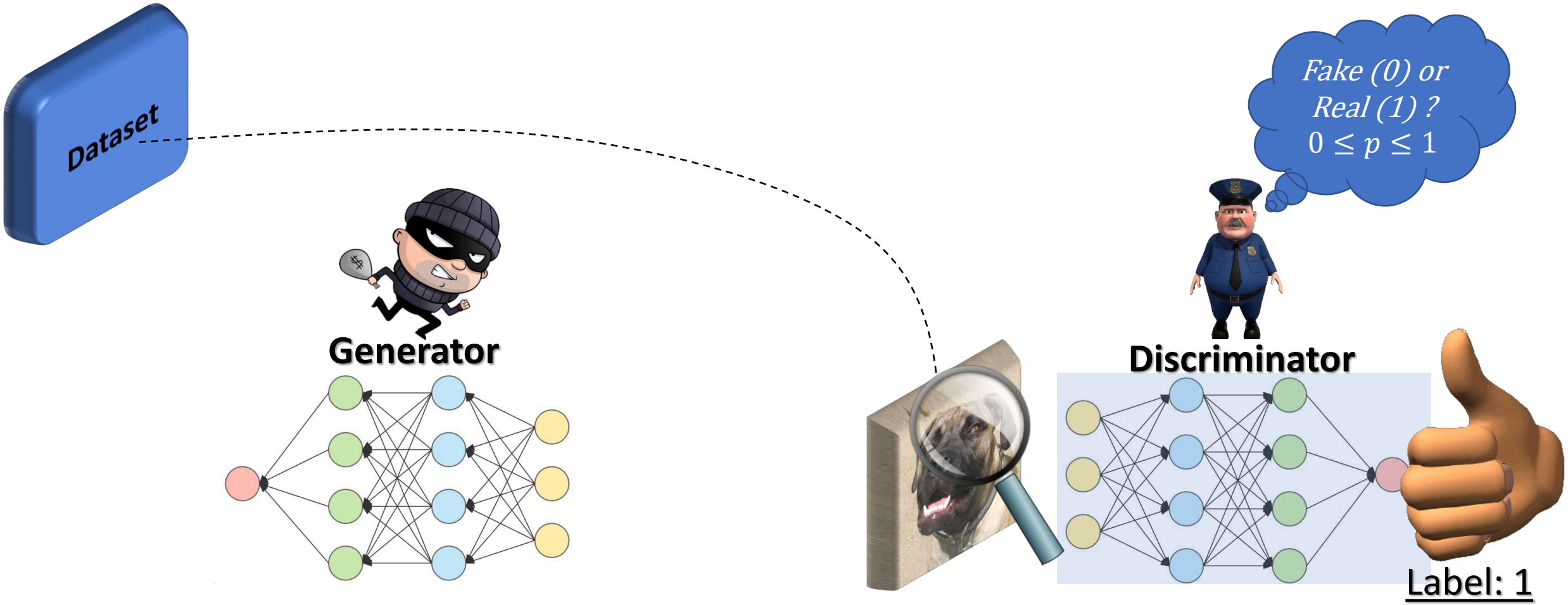
Train D



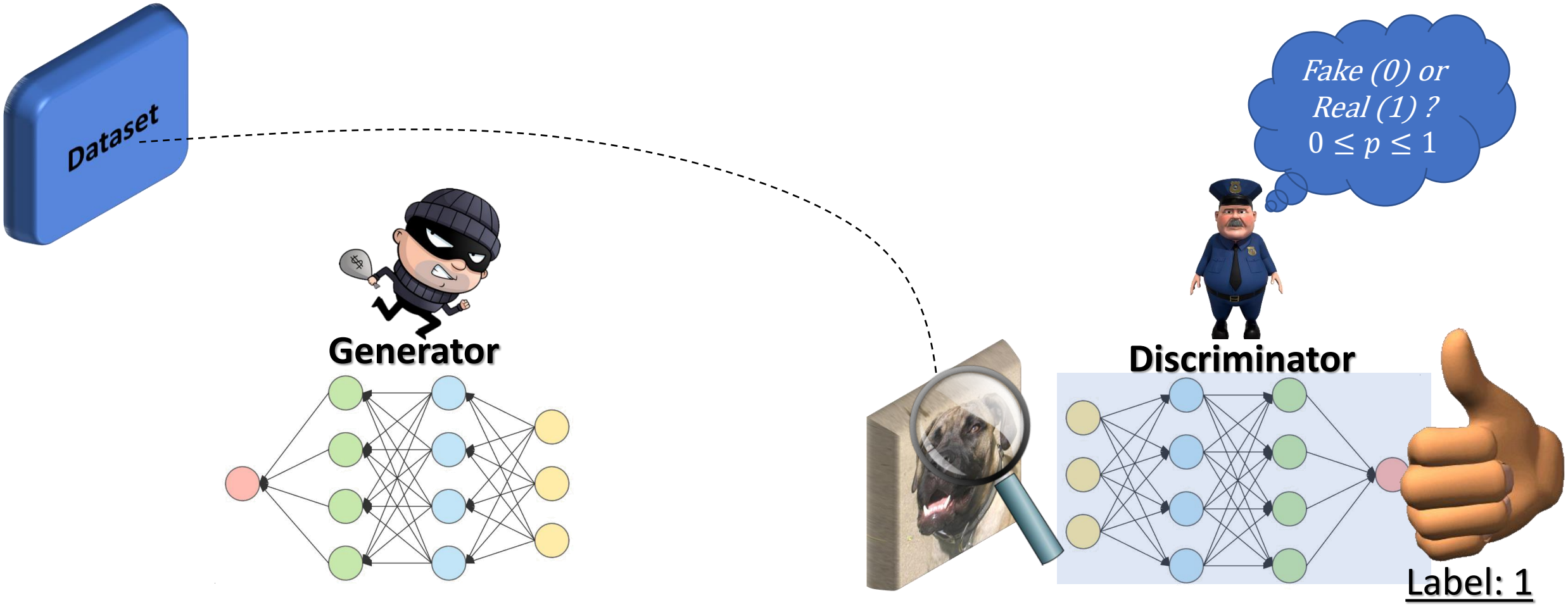
Train D


Dataset
Generator
Fake (0) or
Real (1) ?
$0 \leq p \leq 1$
Discriminator

Train D


Dataset
Generator
Fake (0) or
Real (1) ?
$0 \leq p \leq 1$
Discriminator
Label: 1

Train D



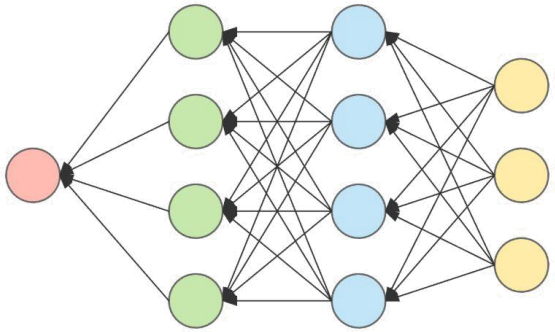
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D





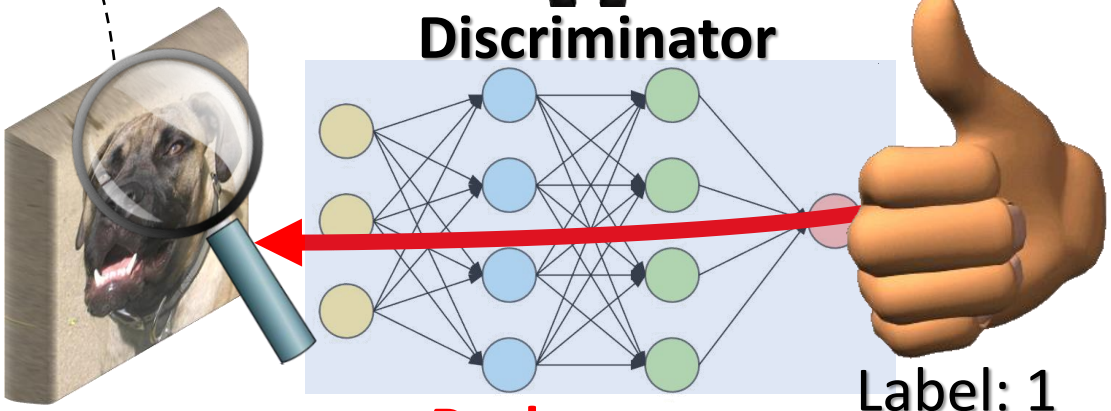
Generator





Discriminator

Fake (0) or Real (1) ?
$0 \leq p \leq 1$



Label: 1

Backprop, update weights

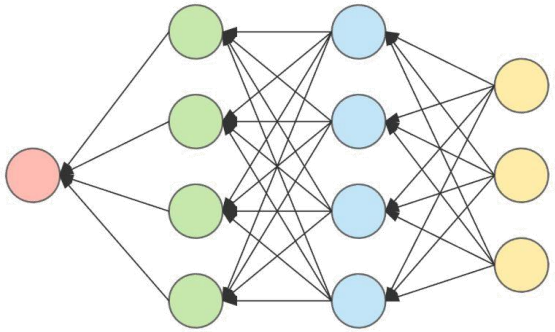
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}} \log(D(x))\}$$

Train D



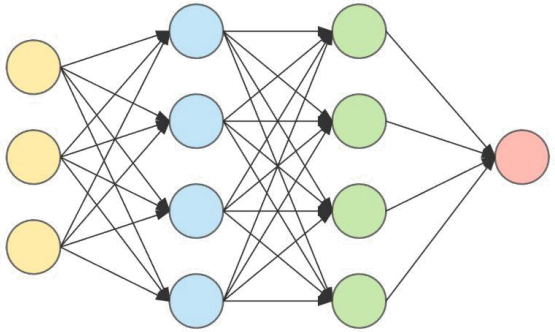


Generator





Discriminator



Maximize log likelihood of true examples

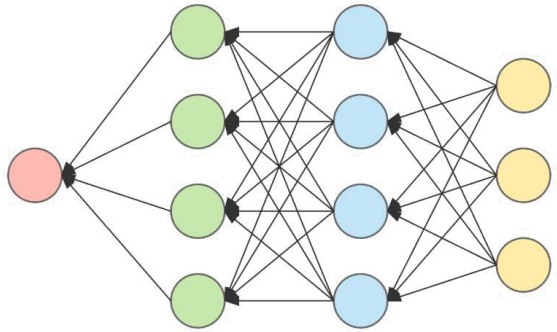
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D



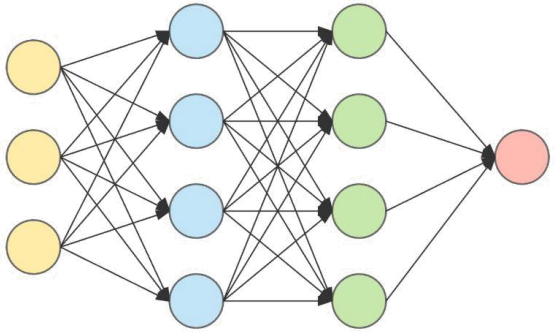


Generator



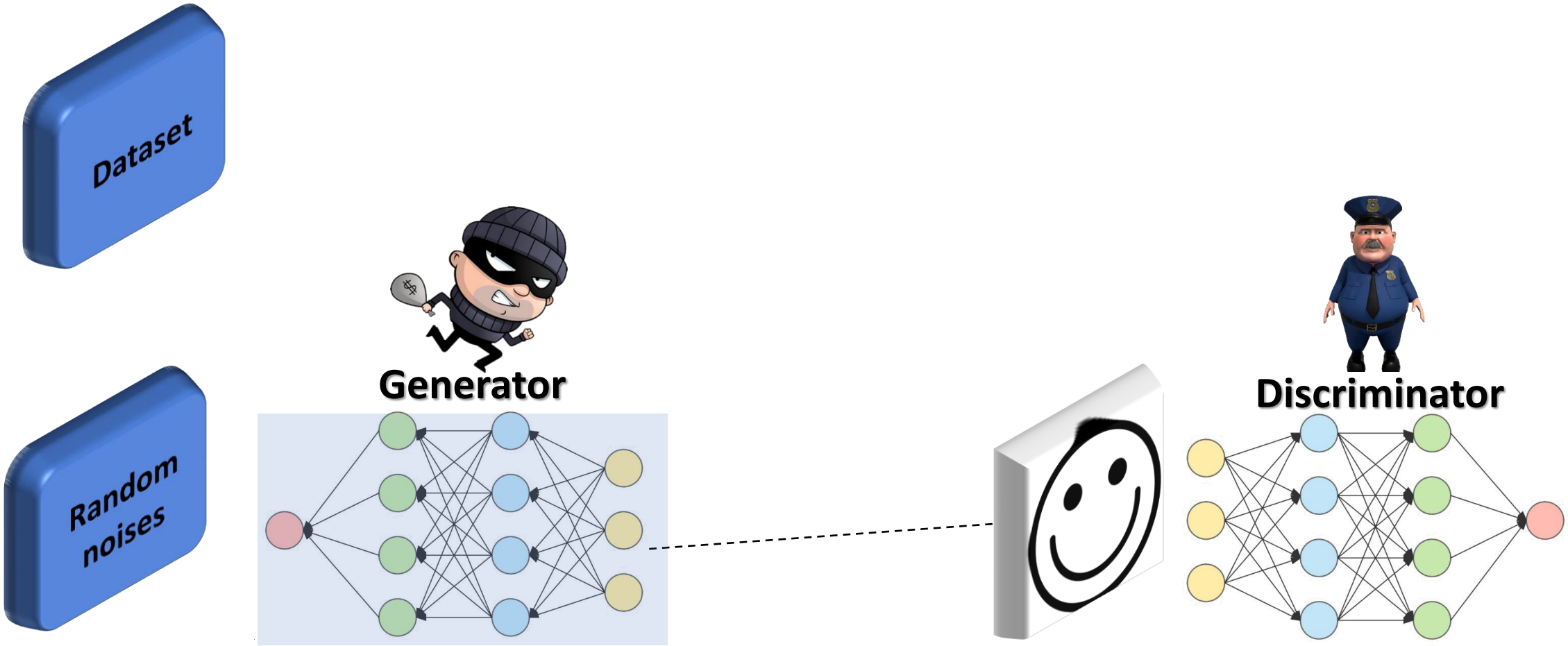


Discriminator



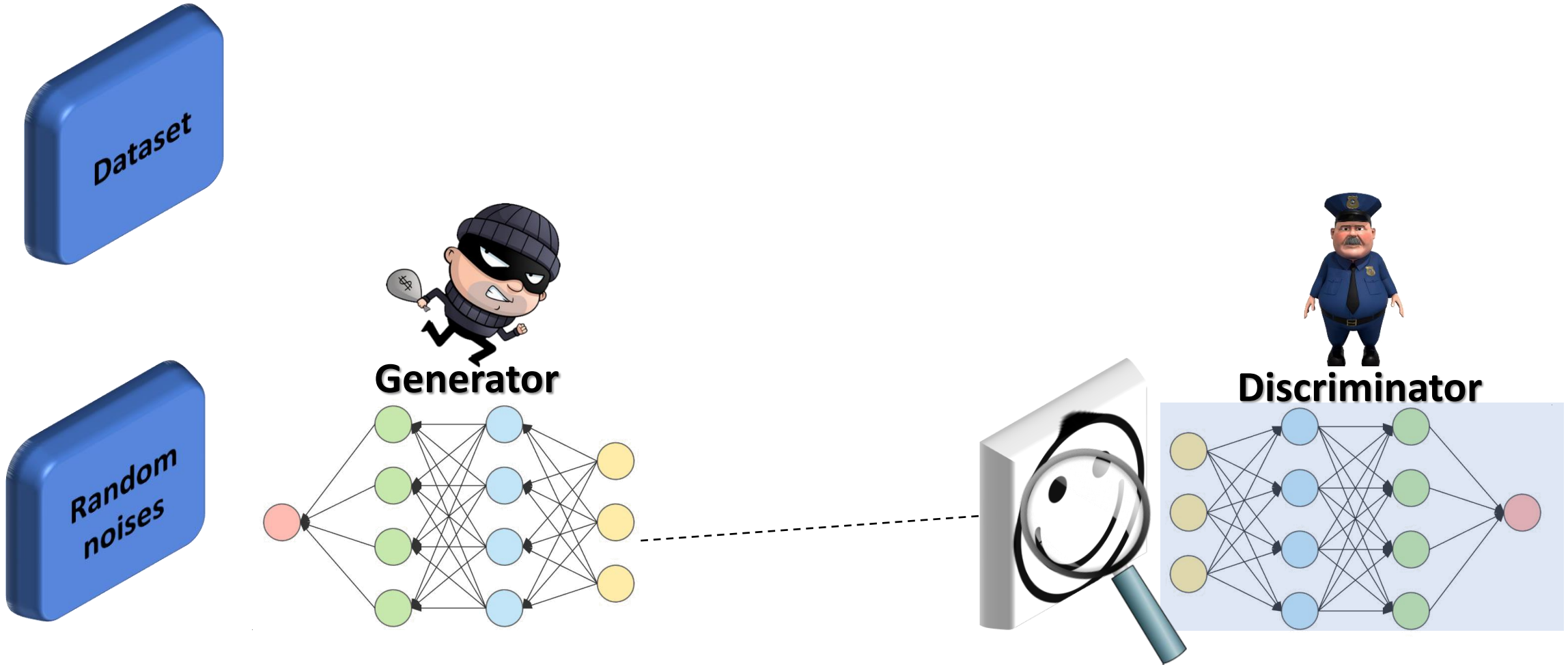
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D



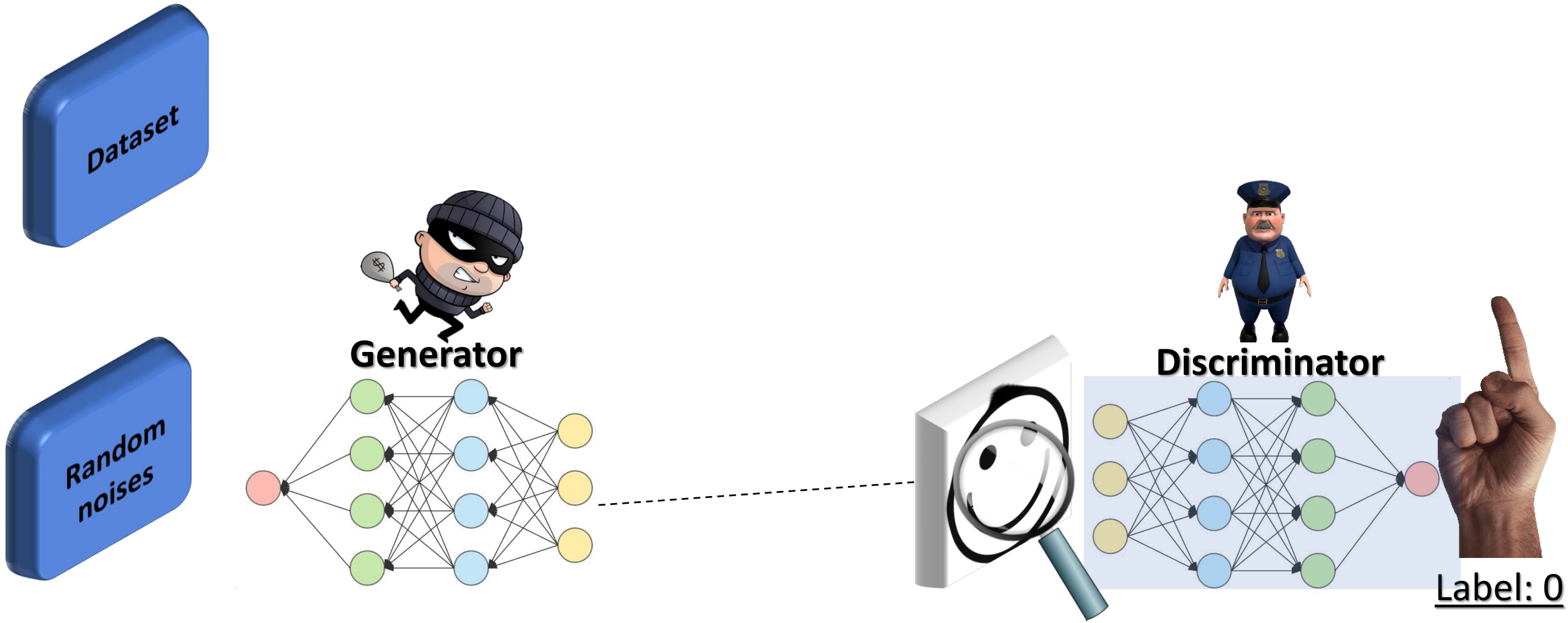
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D



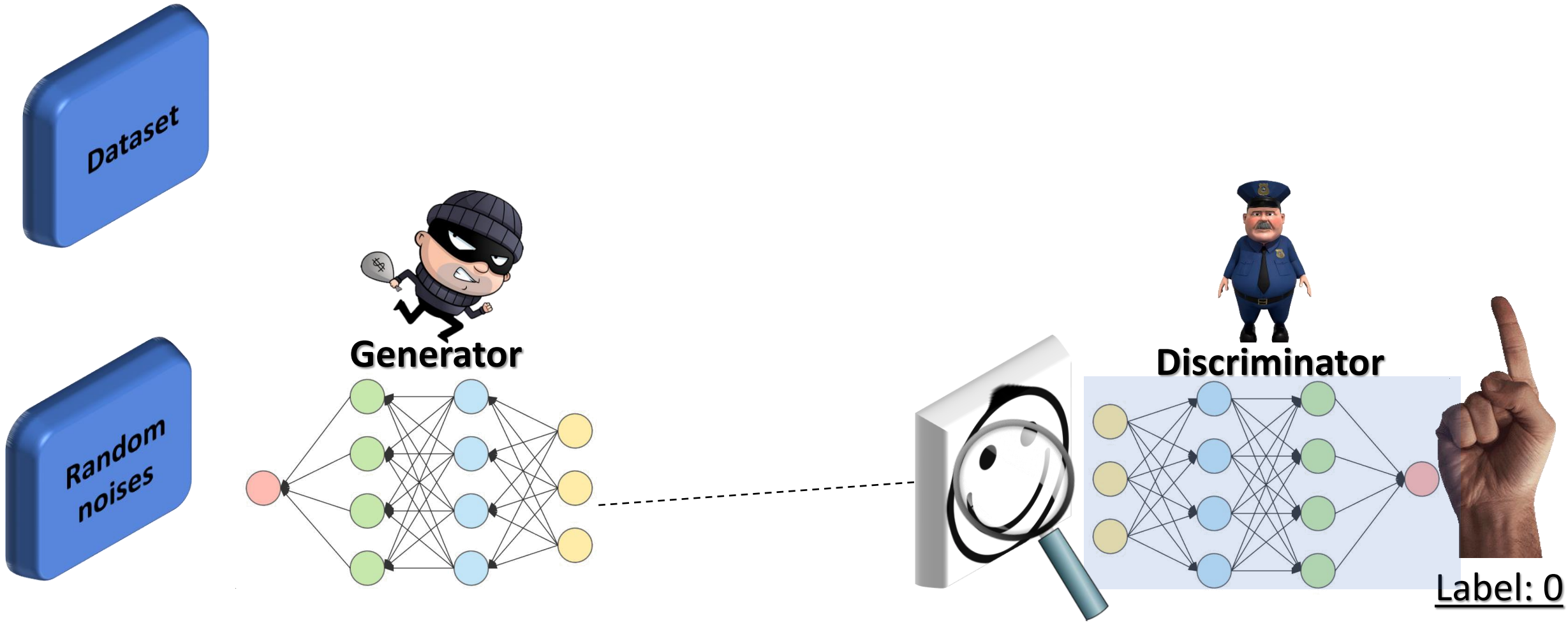
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D



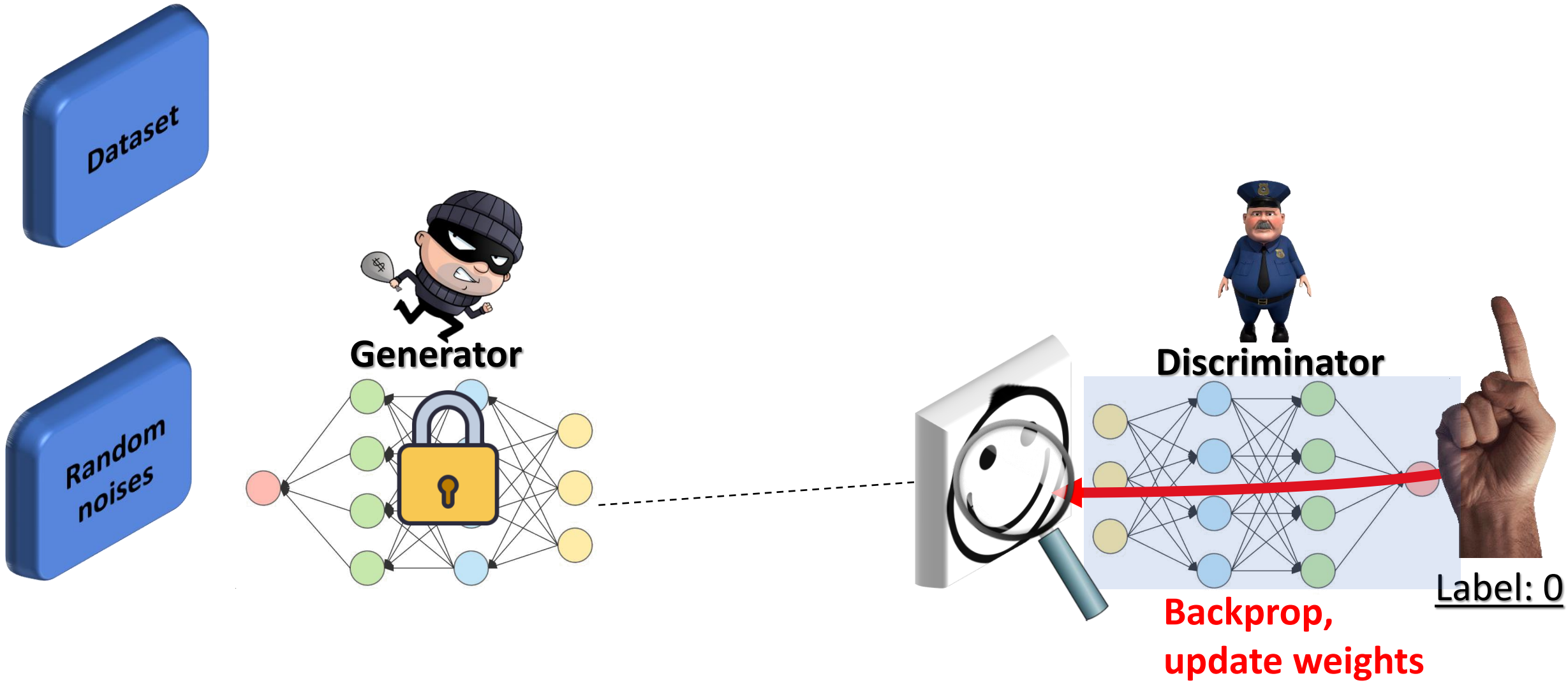
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x))\}$$

Train D



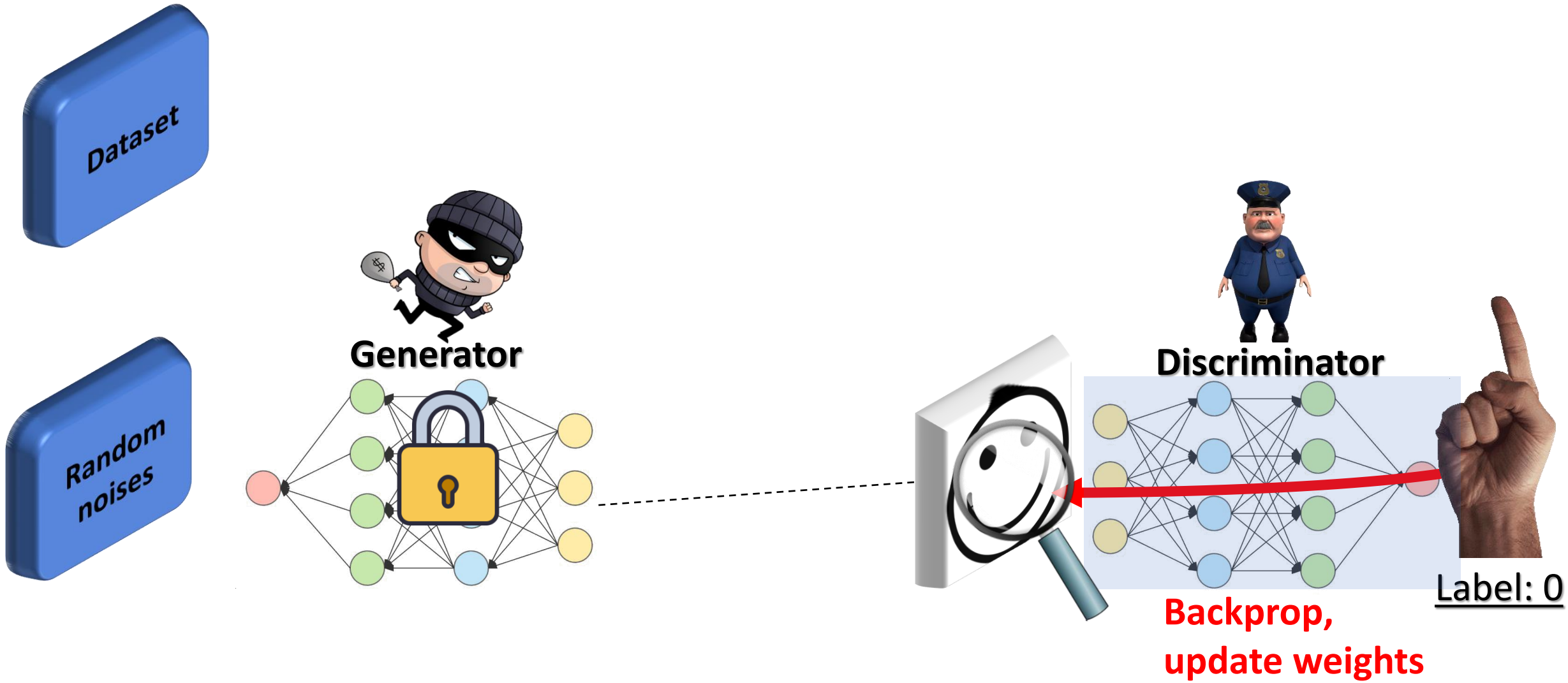
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x)) + \mathbb{E}_{\boldsymbol{z\sim p_z}}\boldsymbol{\log(1-D(G(z)))}\}$$

Train D



$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x)) + \mathbb{E}_{z\sim p_z}\log(1-D(G(z)))\}$$

Train D



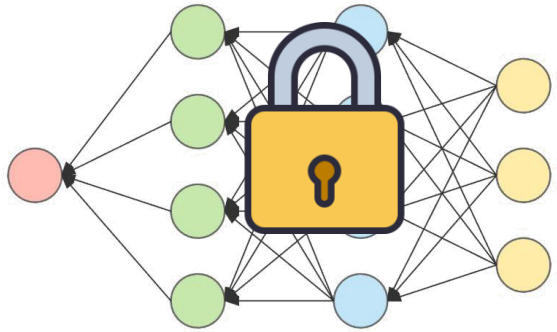
$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x)) + \mathbb{E}_{z\sim p_z}\log(1-D(G(z)))\}$$

Train D





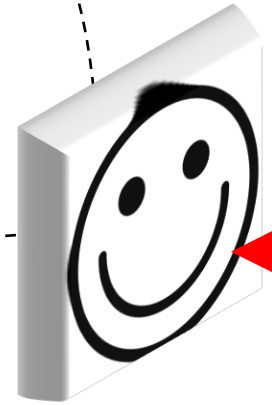
Generator

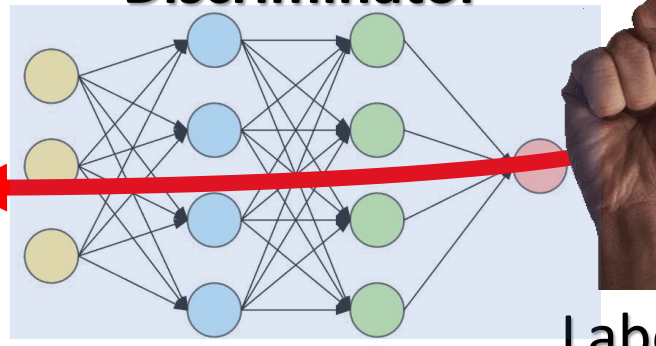


Random noises



Discriminator







Label: 0

Backprop, update weights

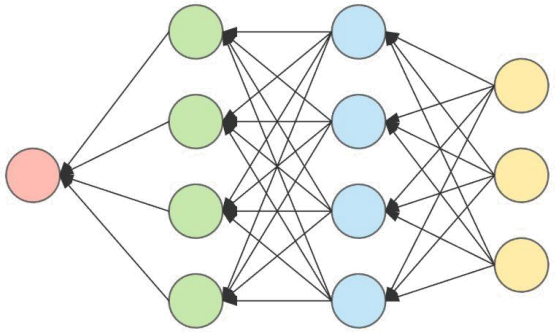
$$\max_D \{\mathbb{E}_{x\sim p_{data}} \log(D(x)) + \mathbb{E}_{z\sim p_z} \log(1 - D(G(z)))\}$$

Dataset

Random noises

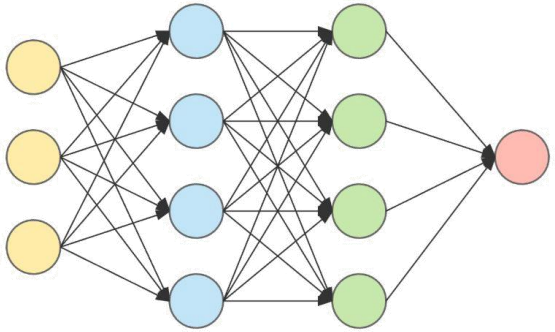


Generator





Discriminator



$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x)) + \mathbb{E}_{\boldsymbol{z\sim p_z}}\boldsymbol{\log(1-D(G(z)))}\}$$

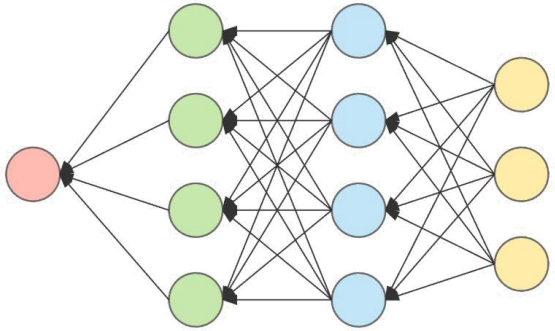
Train G





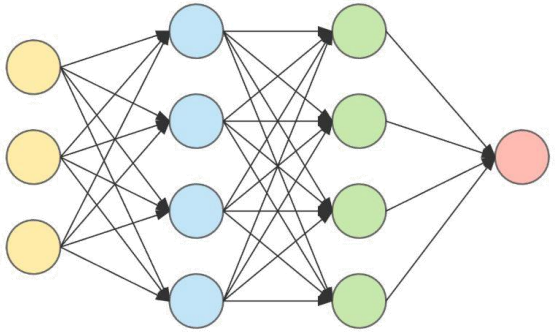


Generator





Discriminator



$$\max_{D}\{\mathbb{E}_{x\sim p_{data}}\log(D(x)) + \mathbb{E}_{\boldsymbol{z\sim p_z}}\boldsymbol{\log(1-D(G(z)))}\}$$

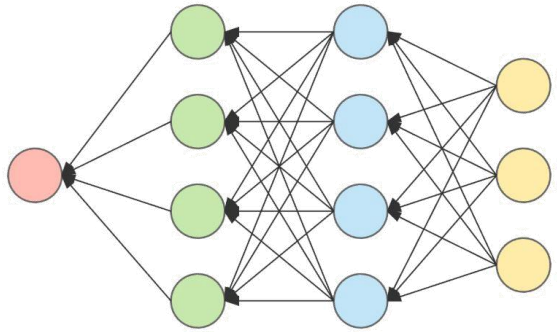
Train G





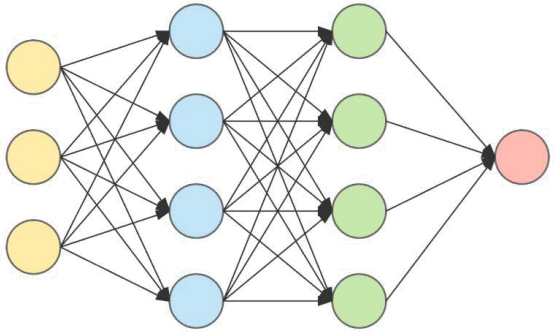


Generator





Discriminator



Minimax game: Make the best cop do the worst mistake!

$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$

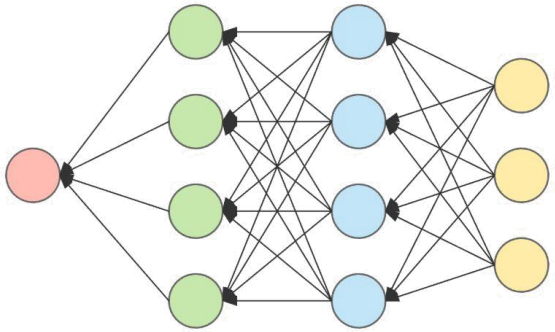
Train G





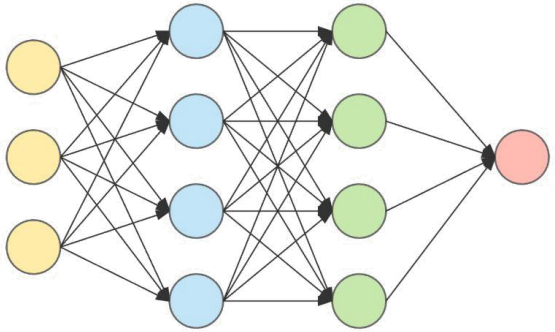


Generator





Discriminator



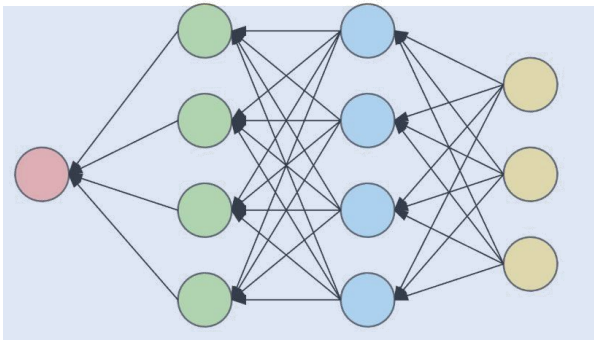
$$\min_{G} \max_{D} \{\mathbb{E}_{x\sim p_{data}} \log(D(x)) + \mathbb{E}_{z\sim p_z} log(1 - D(G(z)))\}$$







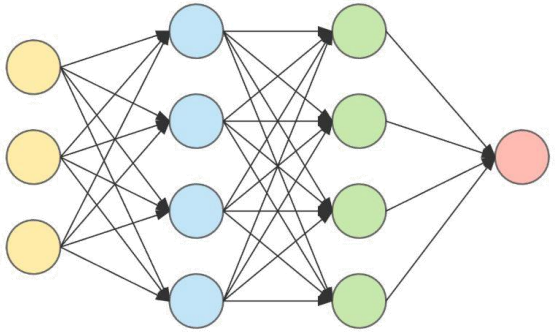
Generator





Discriminator





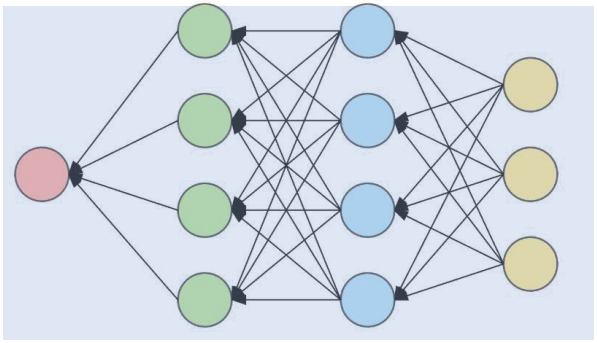
$$\min_{G} \max_{D} \{\mathbb{E}_{x\sim p_{data}} \log(D(x)) + \mathbb{E}_{z\sim p_z} log(1 - D(G(z)))\}$$







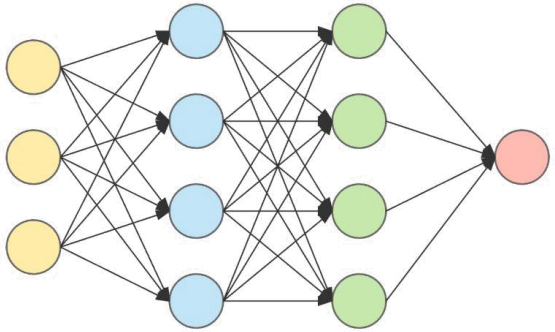
Generator





Discriminator





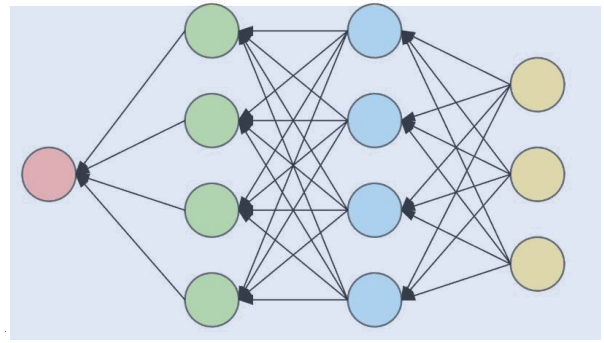
$$\min_{G} \max_{D} \{\mathbb{E}_{x\sim p_{data}} \log(D(x)) + \mathbb{E}_{z\sim p_z} log(1 - D(G(z)))\}$$







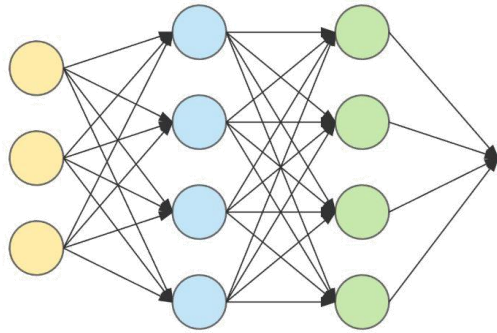
Generator





Discriminator

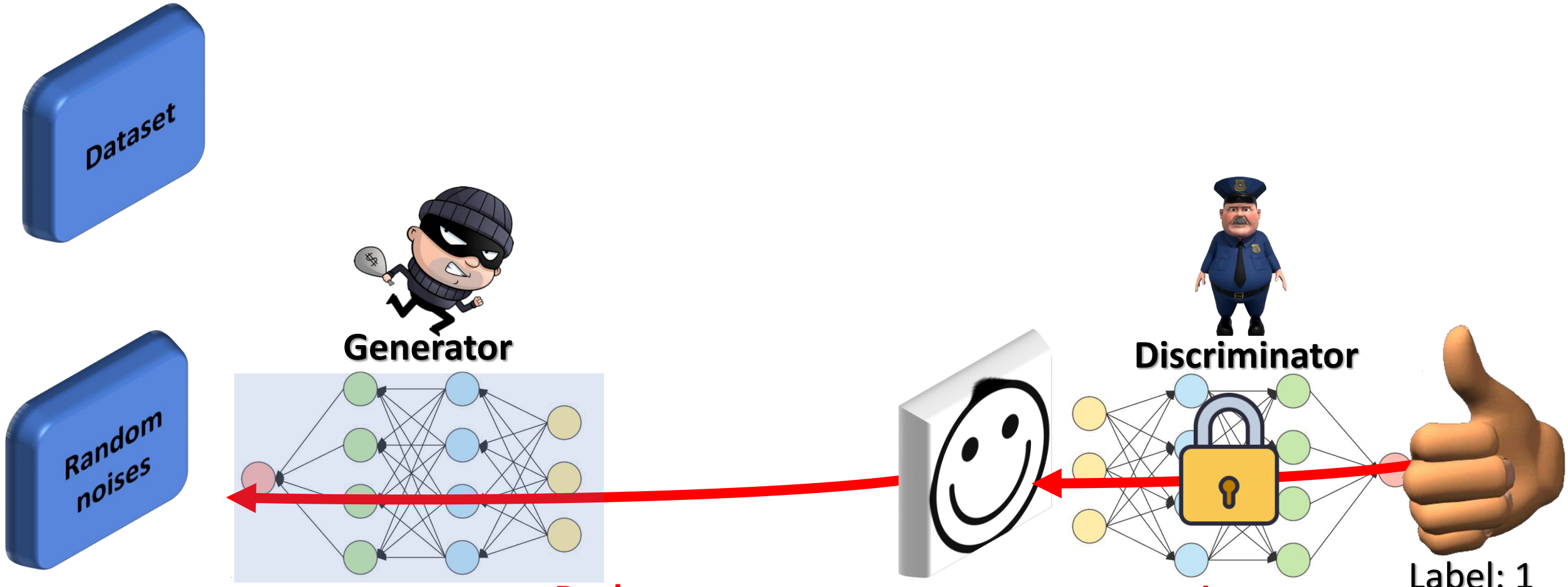






Label: 1

$$\min_{G} \max_{D} \{\mathbb{E}_{x\sim p_{data}} \log(D(x)) + \mathbb{E}_{z\sim p_z} log(1 - D(G(z)))\}$$



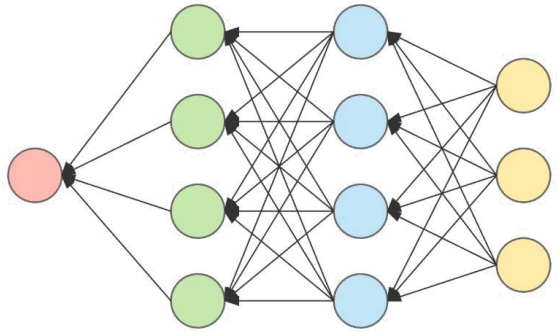
$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$

Dataset

Random noises

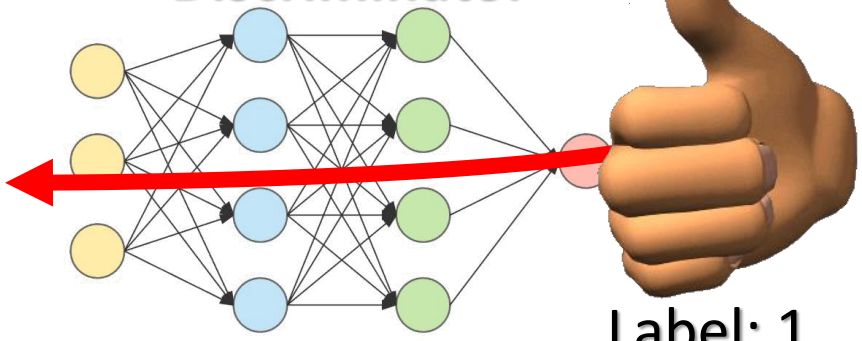


Generator





Discriminator

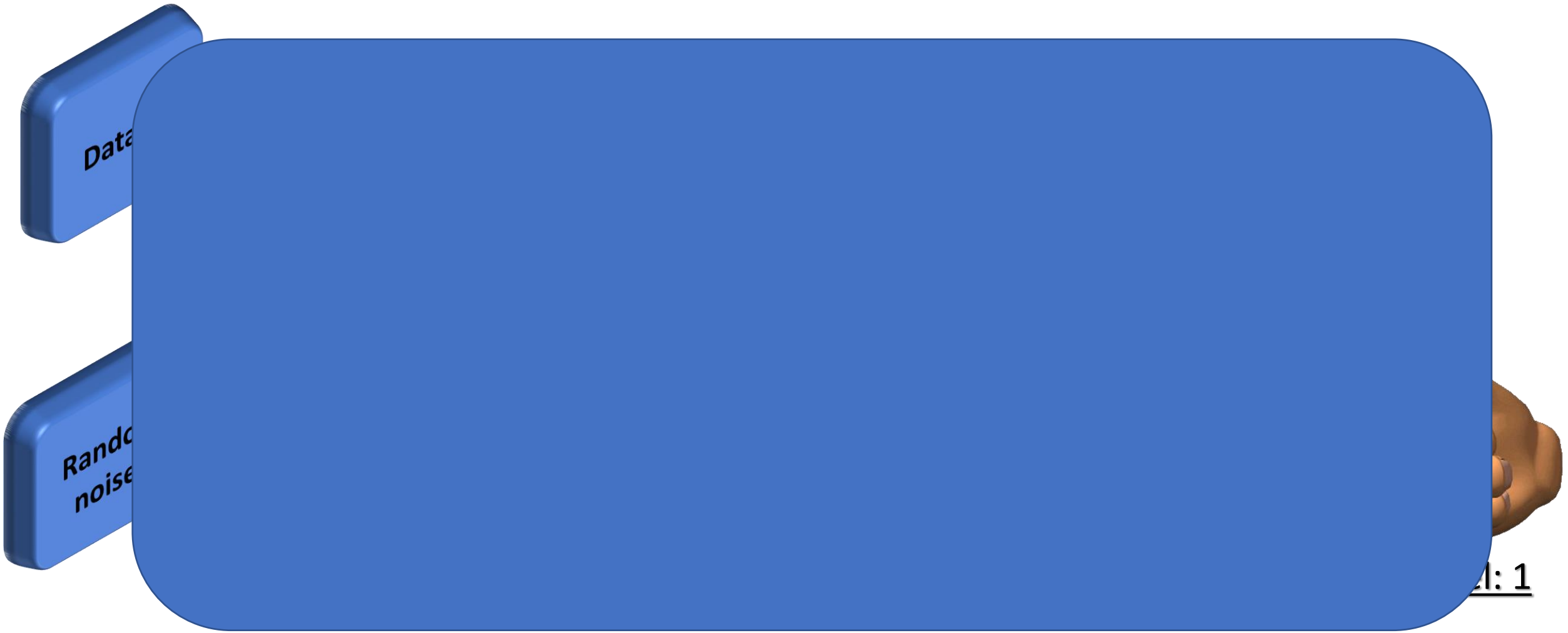


Label: 1

Backprop,

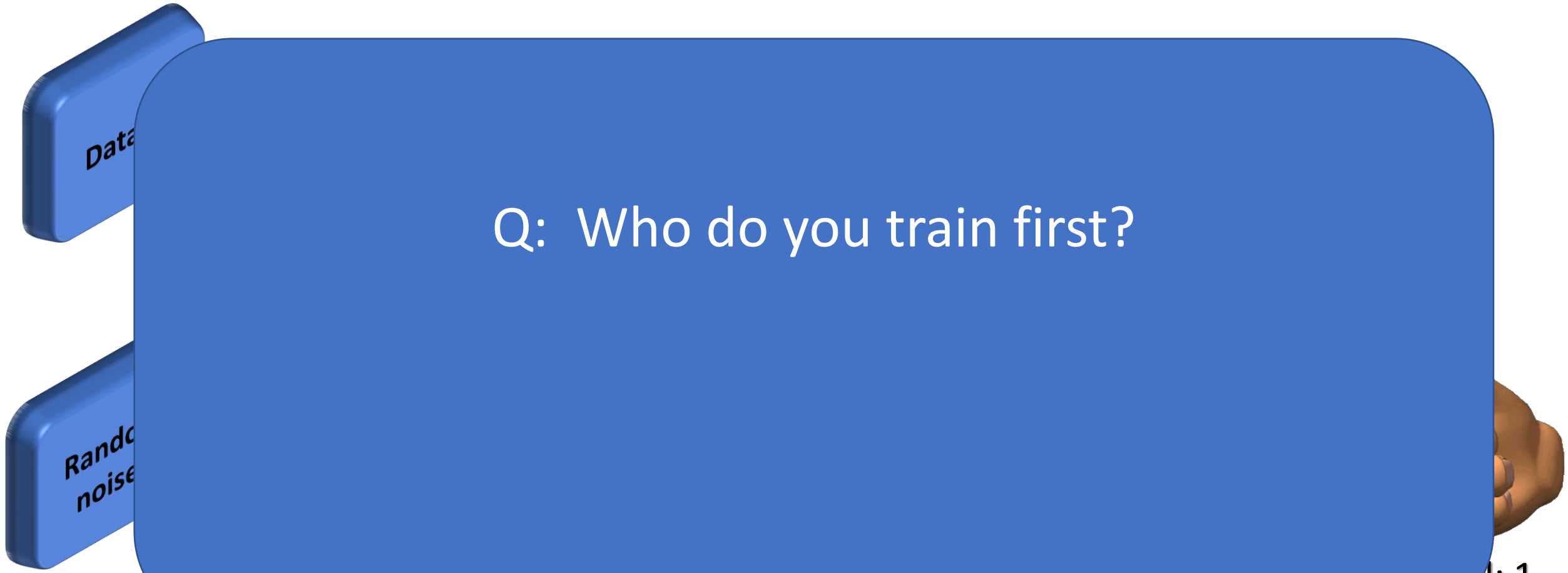
Don't update weights

$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$



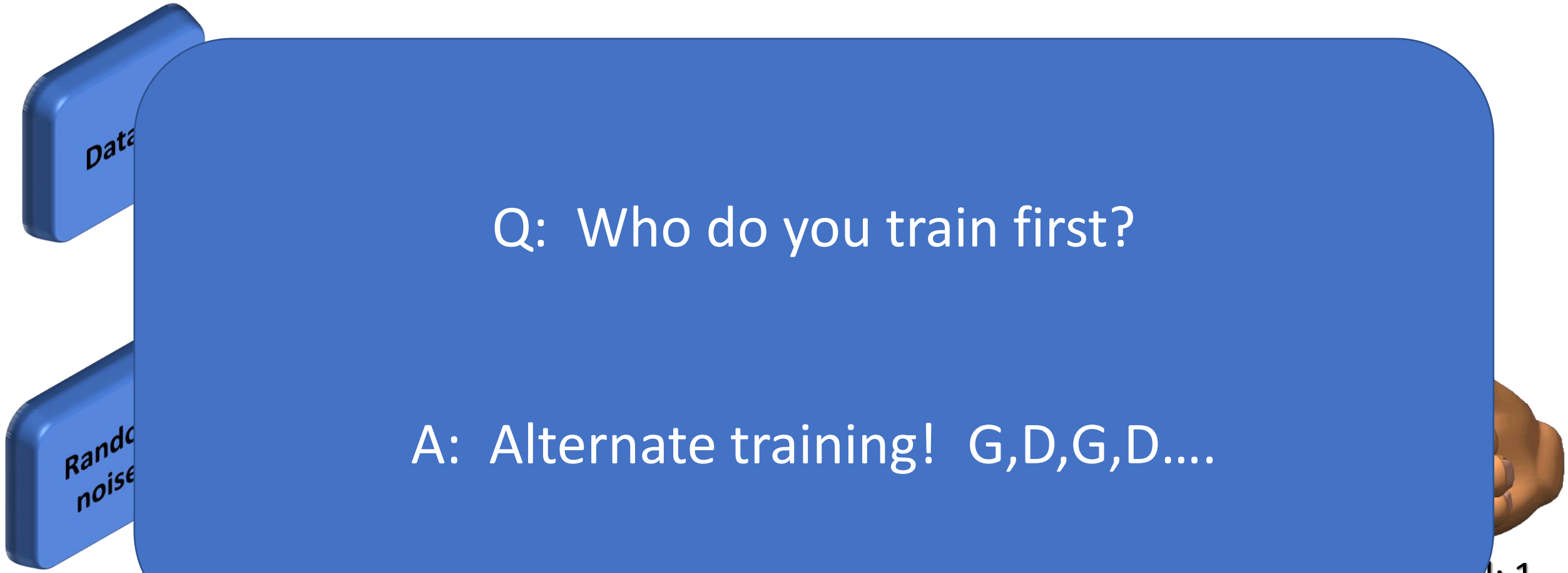
Don't update weights

$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$



Don't update weights

$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$



Don't update weights

$$\min_{G} \max_{D} \{\mathbb{E}_{x \sim p_{data}} \log(D(x)) + \mathbb{E}_{z \sim p_z} log(1 - D(G(z)))\}$$

FAQ1: Why does it work?

- D learns probability! G trains to sample instance with high probability!
- Objective does not determine mapping directly- arrangement of latent space is learned!
- Theory: minimizes JS divergence between generated and real distributions.

FAQ2: Why alternating?

- Gradients are meaningless when game is unbalanced.

- Pre-train D? Negative examples?

- Pre-train G? What loss?
  For G, D is a **learned loss function**



FAQ2: Why alternating?

- Gradients are meaningless when game is unbalanced.

- Pre-train D? Negative examples?

- Pre-train G? What loss?
  For G, D is a **learned loss function**



GANs, Goodfellow 2014



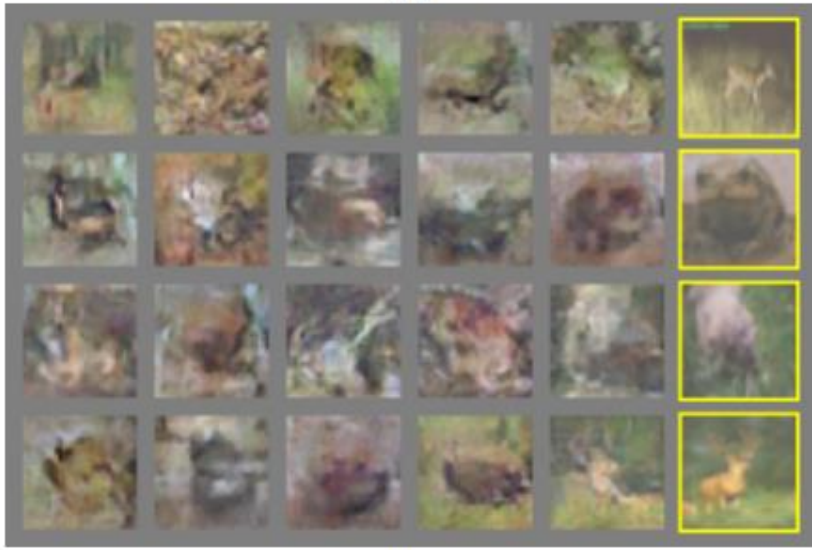
a)



b)

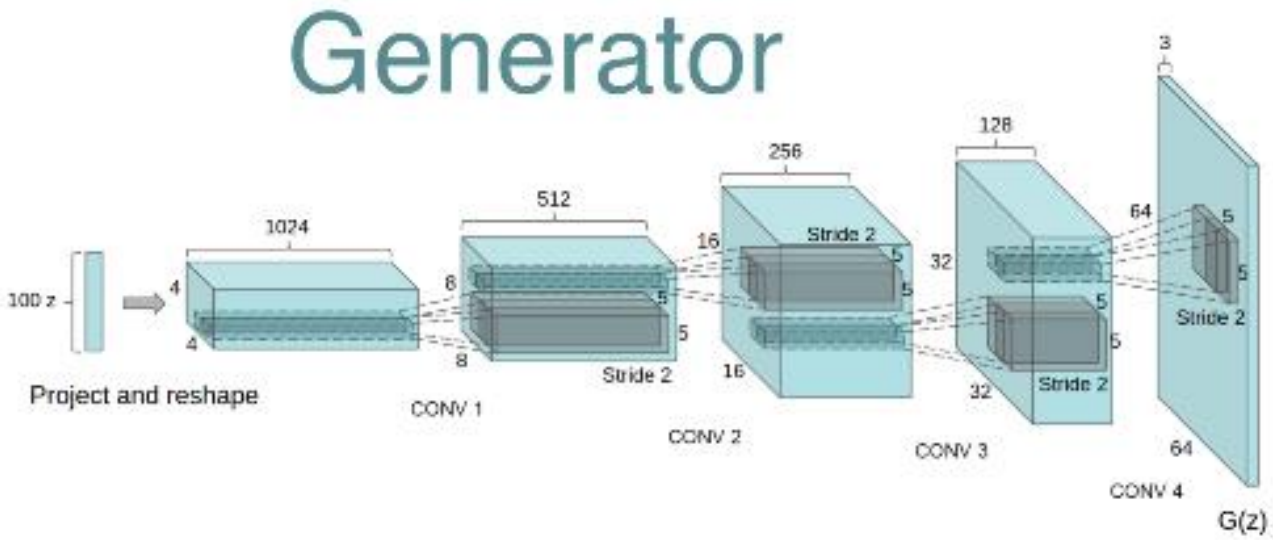


c)



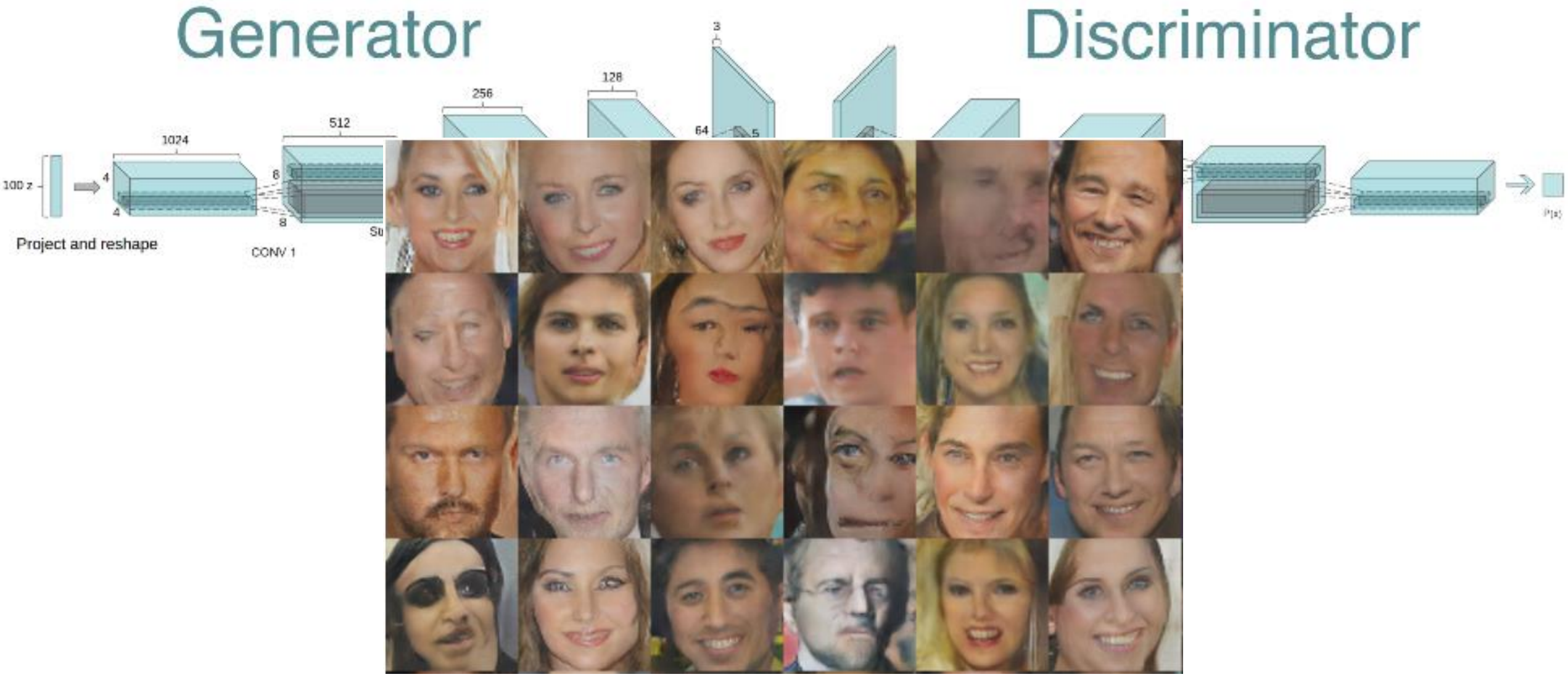
d)

DCGAN Radford 2015





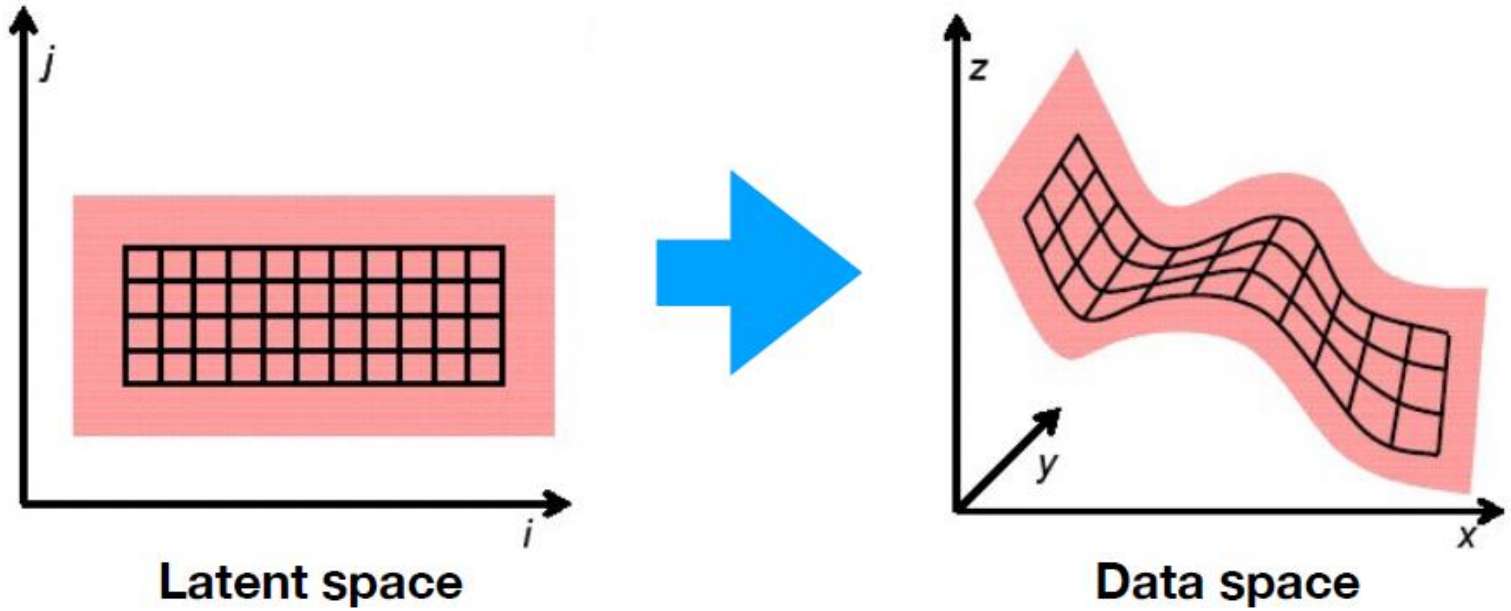
DCGAN Radford 2015



Latent space interpolation

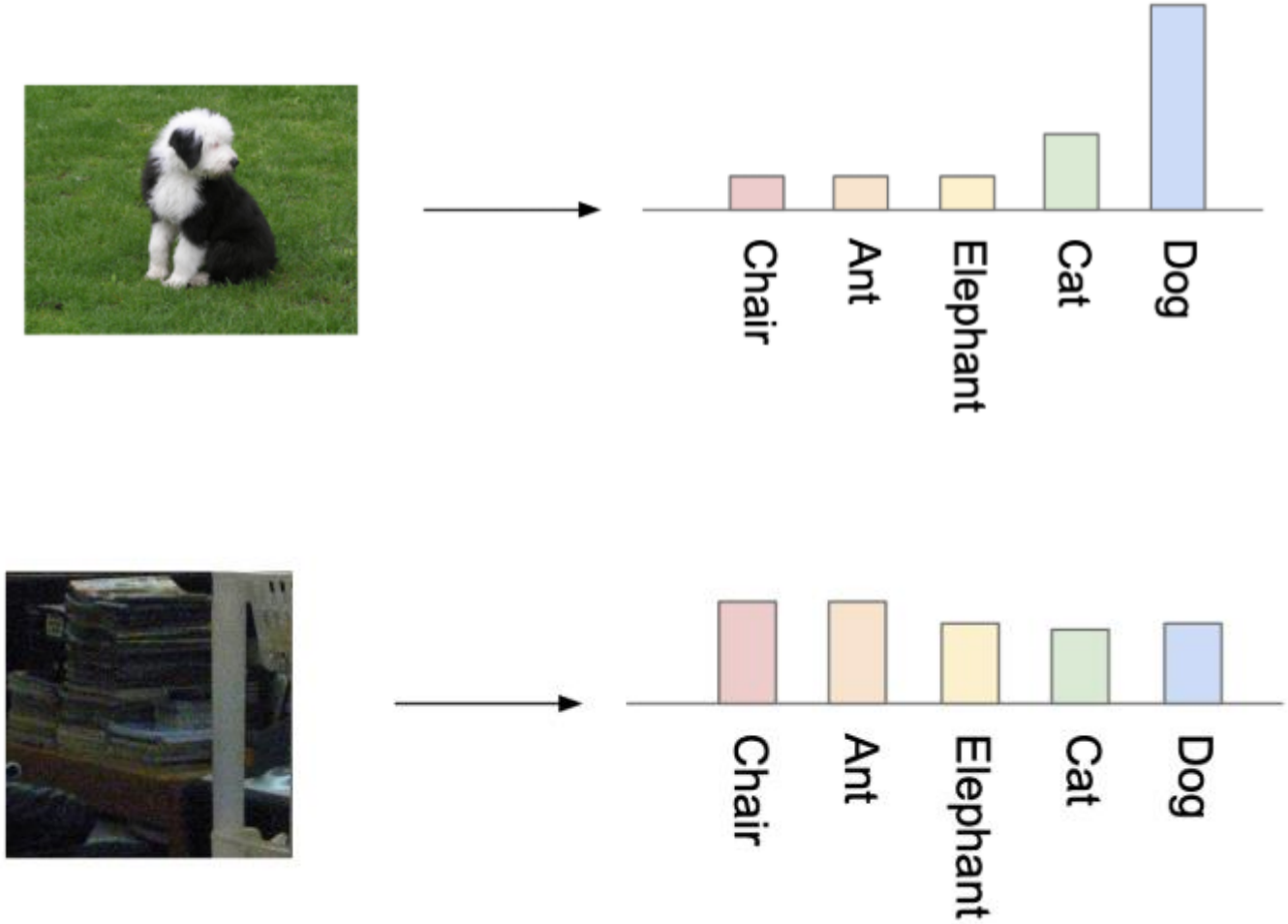


Why does it work?

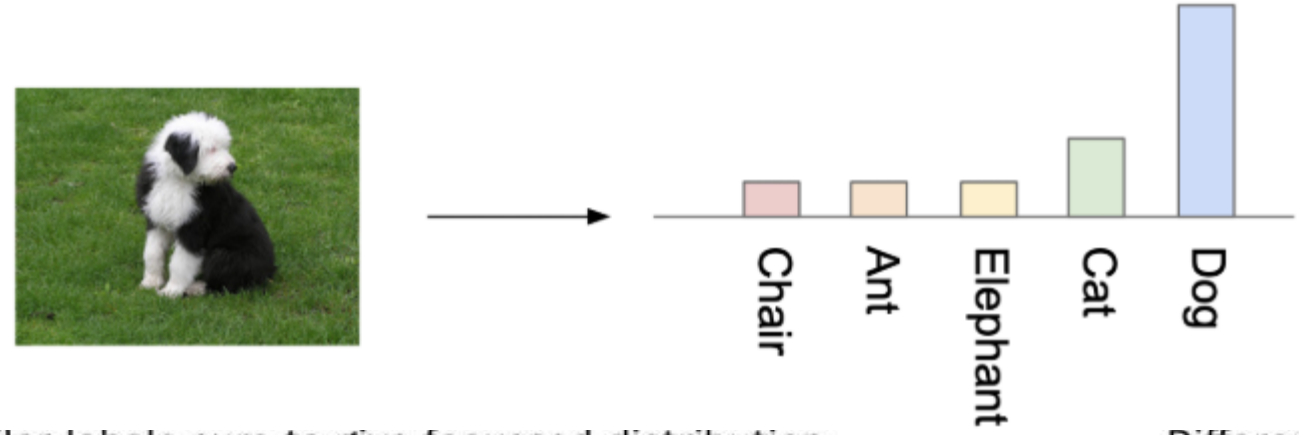


1. Every point is mapped to a valid example.
2. Network is continuous.

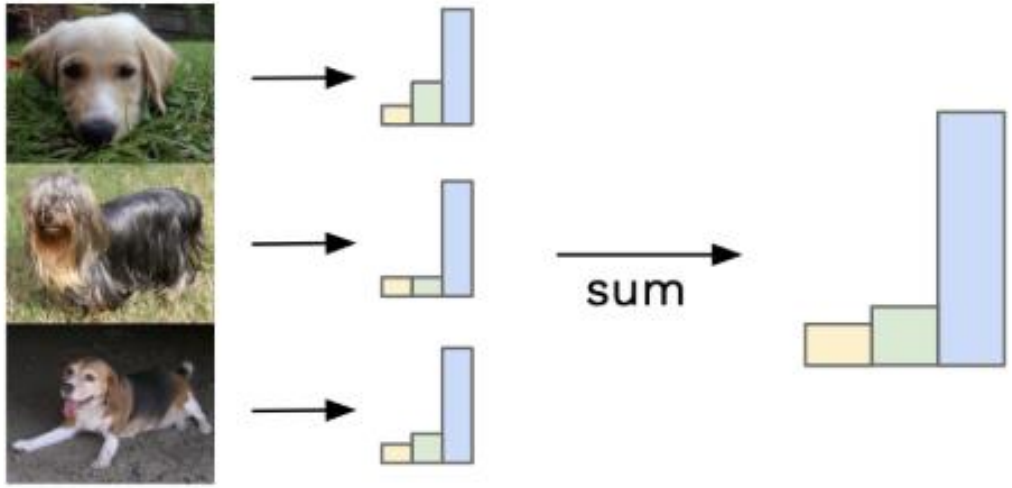
Evaluation metrics: Inception score



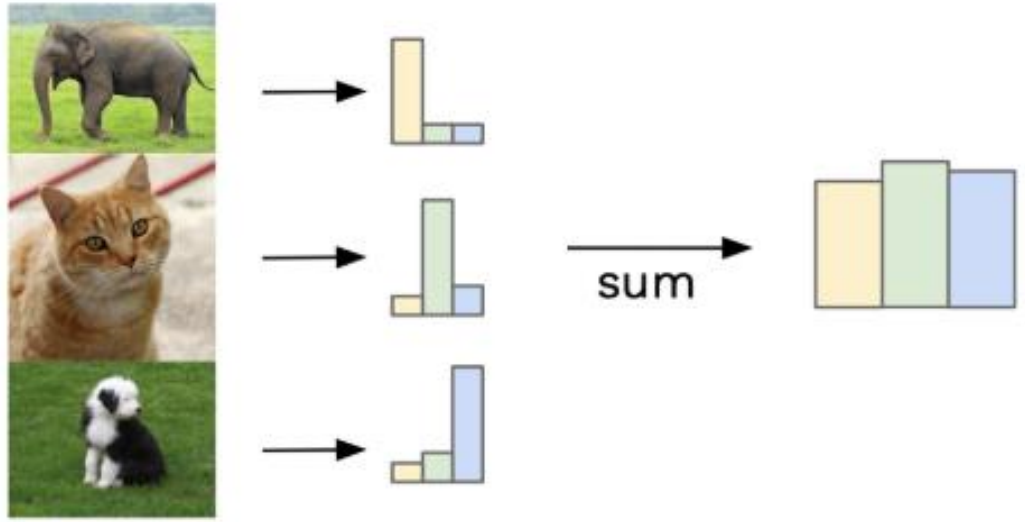
Evaluation metrics: Inception score



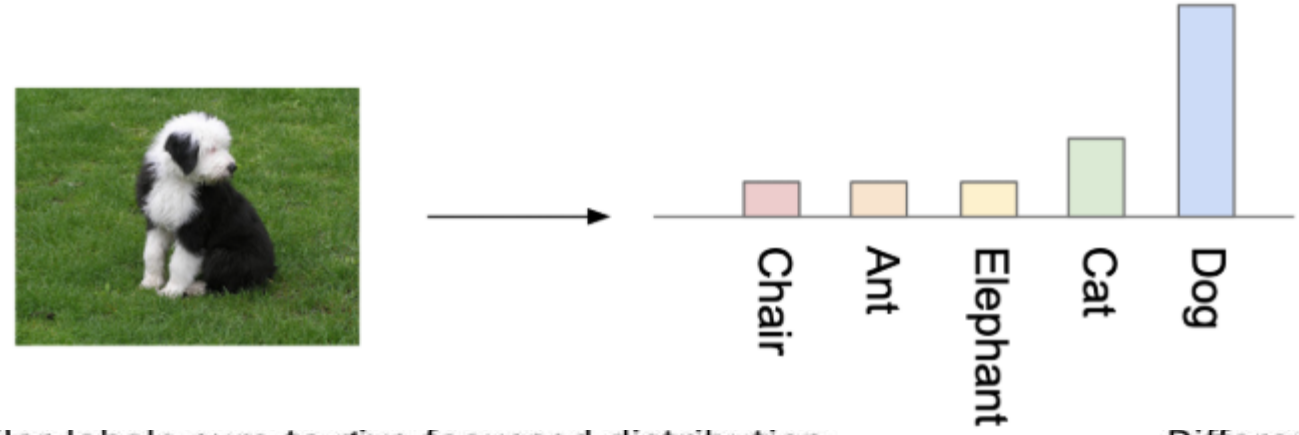
Similar labels sum to give focussed distribution



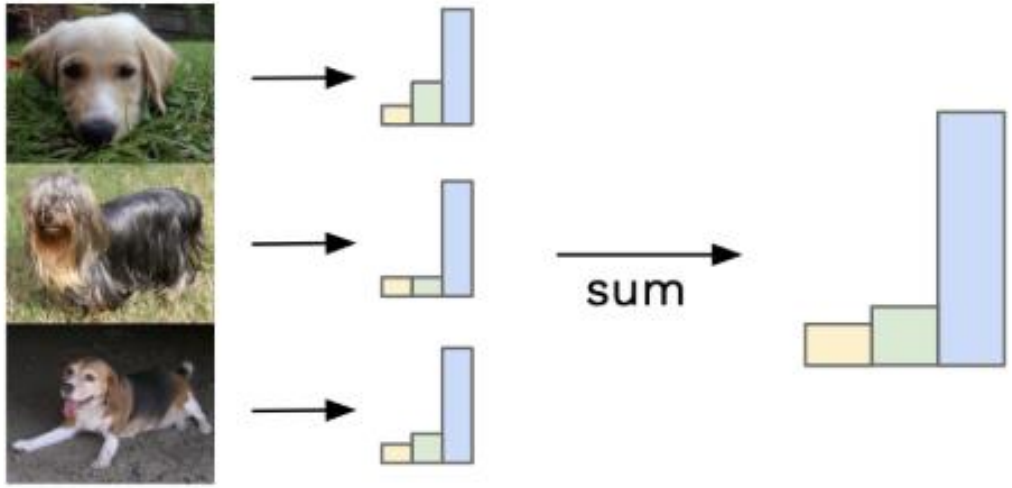
Different labels sum to give uniform distribution



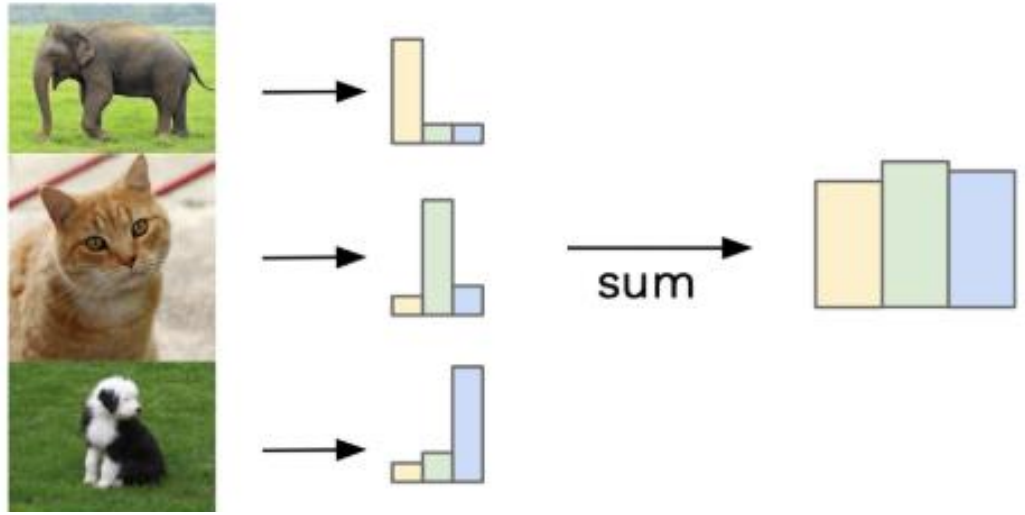
Evaluation metrics: Inception score



Similar labels sum to give focussed distribution



Different labels sum to give uniform distribution



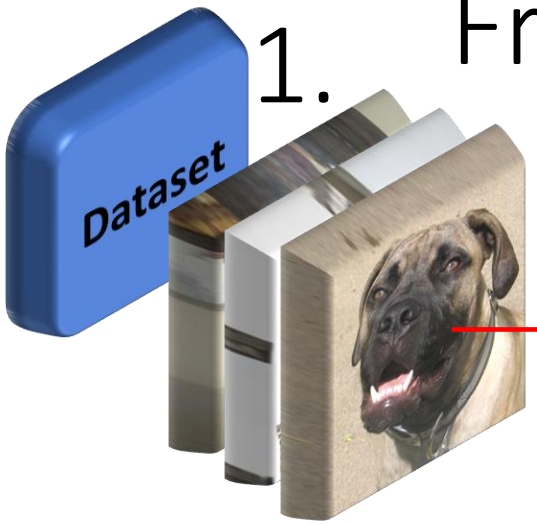
$$\mathbf{IS}(G) = \exp\left(\mathbb{E}_{\mathbf{x} \sim p_g} D_{KL}(p(y|\mathbf{x}) \parallel p(y)) \right)$$

1. Fréchet Inception Distance (FID)

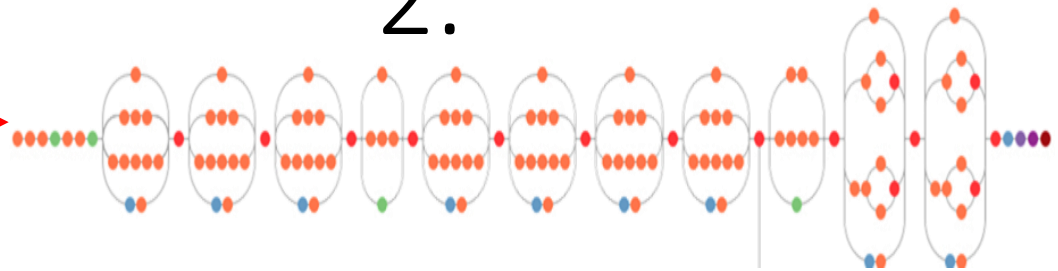


Fréchet Inception Distance (FID)

1.

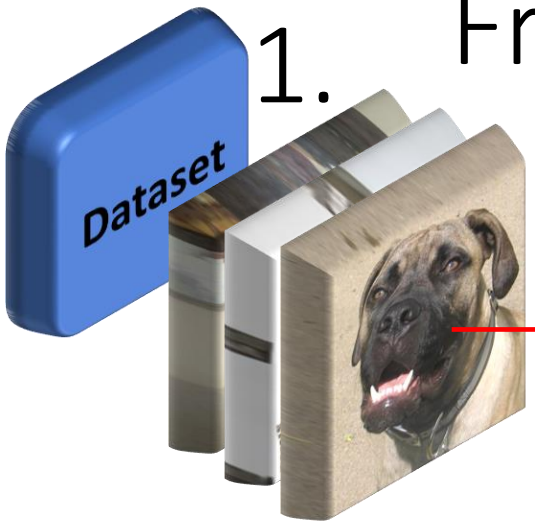


2.

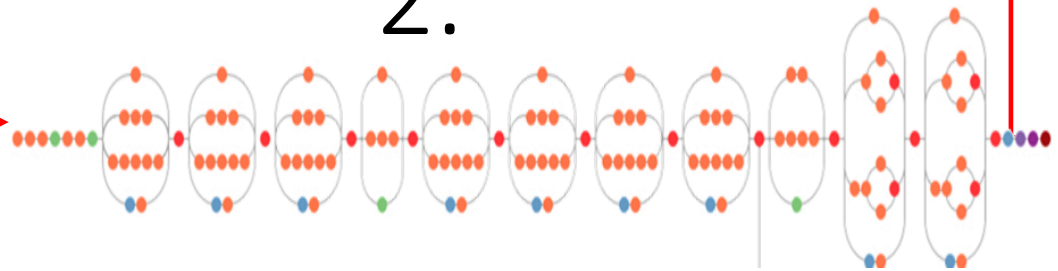


Fréchet Inception Distance (FID)

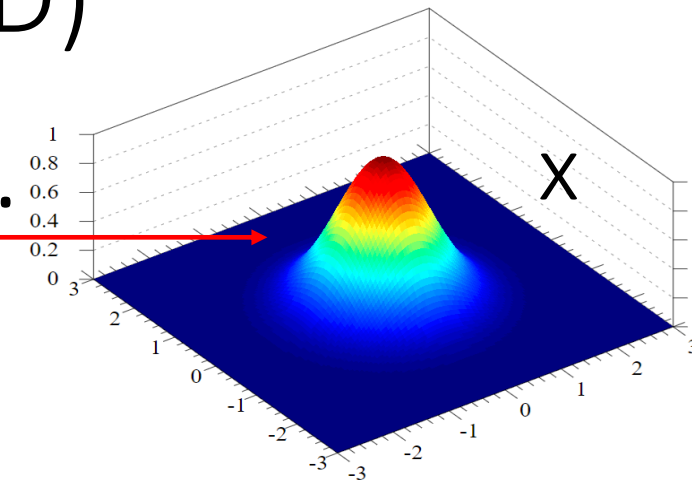
1.



2.

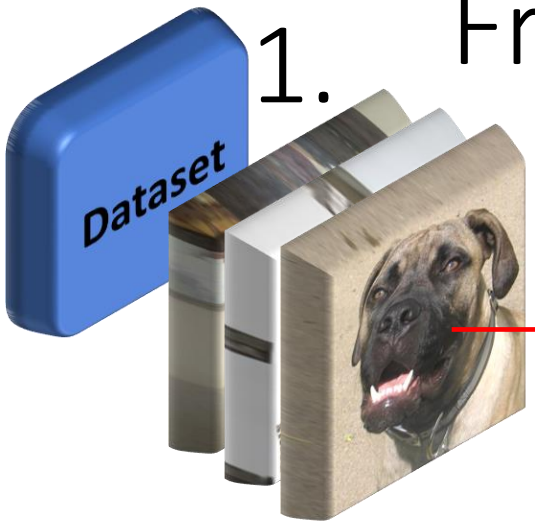


3.

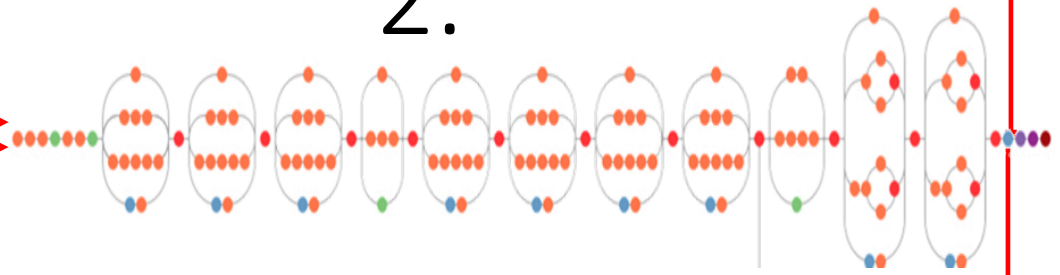


Fréchet Inception Distance (FID)

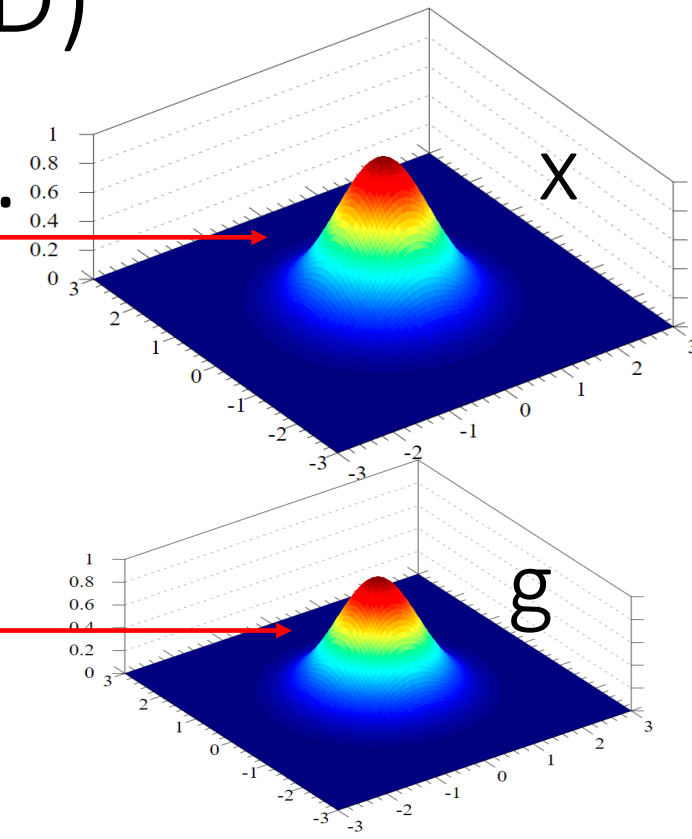
1.



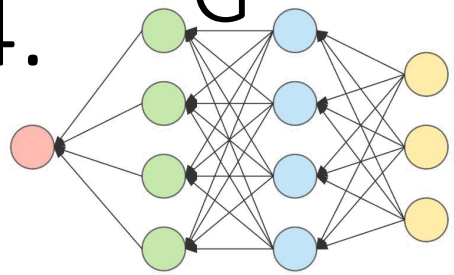
2.



3.



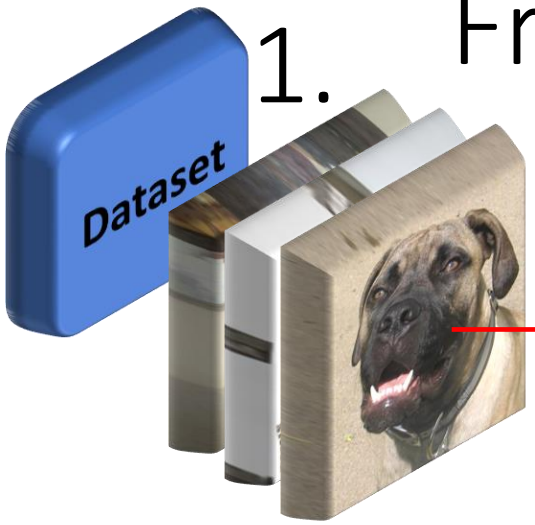
4. G



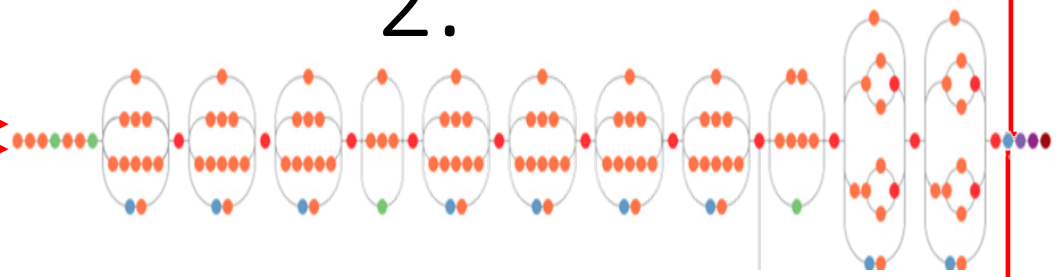
5. $\texttt{FID}(x, g) = ||\mu_x - \mu_g||_2^2 + \text{Tr}(\Sigma_x + \Sigma_g - 2(\Sigma_x \Sigma_g)^{\frac{1}{2}})$

Fréchet Inception Distance (FID)

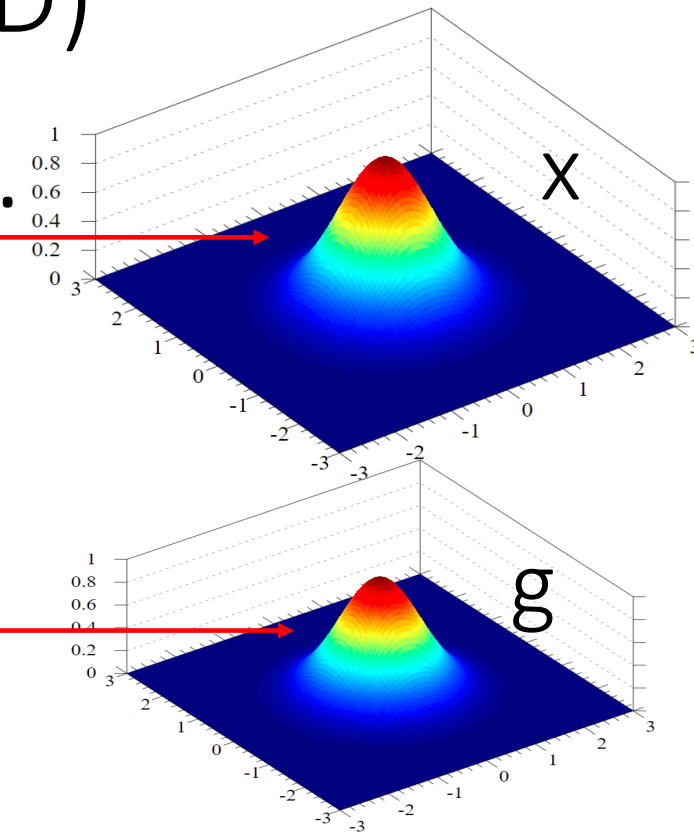
1.



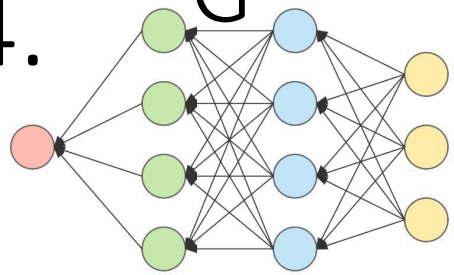
2.



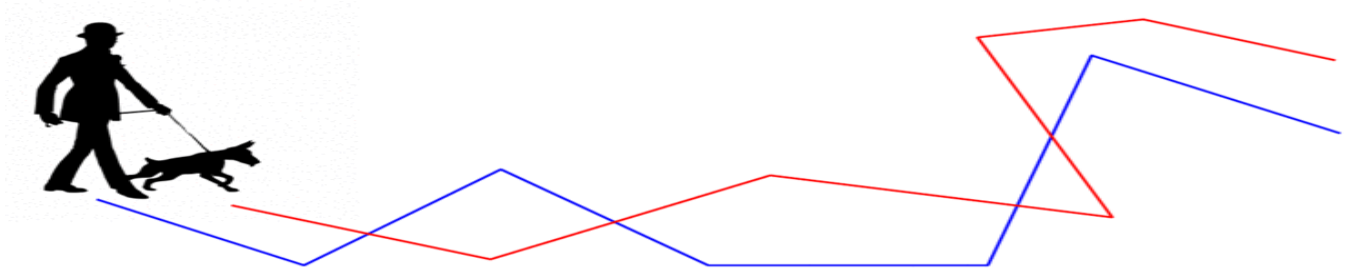
3.



4. G



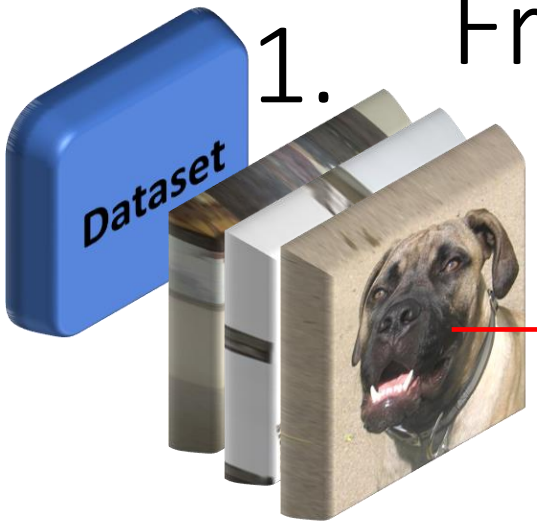
5. $\texttt{FID}(x, g) = ||\mu_x - \mu_g||_2^2 + \mathrm{Tr}(\Sigma_x + \Sigma_g - 2(\Sigma_x \Sigma_g)^{\frac{1}{2}})$



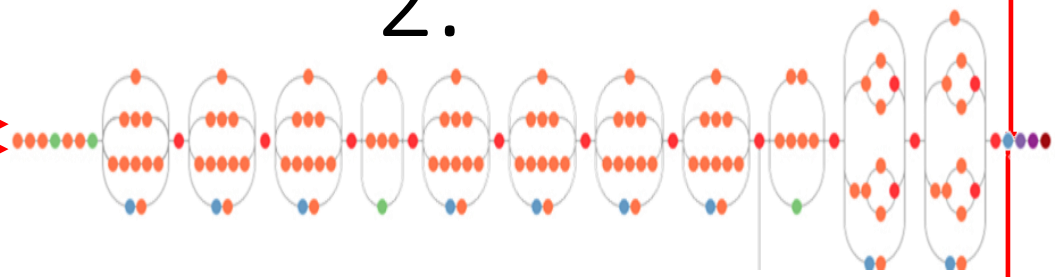
Fréchet Inception Distance (FID)

Depends on the number of samples!

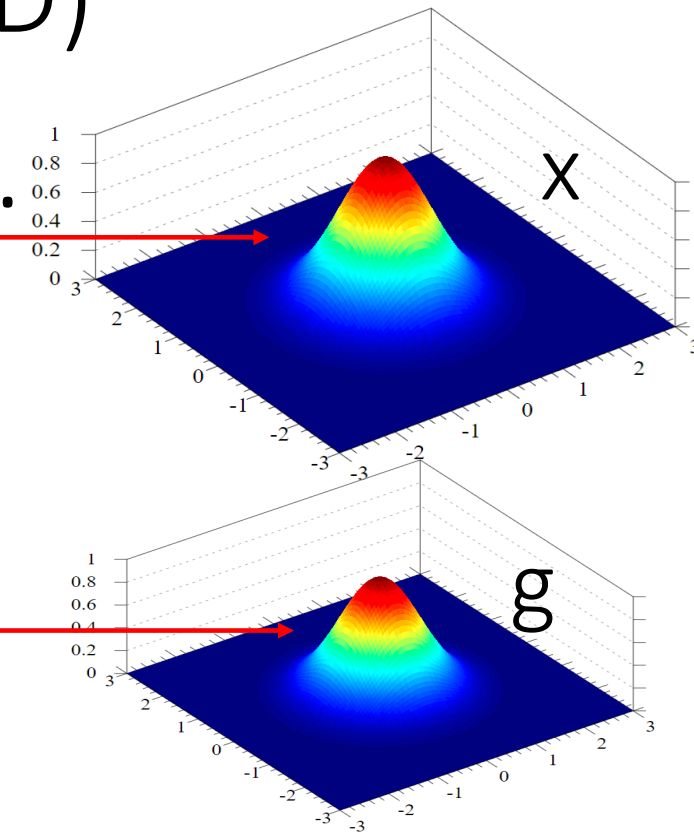
1.



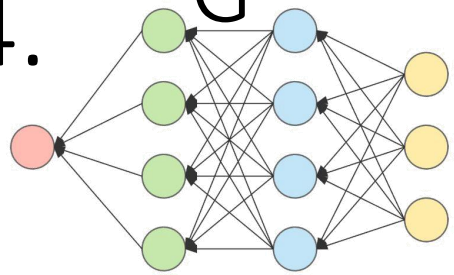
2.



3.



4. G



5. $\texttt{FID}(x, g) = ||\mu_x - \mu_g||_2^2 + \mathrm{Tr}(\Sigma_x + \Sigma_g - 2(\Sigma_x \Sigma_g)^{\frac{1}{2}})$

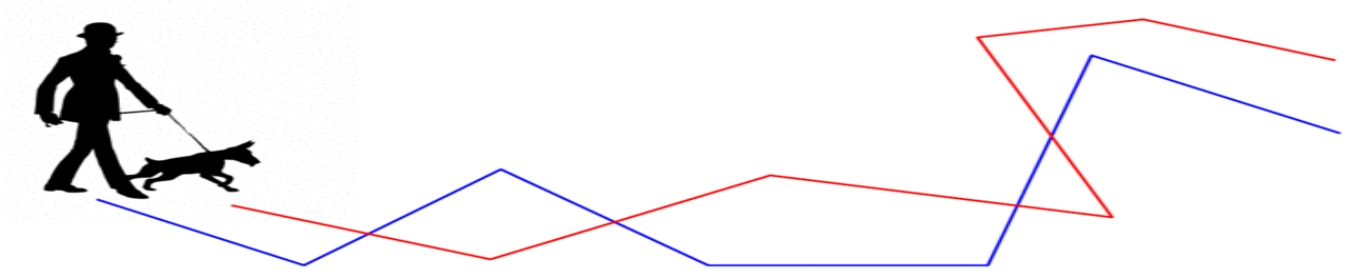
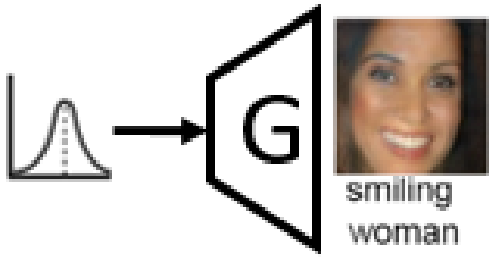
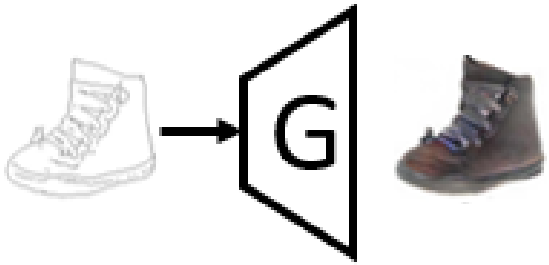


Image to Image translation

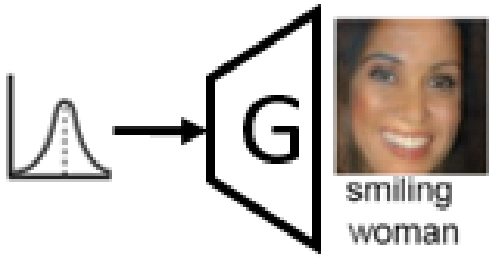


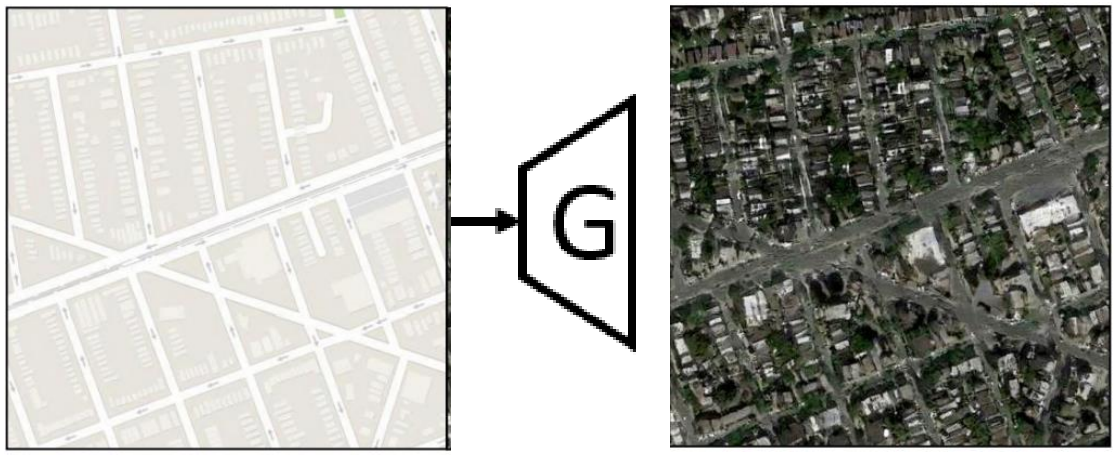


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Image to Image translation

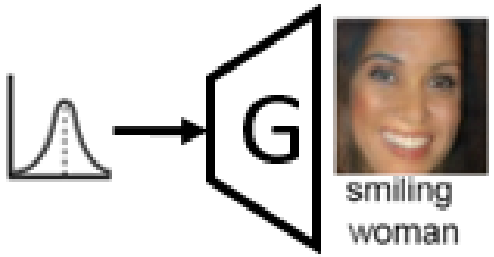


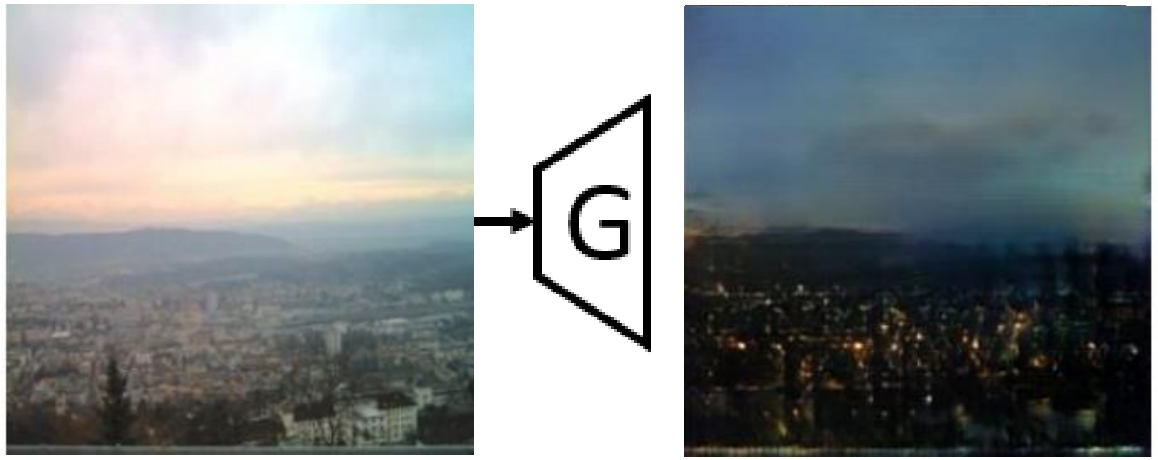


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Image to Image translation

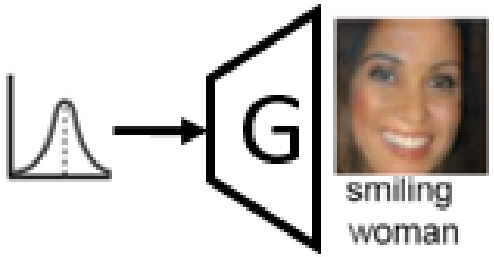


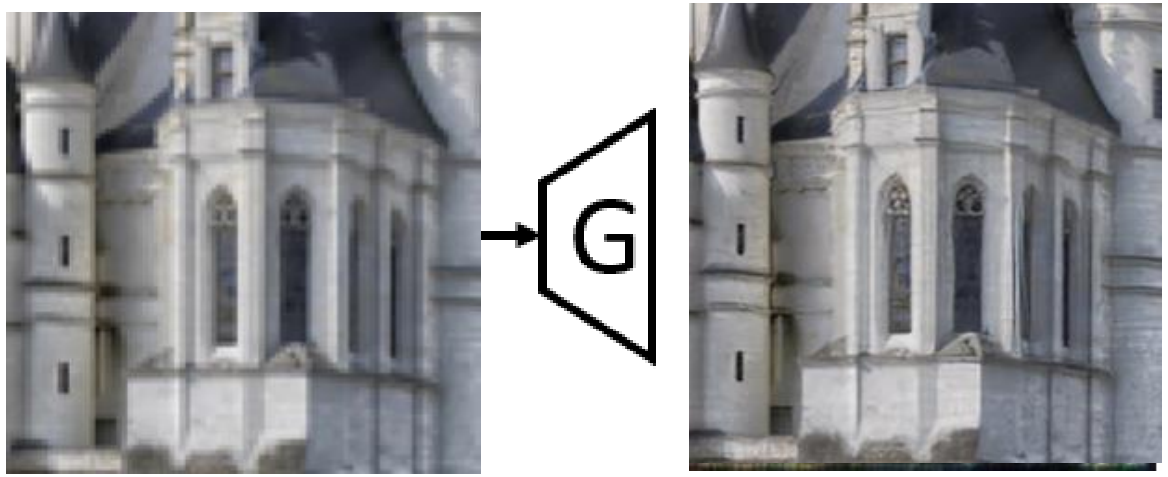


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Image to Image translation

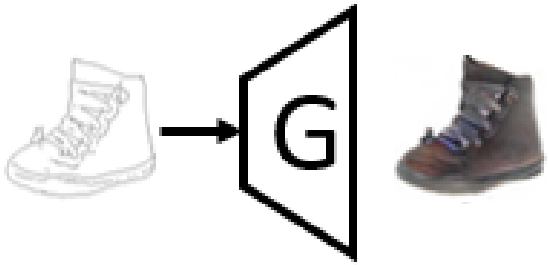




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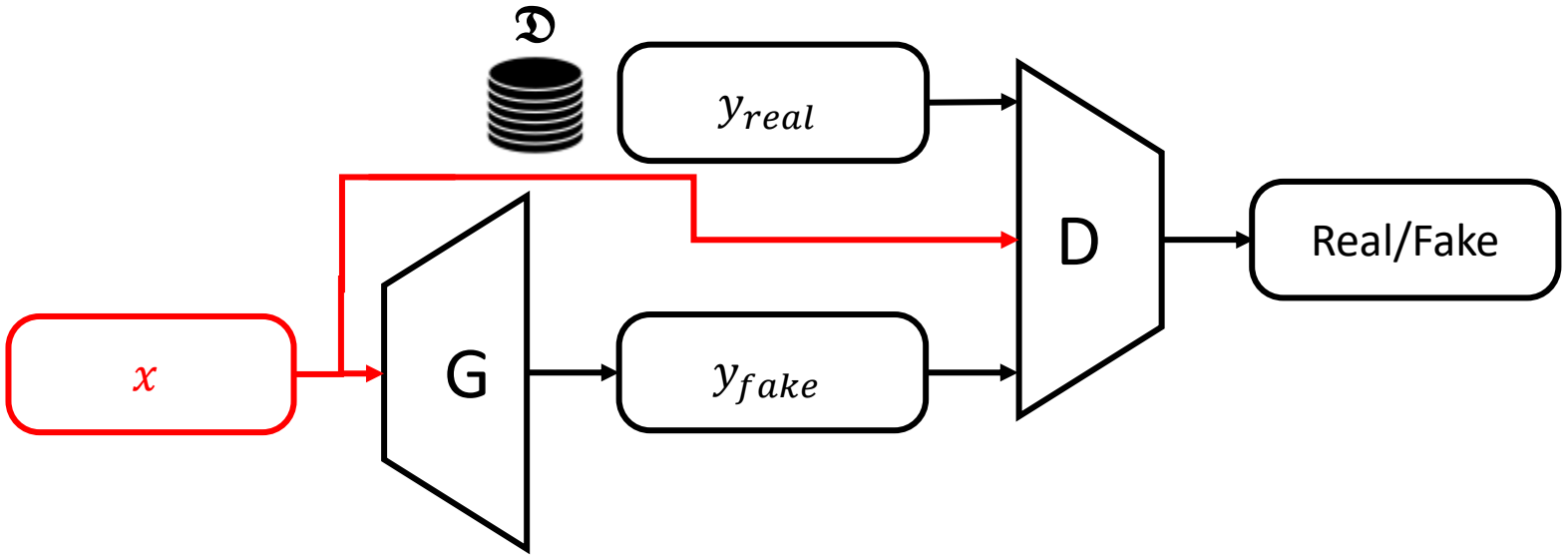
Image to Image translation



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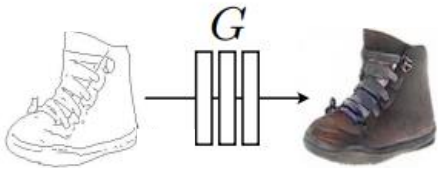
Conditional GAN



$$\mathcal{L}_{C-GAN} = \min_{G} \max_{D} \mathbb{E}[\log D(y, x)] + \mathbb{E}[\log(1 - D(G(x), x))]$$

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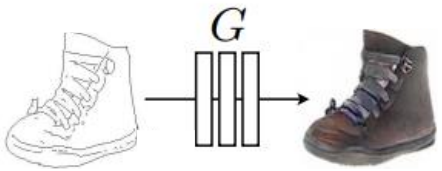
Pix2Pix

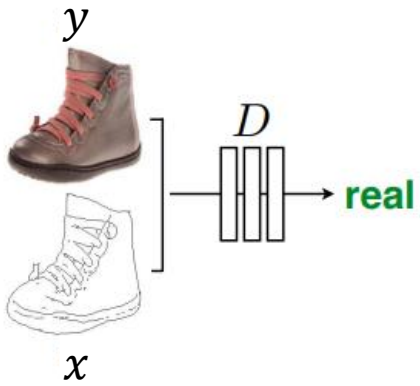


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Pix2Pix

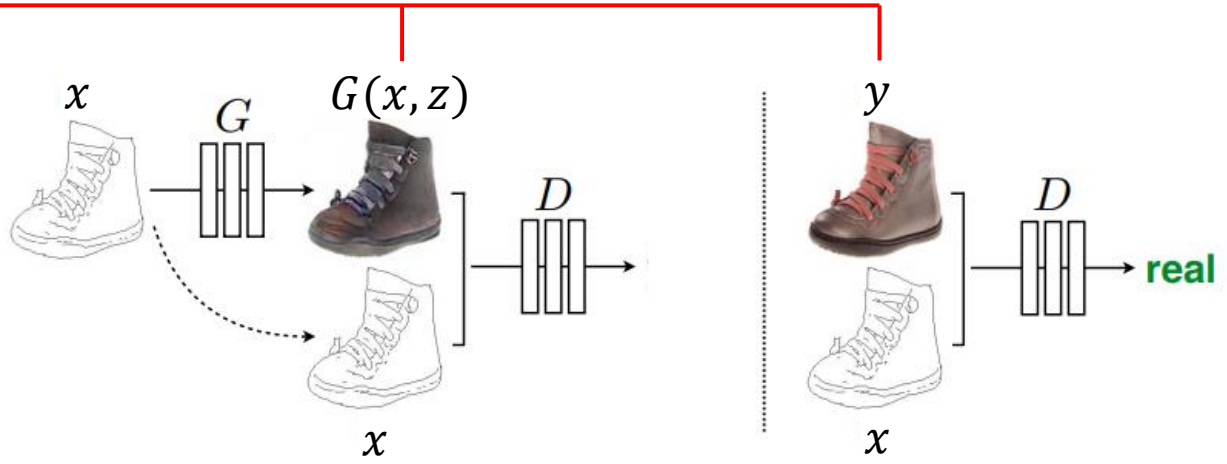




Slide by Sefi Bell-Kligler & Akhiad Bercovich

Isola et al. Nov2016

Pix2Pix



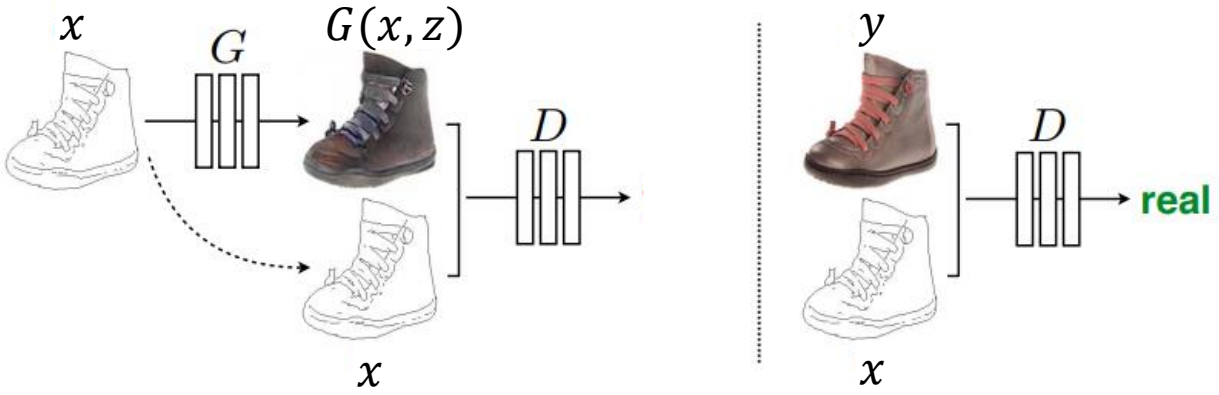
$$\mathcal{L}_{C-GAN} = \min_G \max_D \mathbb{E}[\log D(y, x)] + \mathbb{E}[\log(1 - D(G(x), x))]$$

$$\mathcal{L}_{L1} = \|y - G(x, z)\|_1$$

Isola et al. Nov2016

Slide by Sefi Bell-Kligler & Akhiad Bercovich

Pix2Pix



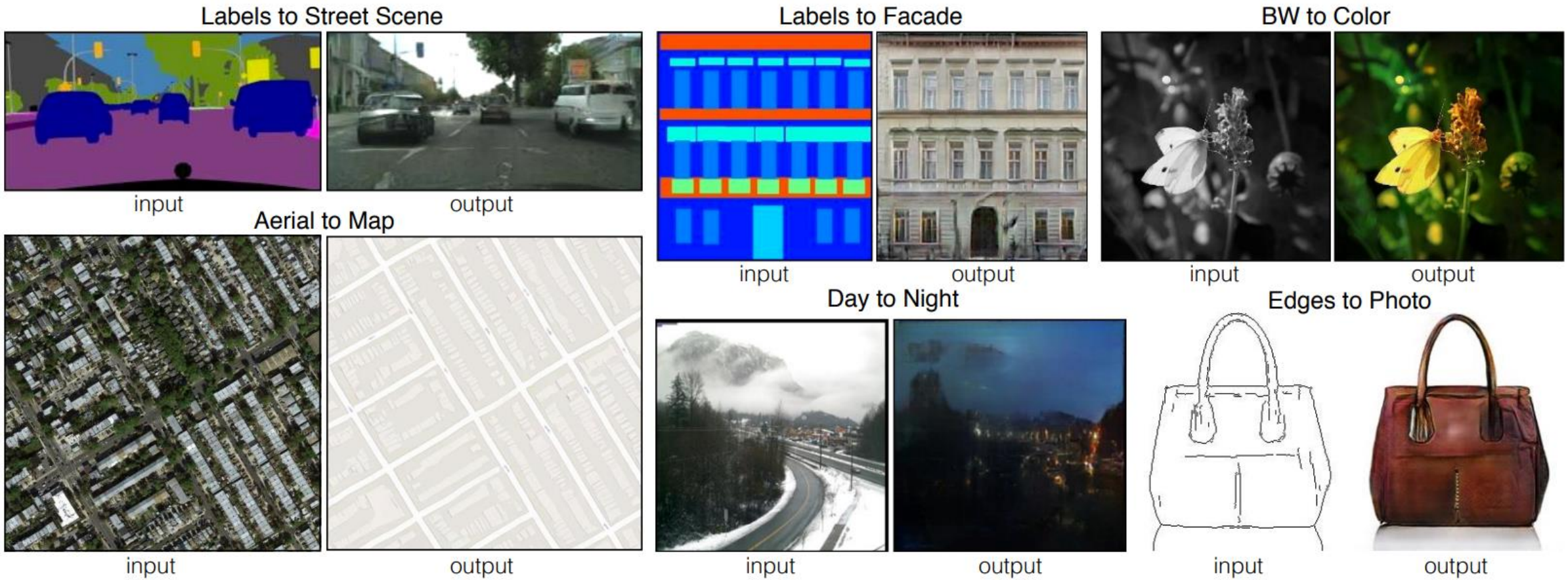
$$\mathcal{L}_{C-GAN} = \min_G \max_D \mathbb{E}[\log D(y, x)] + \mathbb{E}[\log(1 - D(G(x), x))]$$

$$\mathcal{L}_{L1} = \|y - G(x, z)\|_1$$

$$Objective = \mathcal{L}_{C-GAN} + \lambda \cdot \mathcal{L}_{L1}$$

Isola et al. Nov2016

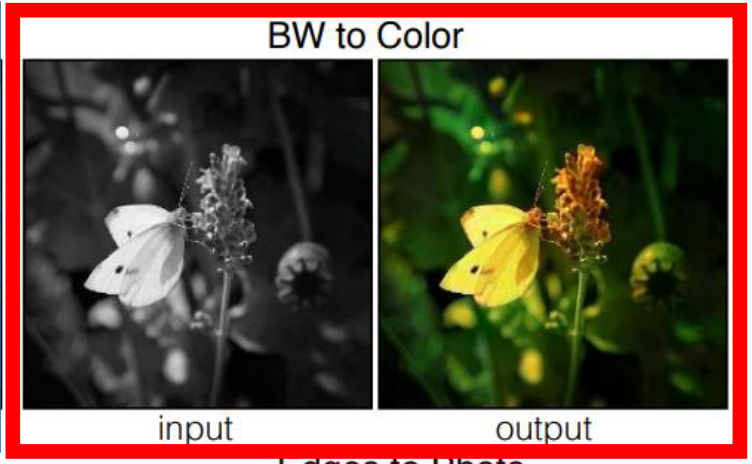
Slide by Sefi Bell-Kligler & Akhiad Bercovich



https://affinelayer.com/pixsrv/

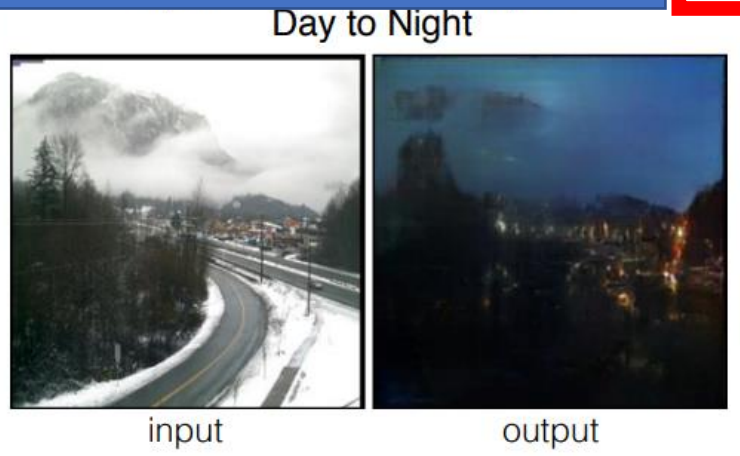
Generative VS Discriminative

What would happen if we train regular supervised mapping?




input


output

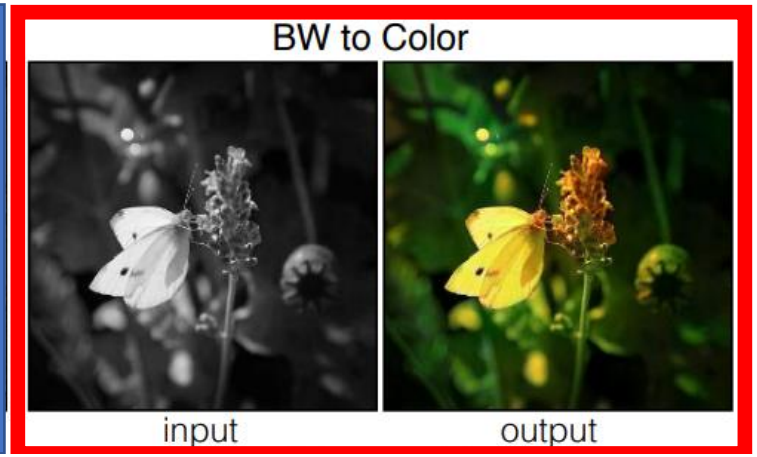


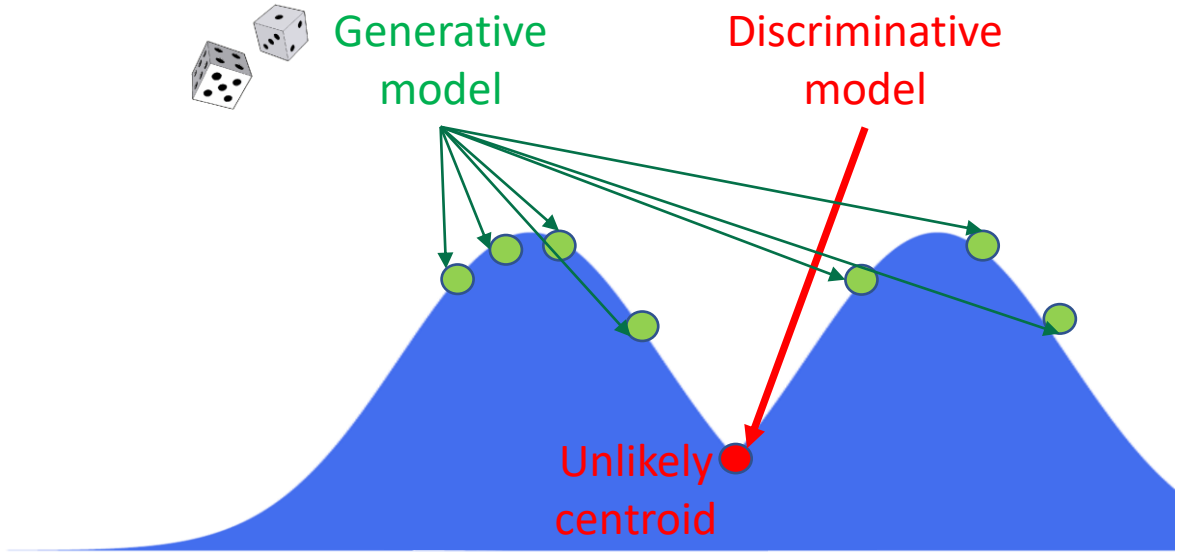


https://affinelayer.com/pixsrv/

Generative VS Discriminative

What would happen if we train regular supervised mapping?





Training GANs is hard

- Stability

Training GANs Be like:



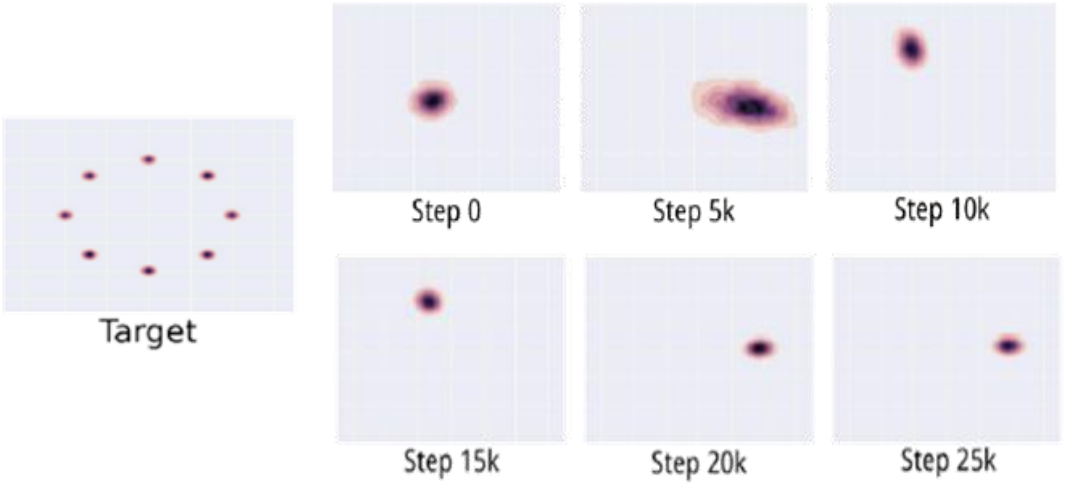


Training GANs is hard

- Stability



- Mode collapse



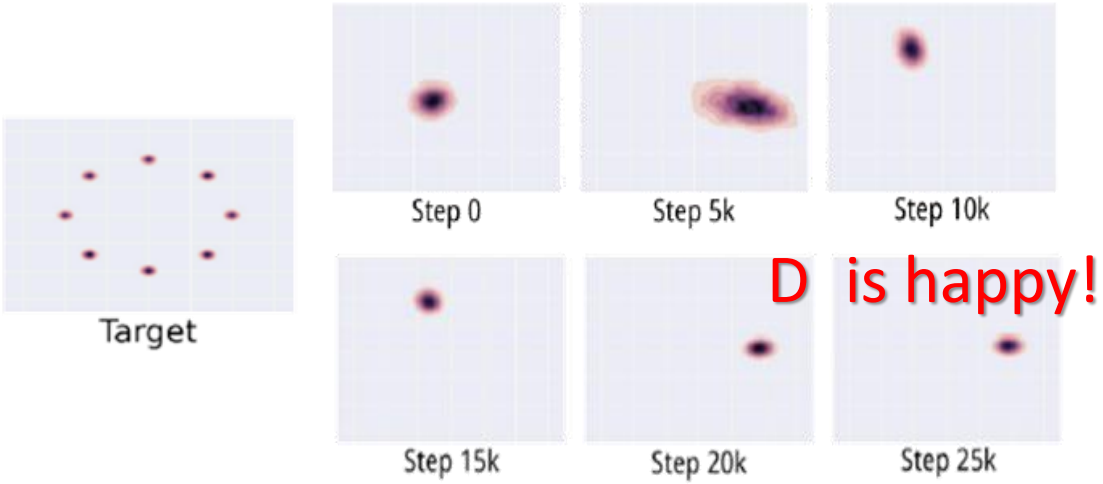


Training GANs is hard

- Stability



- Mode collapse



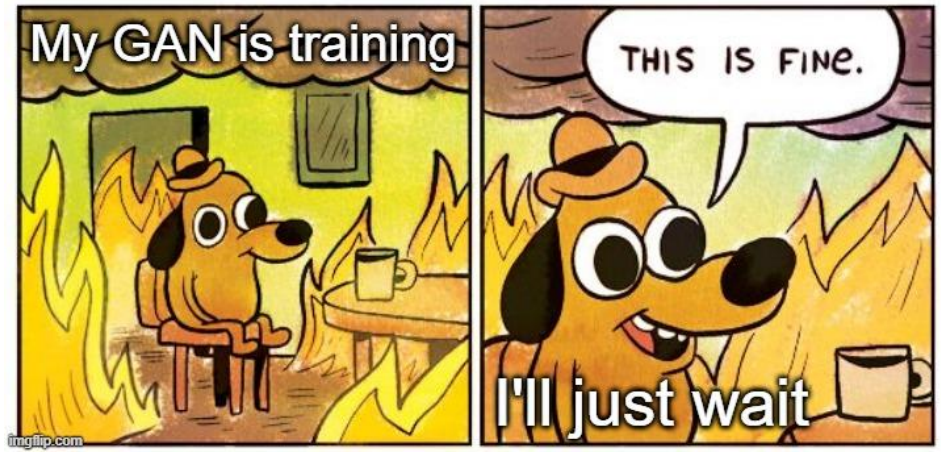


Training GANs is hard

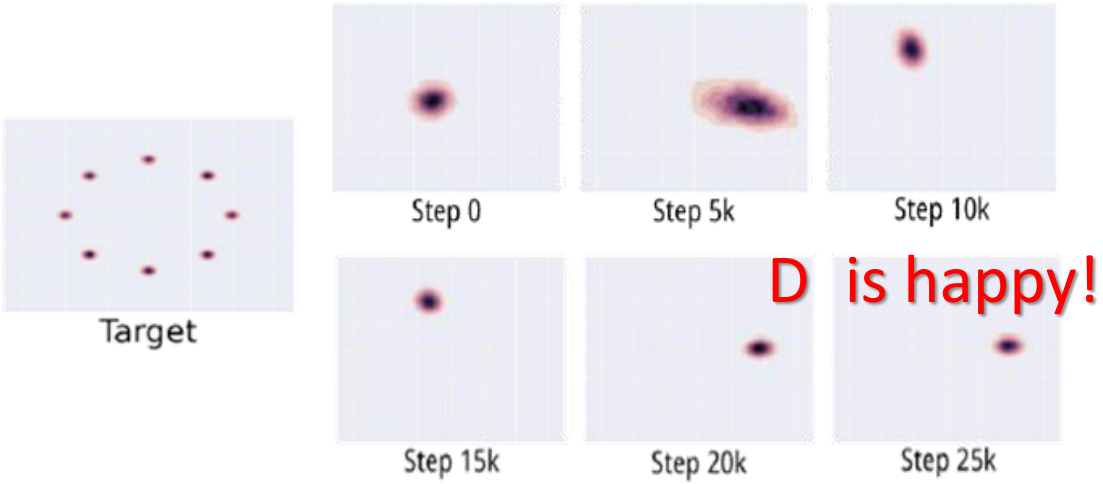
- Stability



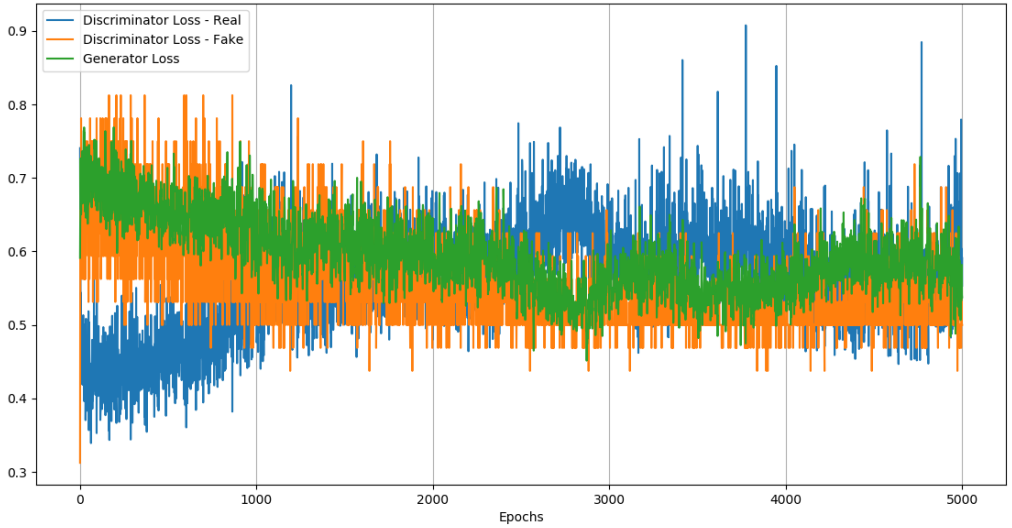
- GANs can over-train



- Mode collapse



- Losses don't reflect quality



Types of GAN losses - Wasserstein GAN

Discriminator

GAN $$\max_D E_{x\sim p_X}[\log D(x)] + E_{z\sim p_Z}[\log(1 - D(G(z))]$$

WGAN $$\max_D E_{x\sim p_X}[D(x)] - E_{z\sim p_Z}[D(G(z))]$$

WGAN: minimize earth mover distance between p_X and $p_{G(Z)}$

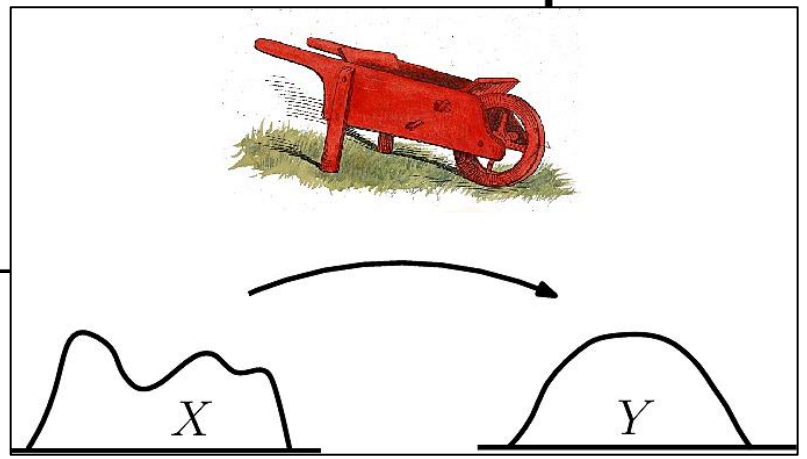
$$EM(p_X, p_{G(Z)}) = \inf_{\gamma\in\prod(p_X, p_{G(Z)})} E_{(x,y)\sim\gamma}[||x - y||]$$

Types of GAN losses - Wasserstein GAN

Discriminator

GAN $\max_D E_{x\sim p_X}[\log D(x)] + E_{z\sim p_Z}[\log(1 - D(G(z))]$

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WGAN: minimize earth mover distance between p_X and $p_{G(Z)}$

$$EM(p_X, p_{G(Z)}) = \inf_{\gamma \in \prod(p_X, p_{G(Z)})} E_{(x,y)\sim\gamma}[||x - y||]$$

Types of GAN losses

GAN	Discriminator Loss	Generator Loss
MM GAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{GAN}} = -\mathbb{E}_{x\sim p_d}[\log(D(x))] - \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{GAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$
NS GAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{NSGAN}} = -\mathbb{E}_{x\sim p_d}[\log(D(x))] - \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{NSGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[\log(D(\hat{x}))]$
WGAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{WGAN}} = -\mathbb{E}_{x\sim p_d}[D(x)] + \mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{WGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})]$
WGAN GP	$\mathcal{L}_{\mathrm{D}}^{\mathrm{WGANGP}} = \mathcal{L}_{\mathrm{D}}^{\mathrm{WGAN}} + \lambda\mathbb{E}_{\hat{x}\sim p_g}[(\|\nabla D(\alpha x + (1-\alpha\hat{x})\|_2 - 1)^2]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{WGANGP}} = -\mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})]$
LS GAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{LSGAN}} = -\mathbb{E}_{x\sim p_d}[(D(x)-1)^2] + \mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})^2]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{LSGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[(D(\hat{x}-1))^2]$
DRAGAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{DRAGAN}} = \mathcal{L}_{\mathrm{D}}^{\mathrm{GAN}} + \lambda\mathbb{E}_{\hat{x}\sim p_d+\mathcal{N}(0,c)}[(\|\nabla D(\hat{x})\|_2 - 1)^2]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{DRAGAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$
BEGAN	$\mathcal{L}_{\mathrm{D}}^{\mathrm{BEGAN}} = \mathbb{E}_{x\sim p_d}[\|x - \mathrm{AE}(x)\|_1] - k_t\mathbb{E}_{\hat{x}\sim p_g}[\|\hat{x} - \mathrm{AE}(\hat{x})\|_1]$	$\mathcal{L}_{\mathrm{G}}^{\mathrm{BEGAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\|\hat{x} - \mathrm{AE}(\hat{x})\|_1]$

Are GANs Created Equal? Lucic&Kurach et al (Google Brain) 2018

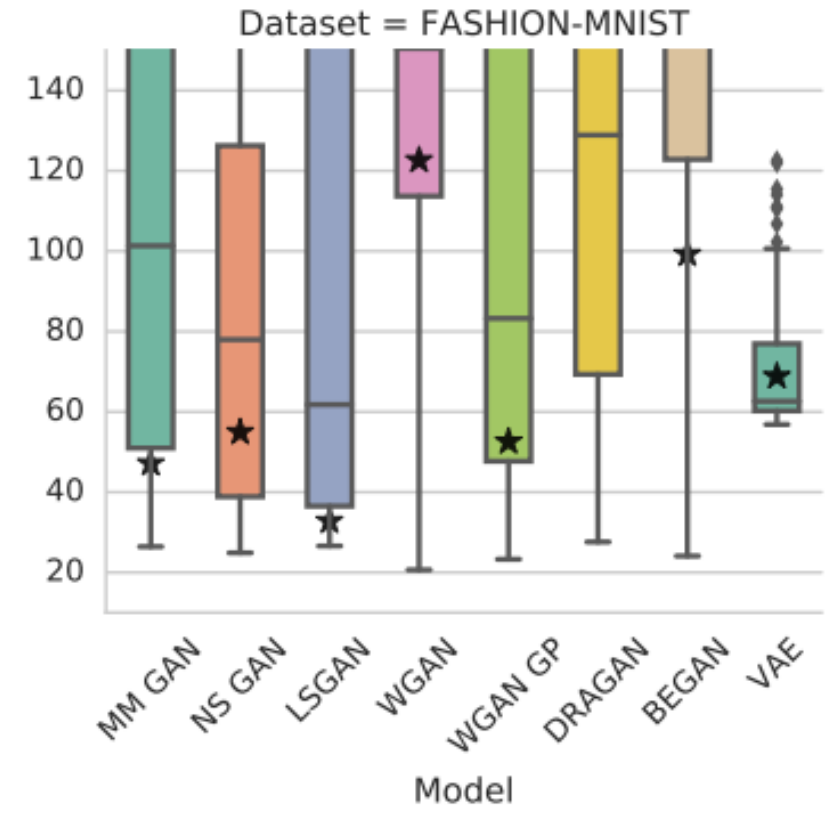
Types of GAN losses

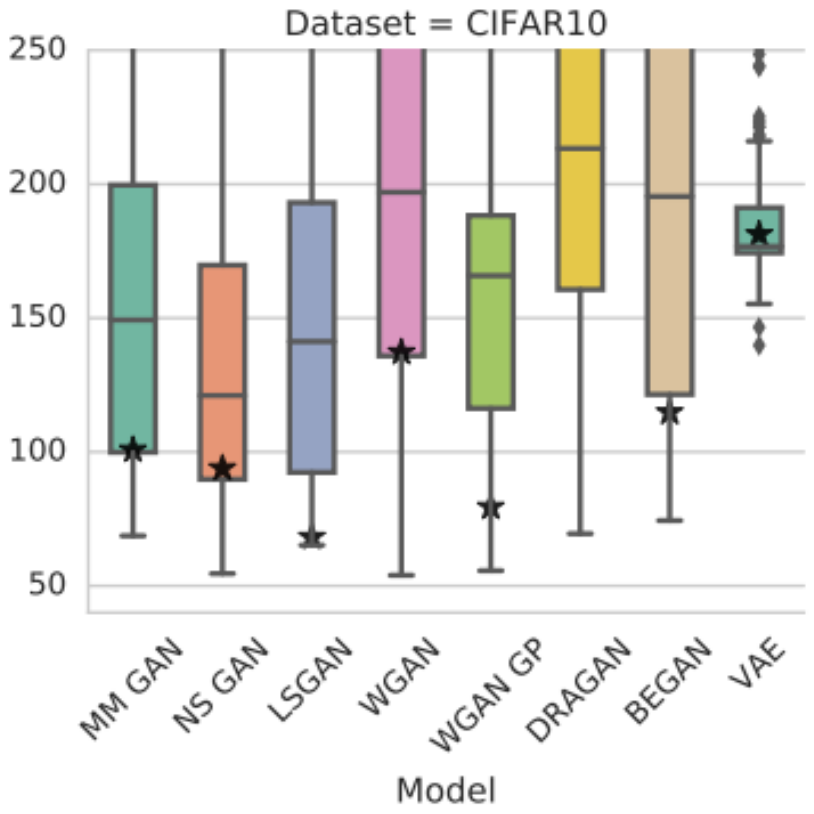


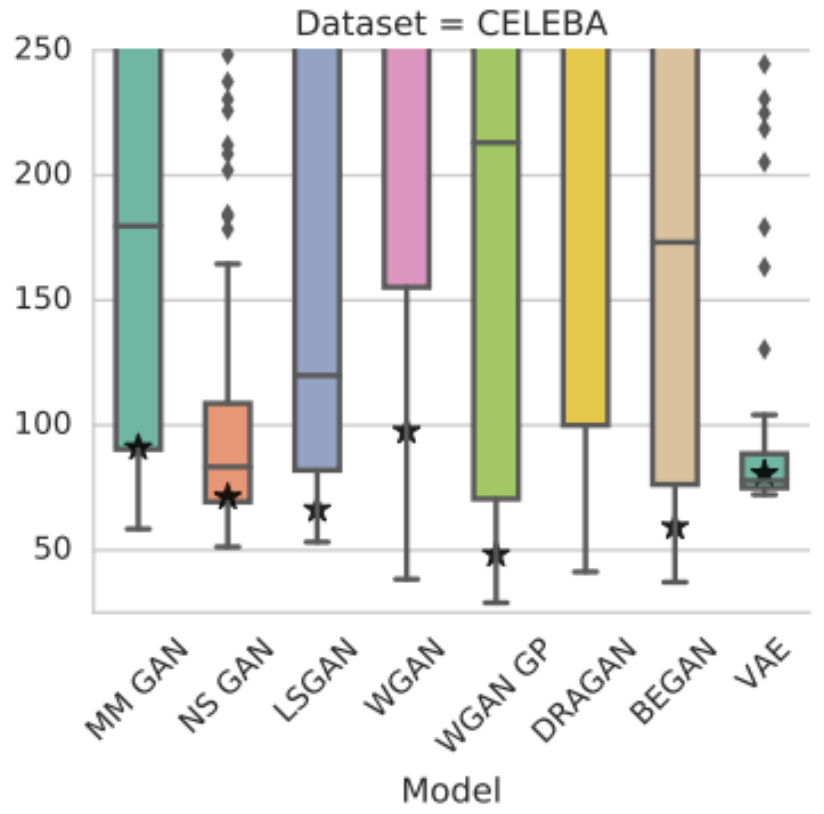
	Generator Loss
$\hat{x}\sim p_g[\log(1-D(\hat{x}))]$	$\mathcal{L}_G^{\text{GAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$
$\mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$	$\mathcal{L}_G^{\text{NSGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[\log(D(\hat{x}))]$
$_g[D(\hat{x})]$	$\mathcal{L}_G^{\text{WGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})]$
$D(\alpha x + (1-\alpha\hat{x})\|\|_2 - 1)^2]$	$\mathcal{L}_G^{\text{WGANGP}} = -\mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})]$
$+ \mathbb{E}_{\hat{x}\sim p_g}[D(\hat{x})^2]$	$\mathcal{L}_G^{\text{LSGAN}} = -\mathbb{E}_{\hat{x}\sim p_g}[(D(\hat{x}-1))^2]$
$)[(\|\|\nabla D(\hat{x})\|\|_2 - 1)^2]$	$\mathcal{L}_G^{\text{DRAGAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\log(1-D(\hat{x}))]$
$- k_t\mathbb{E}_{\hat{x}\sim p_g}[\|\|\hat{x} - \text{AE}(\hat{x})\|\|_1]$	$\mathcal{L}_G^{\text{BEGAN}} = \mathbb{E}_{\hat{x}\sim p_g}[\|\|\hat{x} - \text{AE}(\hat{x})\|\|_1]$

Are GANs Created Equal? Lucic&Kurach et al (Google Brain) 2018

Types of GAN losses

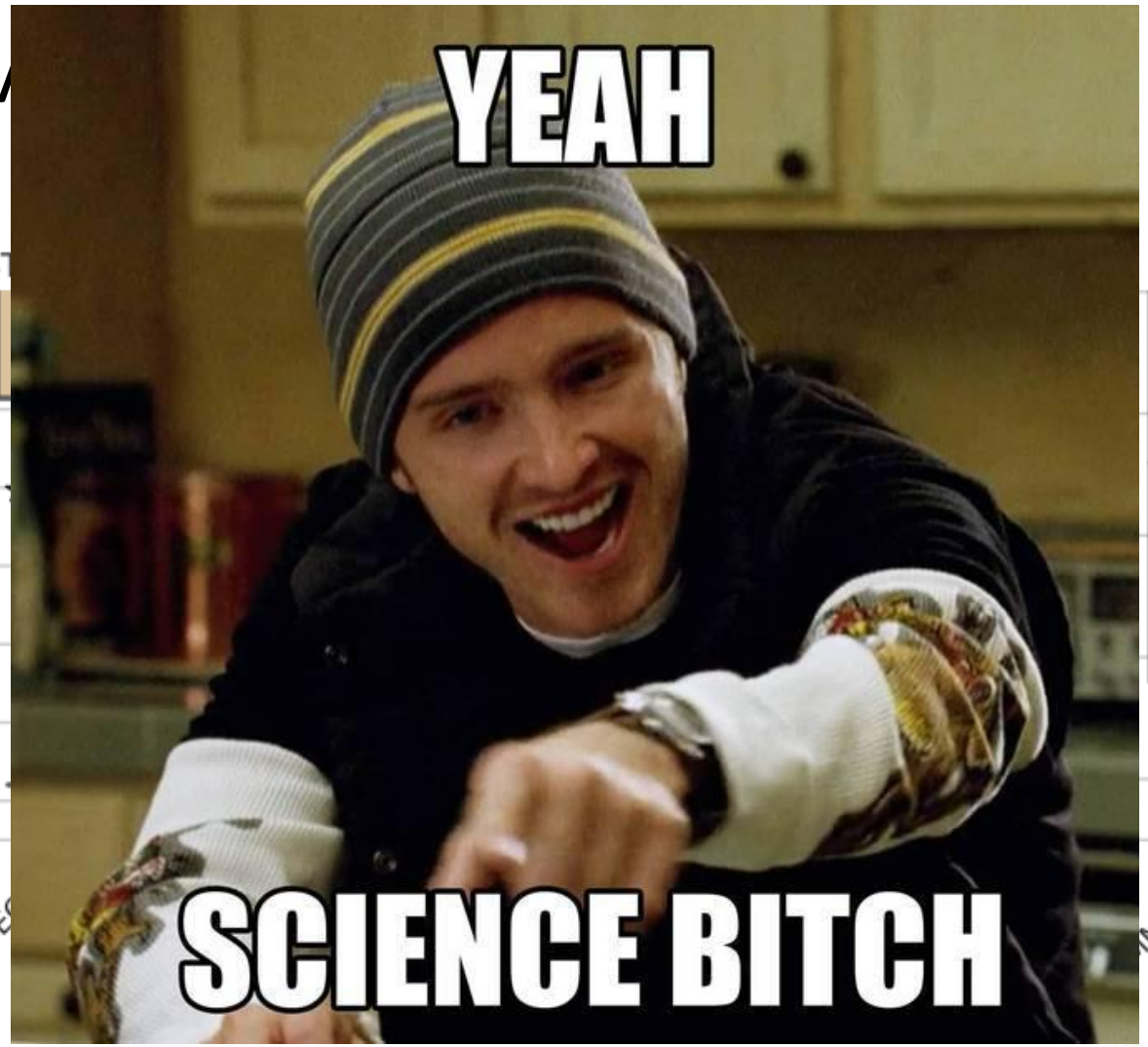


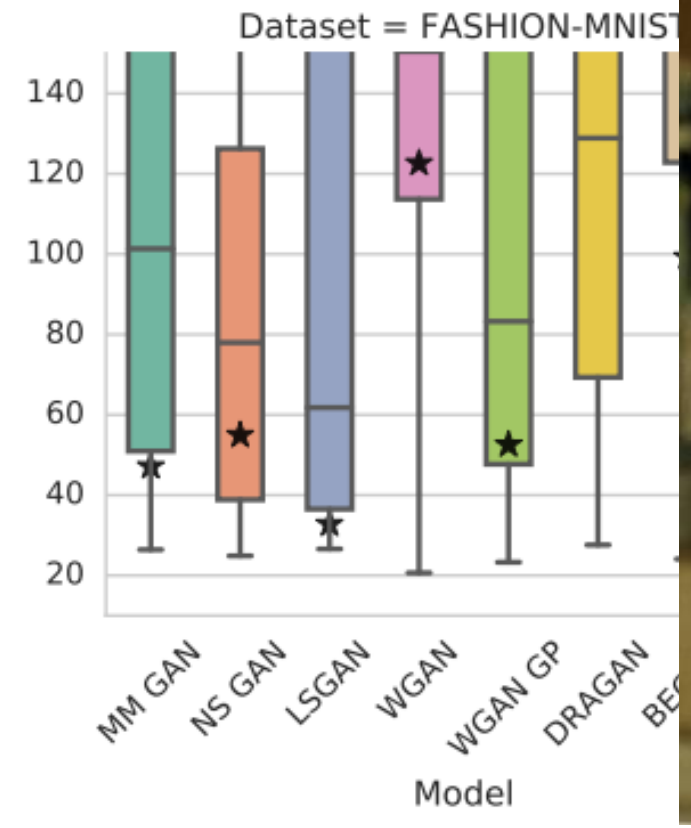


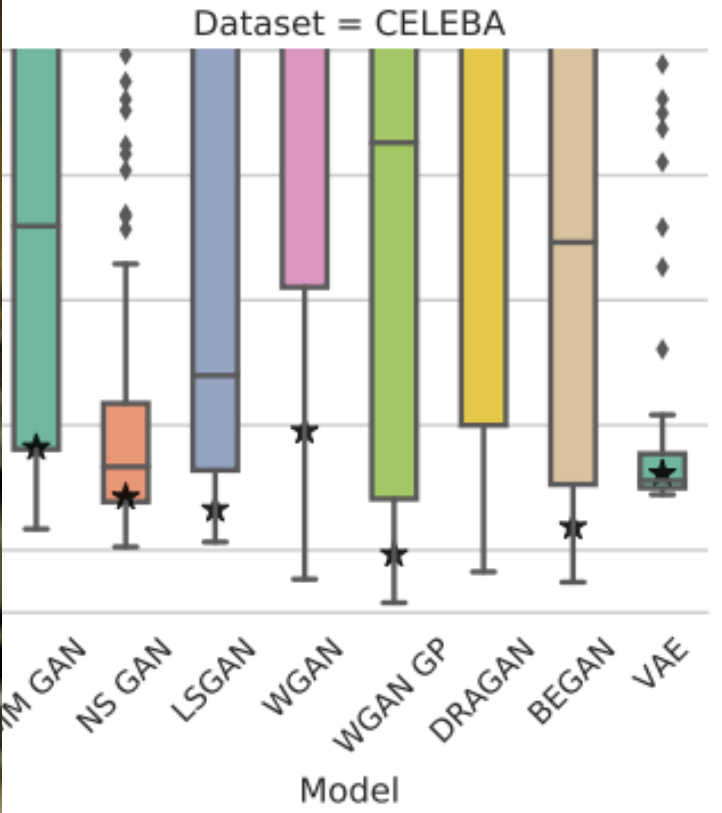


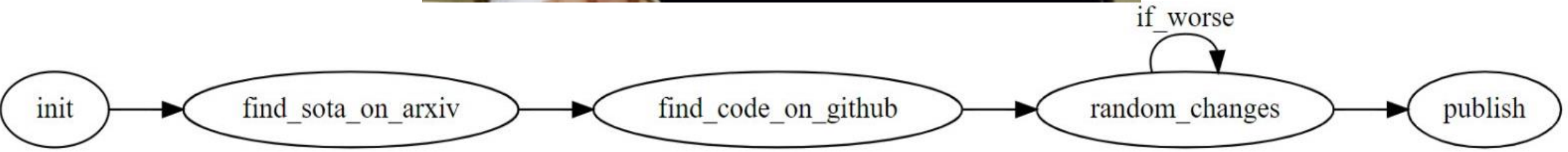
Are GANs Created Equal? Lucic&Kurach et al (Google Brain) 2018

Types of GA



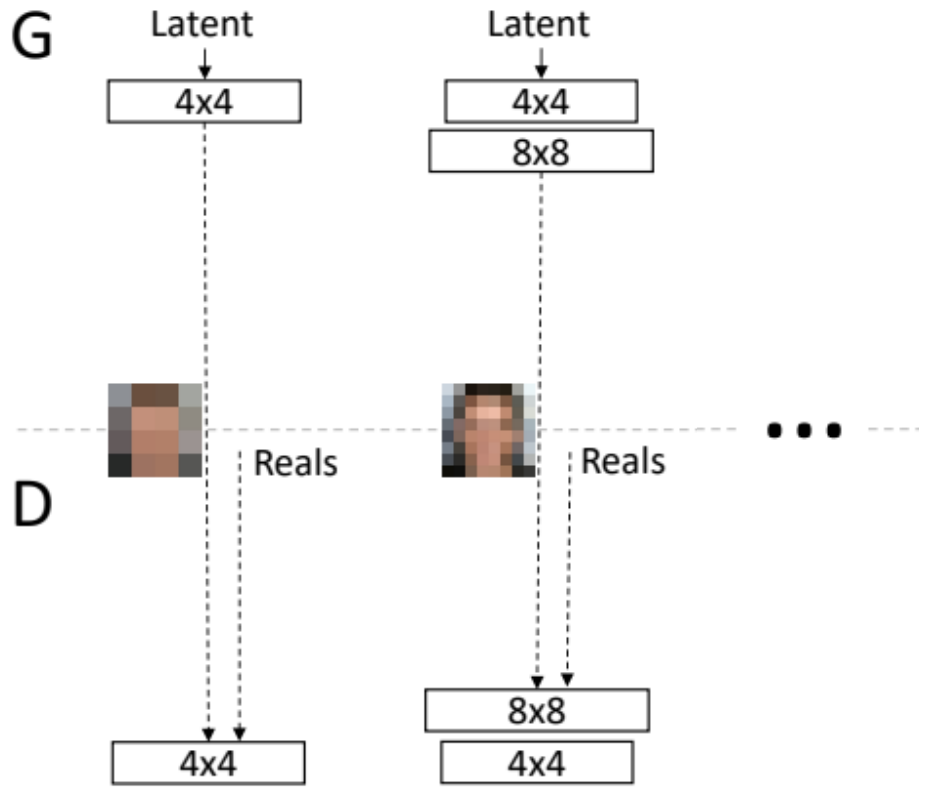






Mark Saroufim

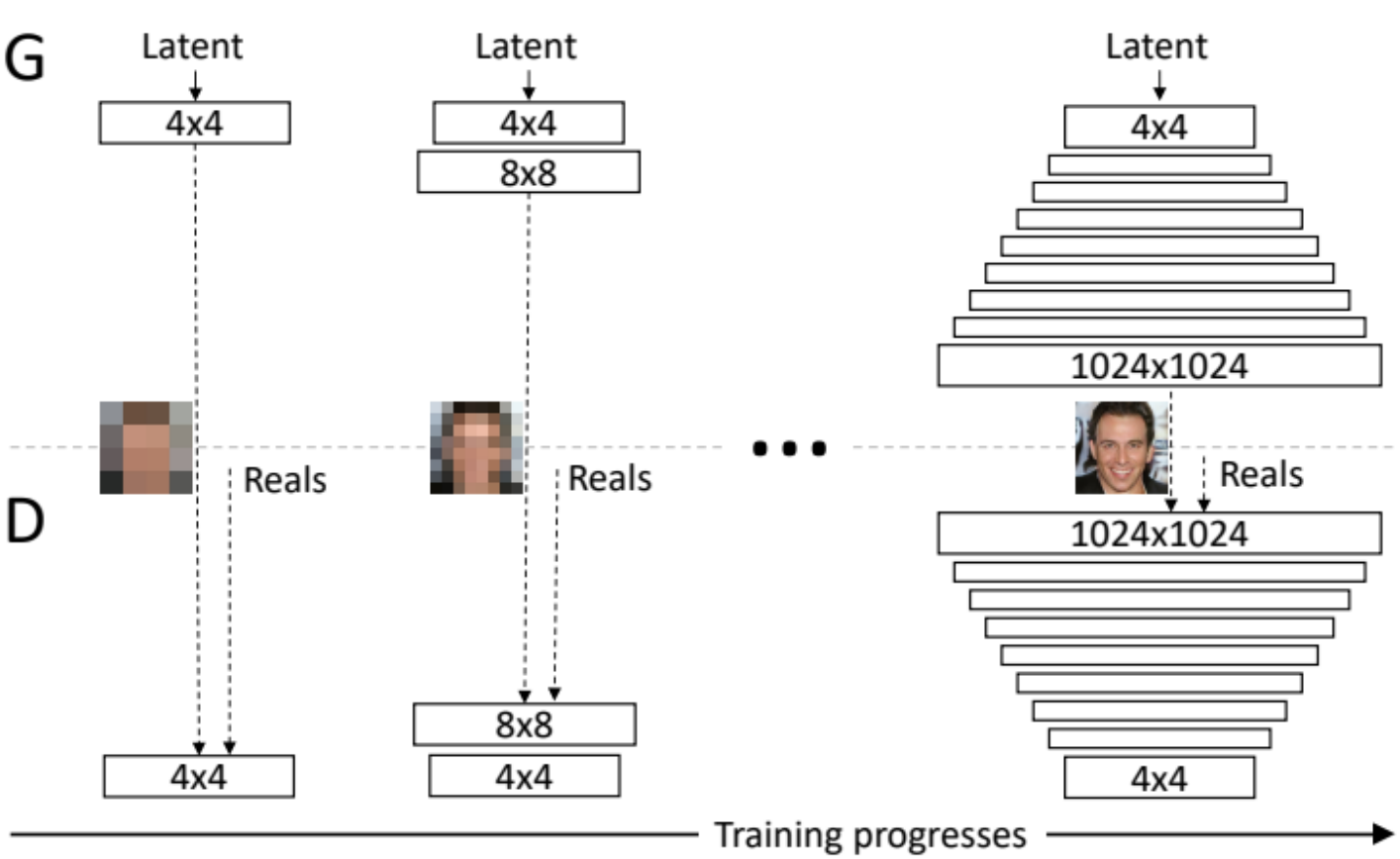
Progressive Grow



Slide by Sefi Bell-Kligler & Akhiad Bercovich

Progressive Growing of GAN, Karras et al., Feb2018

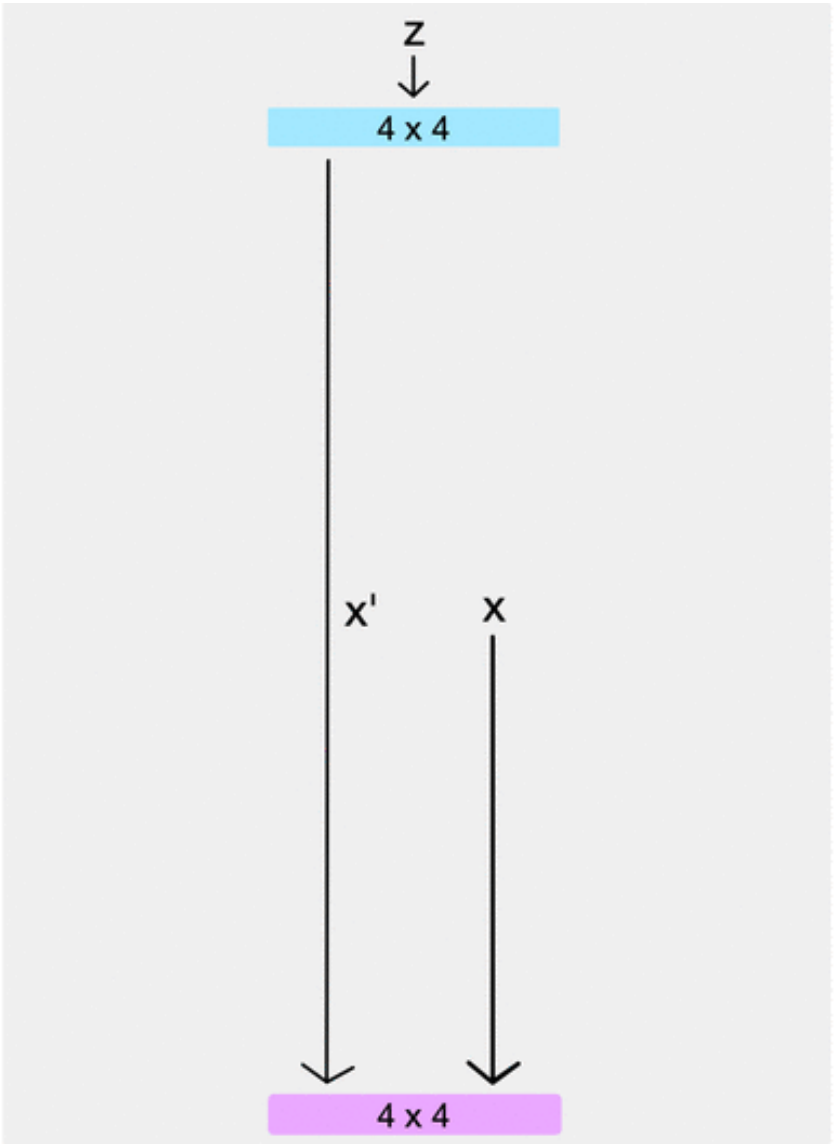
Progressive Grow

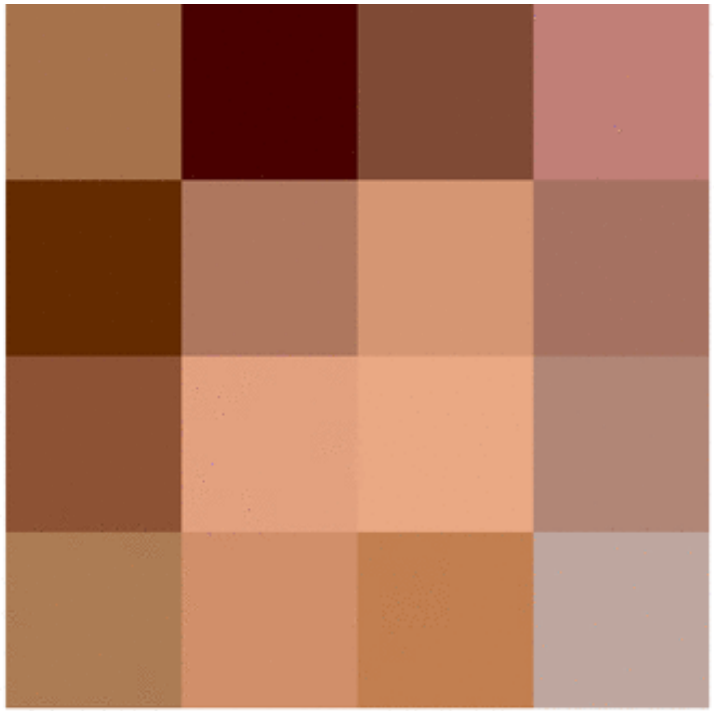




Slide by Sefi Bell-Kligler & Akhiad Bercovich

Progressive Growing of GAN, Karras et al., Feb2018





Training time: 0 days

4x4 resolution

Generator

Discriminator

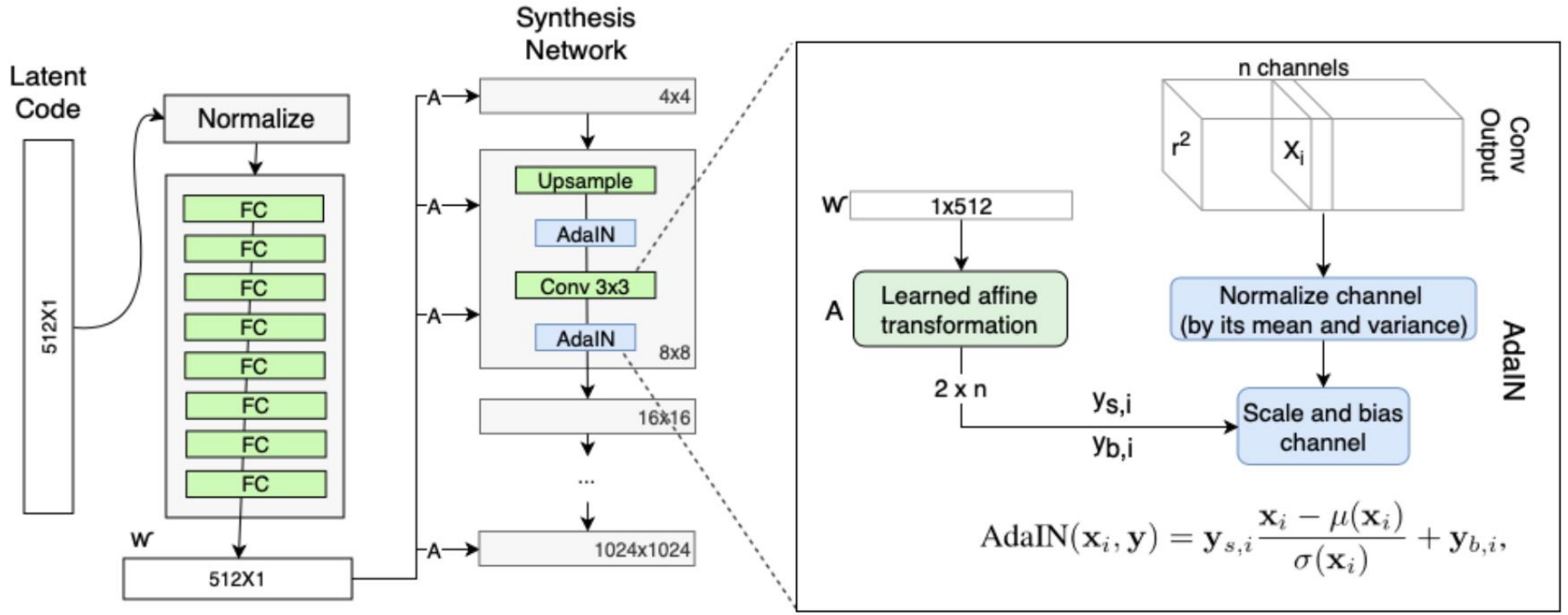
z = random code

x = real image

x' = generated image

Progressive Growing of GAN, Karras et al., Feb2018

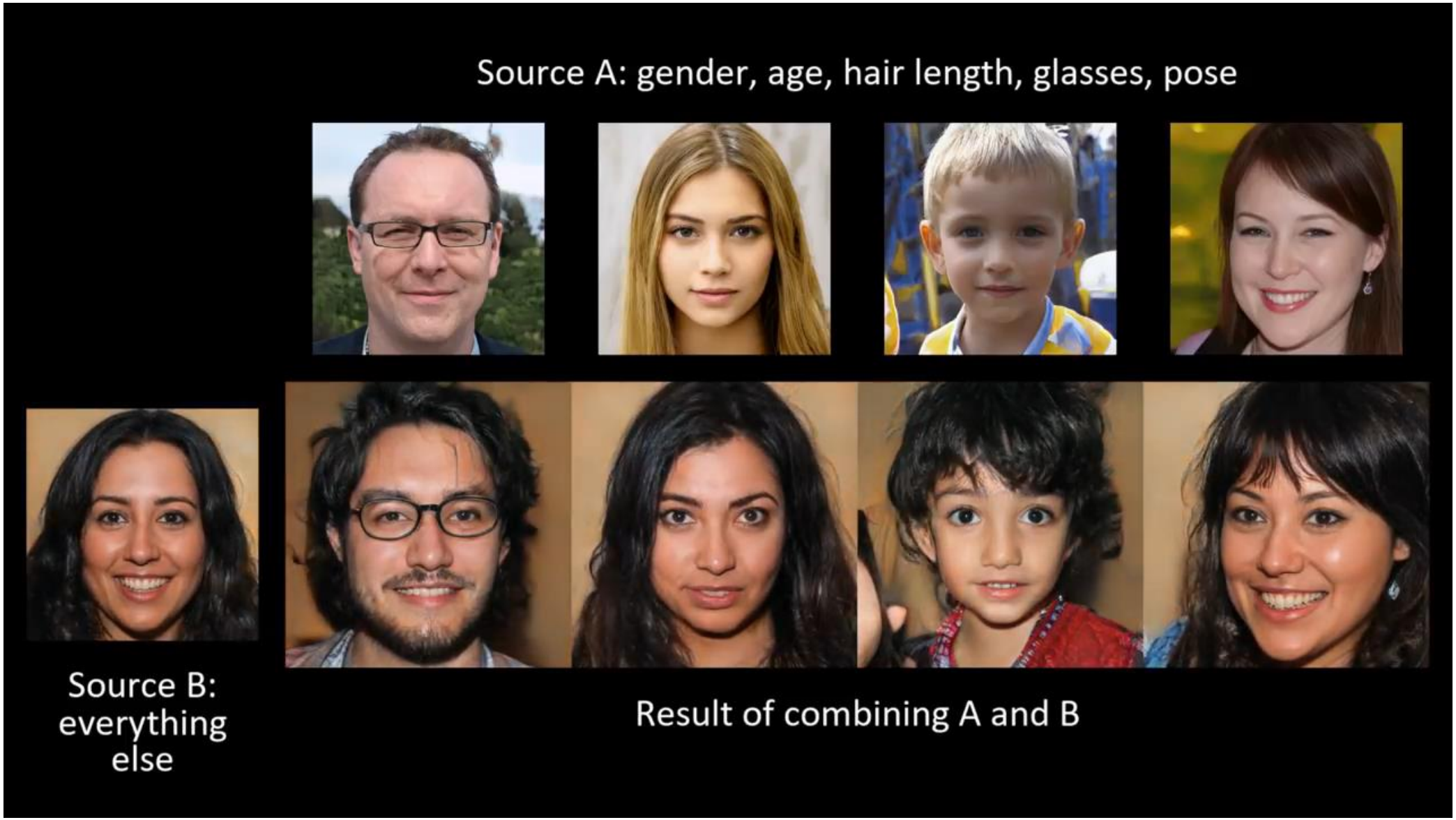
Style Modules (AdaIN)



The generator's Adaptive Instance Normalization (AdaIN)

StyleGAN, Karras et al. NVIDIA 2019

Results


Source A: gender, age, hair length, glasses, pose
Source B: everything else
Result of combining A and B



Animation by Sefi Bell-Kligler & Akhiad Bercovich

https://thispersondoesnotexist.com/

https://whichfaceisreal.com/



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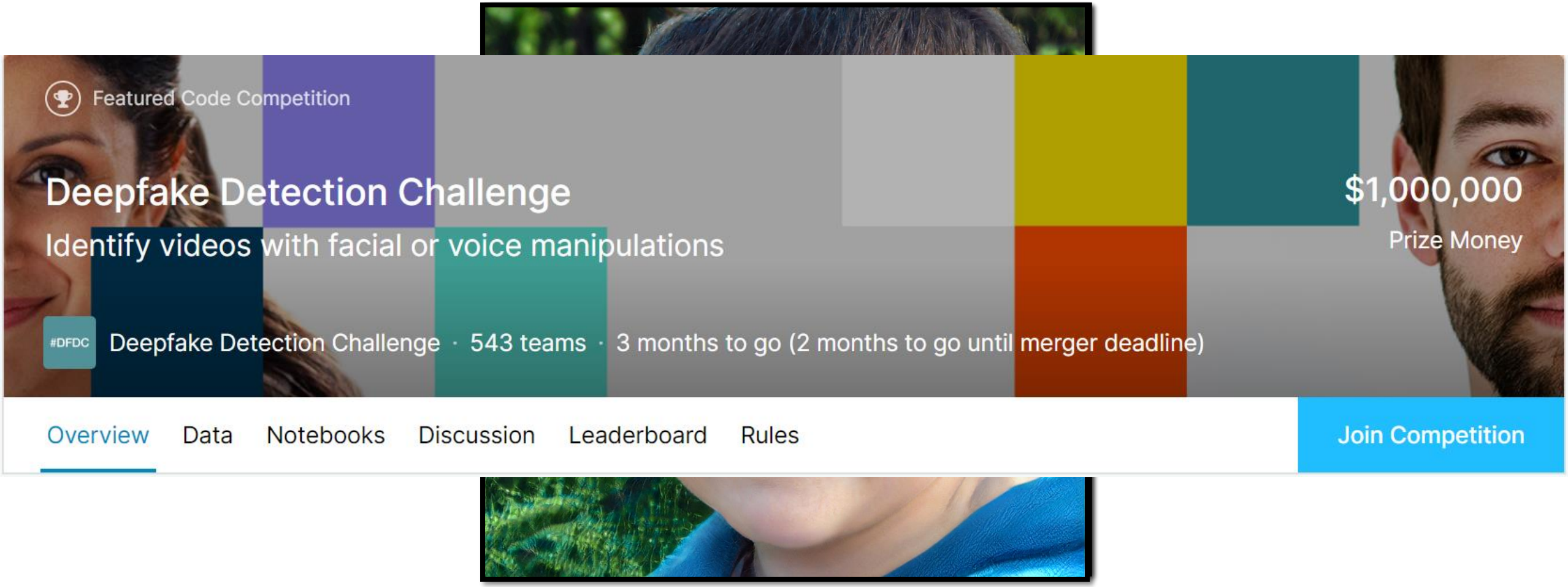
https://whichfaceisreal.com/



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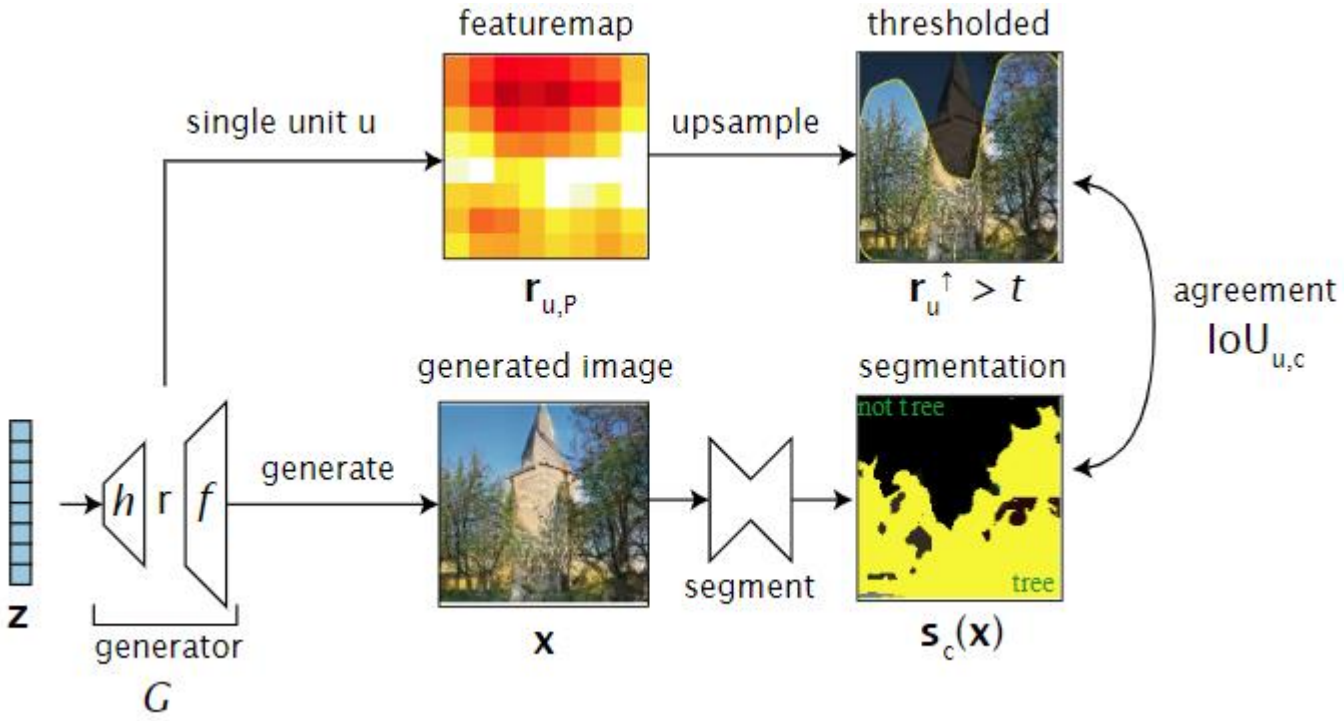


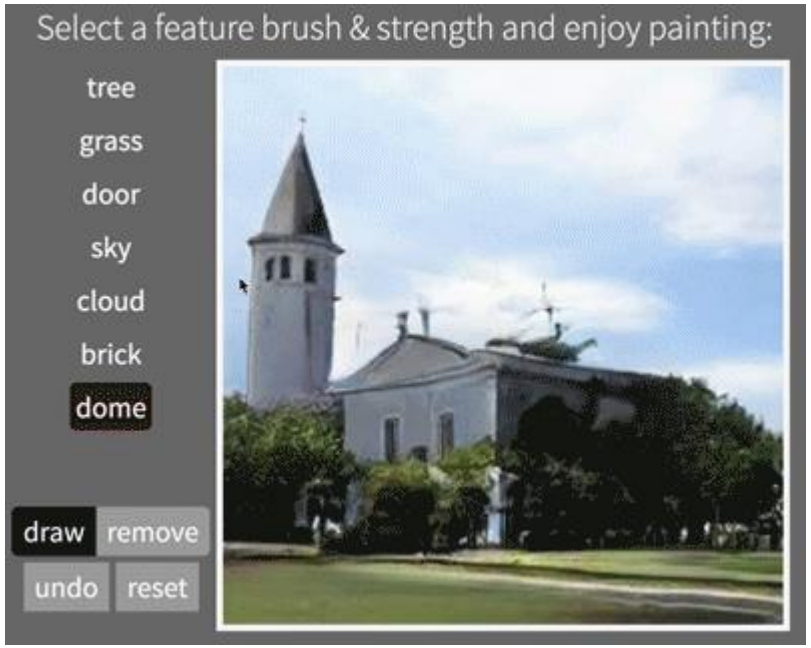
https://thispersondoesnotexist.com/

https://whichfaceisreal.com/

https://thisxdoesnotexist.com/

GAN Dissection

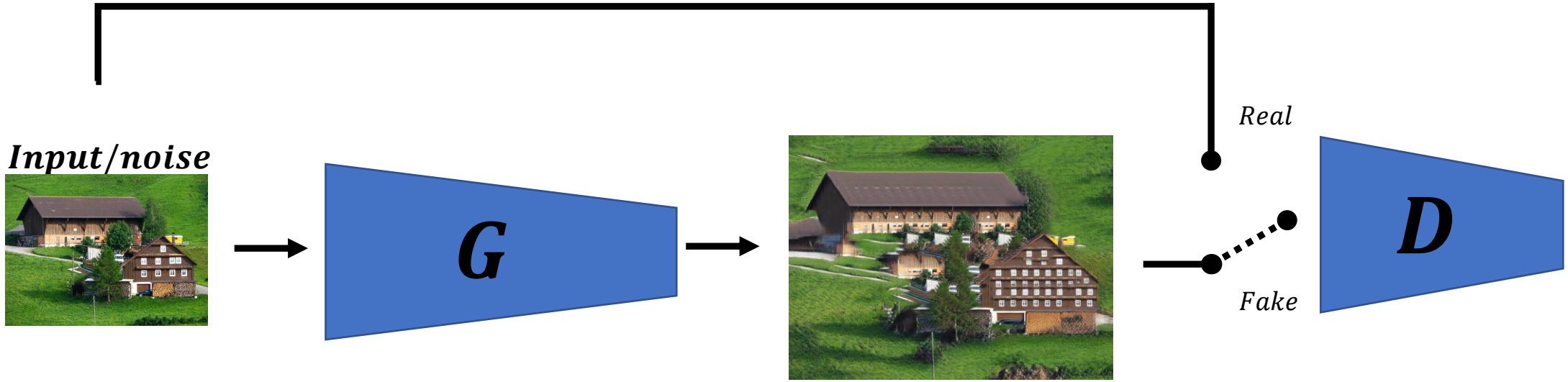




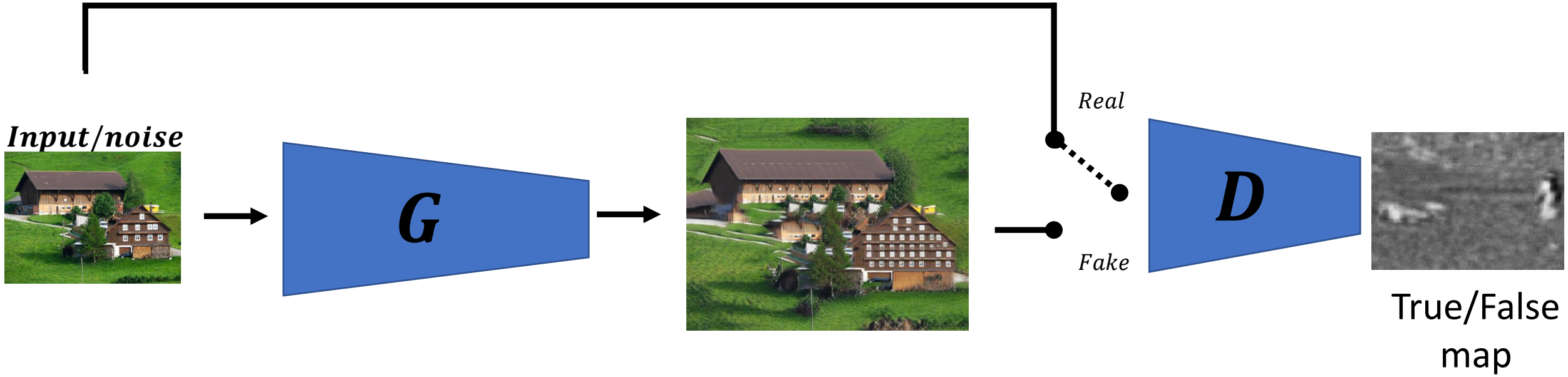
David Bau, Jun-Yan Zhu, Hendrik Strobelt, Bolei Zhou, Joshua B. Tenenbaum, William T. Freeman, Antonio Torralba

http://gandissect.res.ibm.com/ganpaint.html?project=churchoutdoor&layer=layer4

Training a GAN on a single image

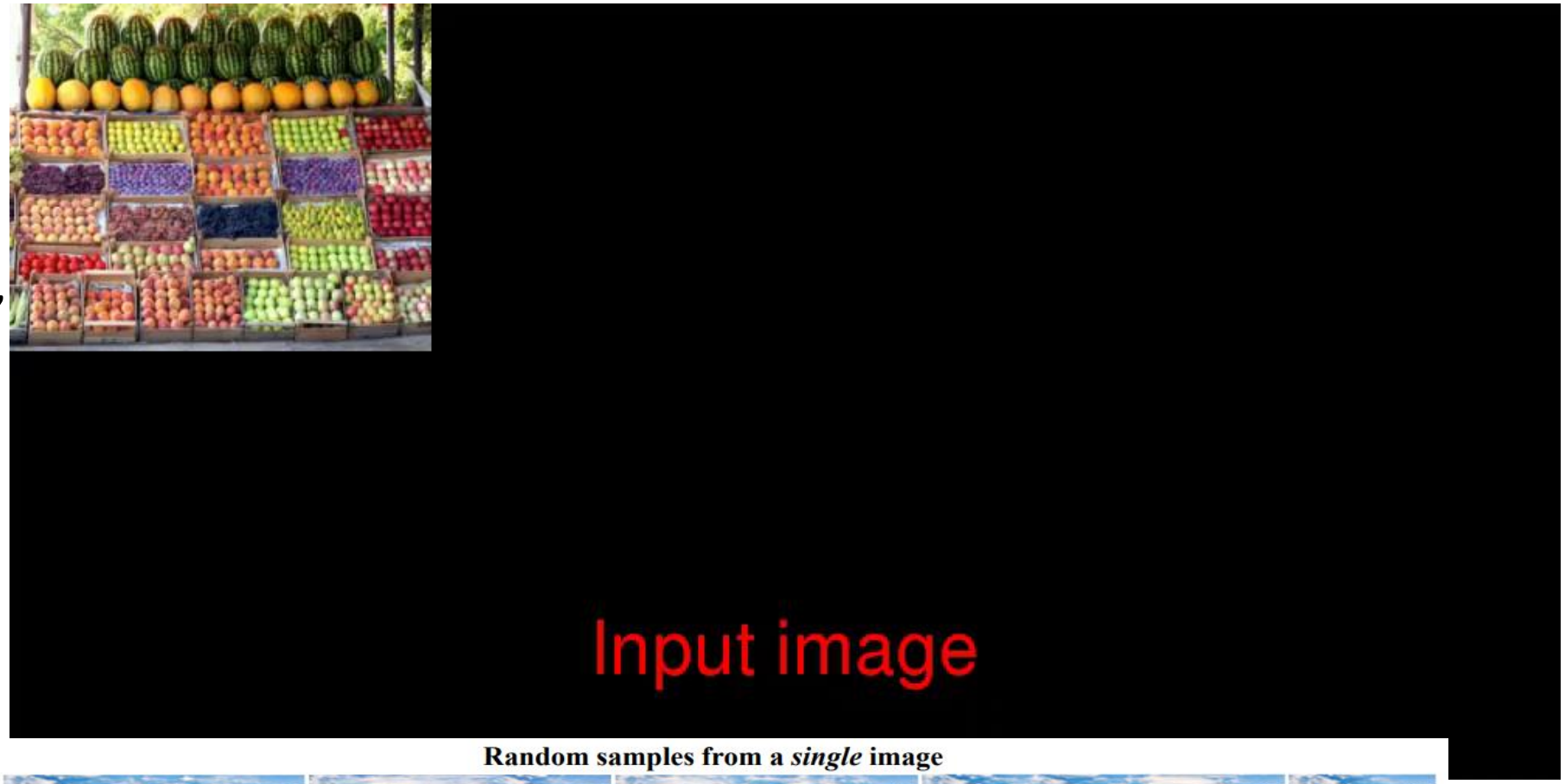


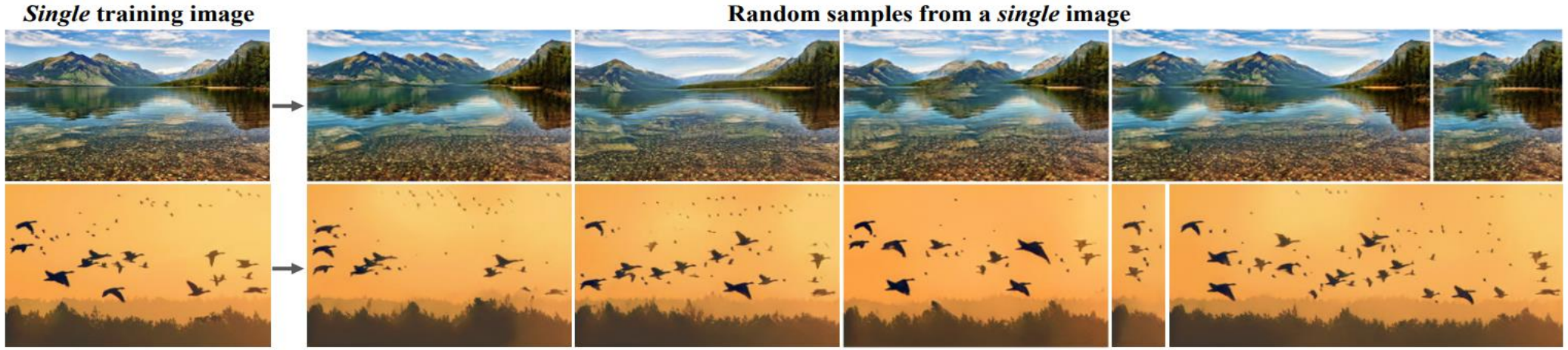
Training a GAN on a single image



Training a GAN on a single image

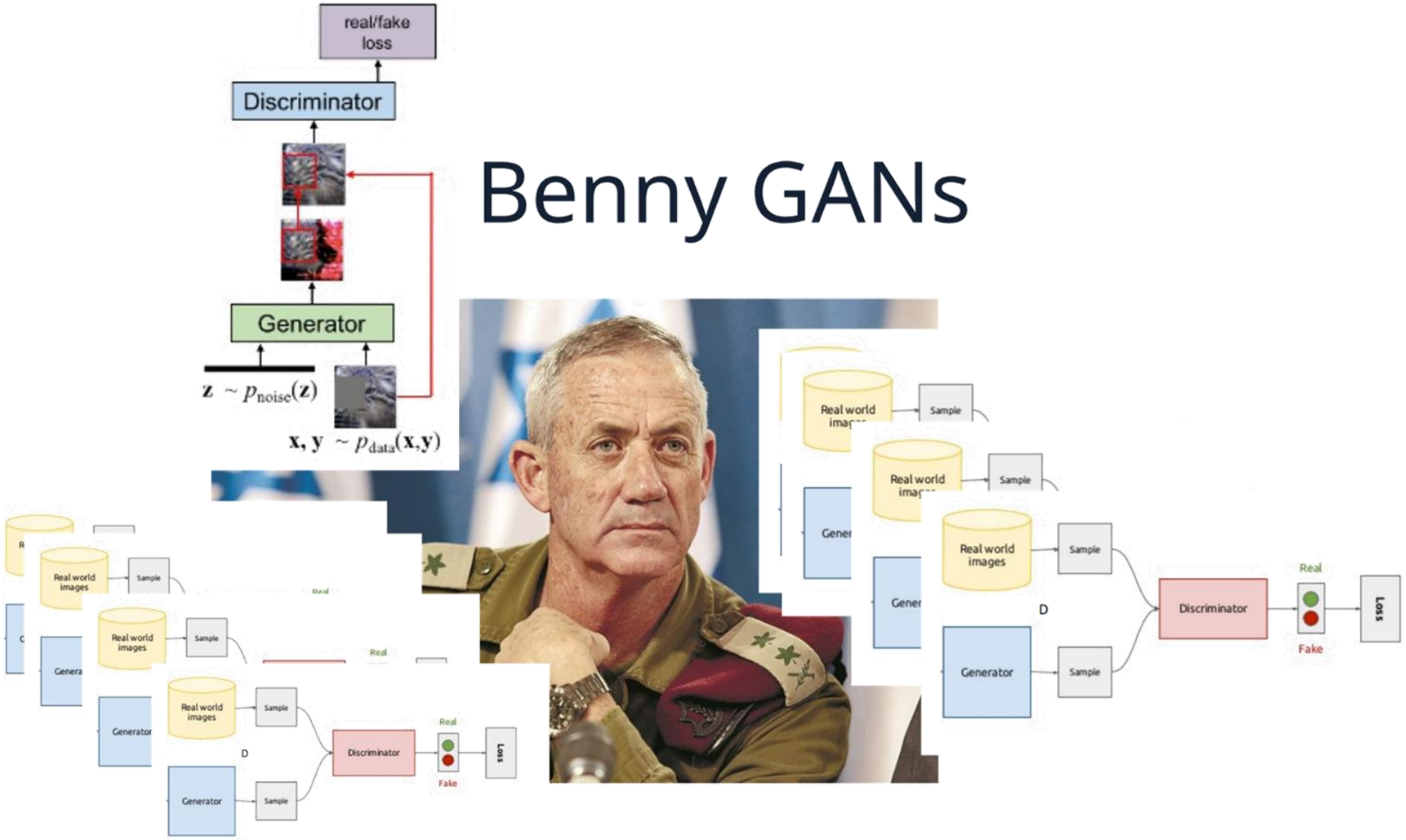
InGAN (Shocher, Bagon, Isola, Irani)





SinGAN (Rott-Shaham, Dekel, Michaeli)

Thanks


real/fake loss
Discriminator
Generator
$\mathbf{z} \sim p_{\text{noise}}(\mathbf{z})$
$\mathbf{x}, \mathbf{y} \sim p_{\text{data}}(\mathbf{x}, \mathbf{y})$
Benny GANs
Real world images
Sample
Generator
Discriminator
Real
Fake
Loss
D